

GSFC NEON Series-1 CMO

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Near Earth Orbit Network Series-1 Project Sounder for Microwave-Based Applications Performance Requirements Document



NOAA/NASA

**Goddard Space Flight Center
Greenbelt, Maryland**

**Near Earth Orbit Network
Series-1 Project
Sounder for Microwave-Based Applications
Performance Requirements Document
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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

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Change History Log

Revision	Effective Date	Description of Changes
Revision –	February 7, 2023	This version incorporates 470-CCR-22-0372 which was approved by the SMBA ERB on December 22, 2022, and by the SMBA CCB on February 7, 2023.
Revision A	December 20, 2024	This version incorporates 470-CCR-24-0402 which was approved by the SMBA ERB on March 20, 2024, and incorporates NS1-CCR-0001 by the NS1 SMBA CCB on the effective date shown. The Document number has changed from 473-SMBA-RQMT-00004 to NS1-SMA-REQ-0001 as the document has transferred from the Management Information System (MIS) to the Technical Data Management System (TDMS). The Document name has changed from SMBA Instrument Performance Specification Document (PSD) to SMBA Instrument Performance Requirements Document (PRD). Revised the following: Requirement identifiers and numbering, Section 1.0 and its subsections, Section 4.0 and its subsections, Sections 5.1, 5.4, 5.5, 5.6, 5.7, Section 7 and its subsections & Section 8.2. Added Table 6.2-2. Moved Acronym List from Appendix A to Appendix B. Added Appendix A VCRM. Updated the following: Tables 2.1-1, Table 2.2-1, Section 3.1, Figure 3.1-1 Table 6.2-1, & Acronym List Editorial changes throughout.
Revision B	April 17, 2025	This version incorporates NS1-CCR-0009 & NS1-CCR-0011 which was approved by the NS1 SMBA CCB on the effective date shown. Figure 5.4.2-1 Pointing Terminology: FROM: Accuracy (Control) TO Accuracy

		Pointing Terminology definitions FROM Pointing Accuracy (Pointing Control) TO Pointing Accuracy Table 5.4.2-1 End-to-End Pointing Knowledge FROM +/-290 arcsec 3sigma TO +/-265 arcsec 3 sigma Table 5.4.2-1 Spacecraft Pointing Knowledge FROM +/-75arcsec 3sigma TO +/-112 arcsec 3sigma Editorial change all instances of climate to weather.
Revision C	August 19, 2025	This version incorporates NS1-CCR-0012 which was approved by the NS1 SMBA CCB on the effective date shown. Update to the PRD in response to contract-type change to Draft SMBA RFP and the de-classification of the SMBA Instrument. The change in procurement strategy has eliminated the IMAR as a stand-alone document. Critical portions of the IMAR are ingested into the SOW and PRD. This CCR incorporates mission assurance design requirements into the PRD and points to SOW sections for other mission assurance requirements. This CCR removes the SMBA instrument classification of Class C and replaces it with a customized risk posture statement. This CCR clarifies and reduces the number of requirements that are to be verified at Satellite level by moving requirements for Aliveness testing, LPTs, CPTs, and GPTs to the SOW. This CCR clarifies requirements about science performance, flight Software soft reset, orbital debris, and radiation. It also updates Section 7 subsection titles and the acronym list. This CCR adds a new Spacecraft-to-Instrument command & data transfer interface requirement. Editorial changes throughout.
Revision D	January 15, 2026	This version incorporates NS1-CCR-0015 which was approved by the NS1 SMBA CCB on the effective date shown. Updates to the SMBA PRD are in response to a) industry partner responses to Draft RFP posted 4/18/2025 (NS1-SMBA-REQ-0001 Rev. B) and to Draft RFP Update posted 9/5/2025 (NS1-SMBA-REQ-0001 Rev. C); b)

		internal NASA technical discussions; and c) NASA program and project changes This CCR affect all sections of the PRD. Major performance, function, and test requirement revisions are as follows: Relaxed the V-band narrowest hyperspectral resolution to only be around the peaks of the oxygen absorption spectrum (Table 4.1-2). No longer requiring heritage channels in unprotected spectrum (Table 4.1-3) Removed 51.76 GHz (ATMS Ch 4) and modified lowest F-Band channel from 114.5 GHz to 114.75 GHz Added heritage channels to cover all of MWS channels (Table 4.1-3), which are 53.246 ± 0.08 GHz (MWS Ch 5), 53.948 ± 0.081 GHz (MWS Ch 7), and 89 GHz (MWS 17) Added beam polarization requirements to heritage changes (Table 4.1-3) and all SMBA channels (Section 4.8.1) Refined the aspects of the RFI detection capability that are required to be reprogrammable (SMBA-7) Requiring RFI detection packet instead of binary flag (SMBA-13, SMBA-14) Removed RFI metrics requirement, since metrics will be computed by NOAA Ground System (NGS) Aligned System Calibration Uncertainty definition, contributors, and accounting with NIST, BIPM, and JPSS to allow SMBA designers freedom to develop their own calibration uncertainty, accuracy, precision without bias towards a particular design (Section 4.3, Table 4.1-4) Redefined Radiometric Sensitivity (NEDT) as having two contributions: white thermal noise and $1/f$ noise (Section 4.4, Section 4.7, Table 4.1-4) and allowing the white thermal noise contribution to be scalable for bandwidth and integration time. Deleted vertical reporting interval requirement (old SMBA-43) and agreed with industry partners that it is out of scope for the instrument Resolved the TBDs for SRF requirements (Section 4.9) Storage period from 10 years to 13 years (SMBA-109) Instrument storage maintenance period from 30days to 12 months (SMBA-110) Satellite storage maintenance period from 30days to 12 months (SMBA-118)
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		Added instrument fundamental resonant frequency requirement (> 100 Hz) due to varying loads applied by different Flight Elements (SMBA-149) Bus voltage requirement became a “should” statement to avoid driving instrument design (SMBA-177) EMI/EMC Testing no longer requires CS116 test due to no identified mechanism for this threat at the satellite level Rescoped the required TVAC cycles from 12 cycles before launch to 8 cycles before instrument delivery (SMBA-257) Relaxed TVAC thermal balance testing of all units to only requiring testing for FM1 and any flight units with design changes that affect thermal performance (SMBA-268) Other changes clarify the PRD requirement set and do not significantly change the PRD intent (e.g., revising requirement language, adding context, restructuring sections, and moving requirements to SOW/CDRL). The requirements are renumbered in sequential order. Editorial changes throughout.
Revision E	March 11, 2026	This version incorporates NS1-CCR-0018 which was approved by the NS1 SMBA CCB on the effective date shown. Updates to the SMBA PRD Rev E are in response to a) industry partner responses to Industry Day 1/21/2026, where changes from NS1-SMBA-REQ-0001 Rev. C to Rev. D were summarized; b) internal NASA technical discussions; and c) NASA program and project changes . This CCR affects Section 4 and Section 6 of the PRD. Major requirement revisions are summarized as follows: Extended the J-band range to include up to the protected spectrum up to 238 GHz and it aligns with Table 6.2-2 RS to In-band transmitters which requires analysis up to 238 GHz. Relaxed the V-band narrowest hyperspectral resolution requirement to be around the two heritage channel absorption peaks with an objective to extend the narrow

		resolution to additional absorption peaks as design permits. (Table 4.1-2). Removed the linear polarization direction for heritage channels (Table 4.1-3 and Rev D SMBA-35). Modified radiometric sensitivity requirements by adding NEDT requirement for heritage channels (SMBA-3, Table 4.1-3 & notes, Section 4.4 text) and adjusting Thermal Noise Contribution to NEDT values (Table 4.1-4 & notes). Modified text and added Equation 6 for 1/f noise ratio computation. (Section 4.7) Relaxed SRF requirements (Section 4.9) to align with test equipment capabilities. Updated Table 6.2-1 & notes, SMBA-188, SMBA-189, and SMBA-190 to address instrument component survival at the Launch Site and instrument operate-through exposures from external transmitters while at the Launch Site & On-Orbit. Launch Site includes Factory, Transport, PPF, Launch Vehicle, and Ascent. Reduced upper end of spectrum to be tested and/or analyzed from 325 GHz to 238 GHz to align with Table 4.1-1. Changed TBS to TBD for Table 6.2-2 for spectrum where the instrument may be sensitive in unprotected spectrum and may be exposed to external transmitters. Updated Appendix B Acronyms The requirements and equations are renumbered in sequential order. Editorial changes throughout.
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Table of Deviations/Waivers

Item No.	Deviation/Waiver	Waiver Approved	Effectivity

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date
1	2.1	Table 2.1-1 Applicable Document List - NS1-SMBA-xxx-xxxx - NEON Series-1 Spacecraft to SMBA Instrument Interface Control Document (TBS)	Instrument Provider	CDR
2	2.1	Table 2.1-1 Applicable Document List - NS1-SMBA-xxx-xxxx - NEON Series-1 Spacecraft to SMBA Instrument Mechanical Interface Control Document (TBS)	Instrument Provider	CDR
3	4.8.4	SMBA-41 Channels shall have peak- to-peak gain variation within each passband no greater than 1.0 dB (TBR) for K, Ka, and V bands, 1.5 dB (TBR) for W, F, D, G, and J bands.	ISE	PDR
4	4.8.5	SMBA-47 The total scan period shall be TBS seconds $\pm$ TBS msec.	Instrument Provider	PDR
5	5.1	SMBA-104 Allowable heat transfer to the spacecraft during science operations is not to exceed 175 W (TBR) at an interface temperature of +15 °C (TBR).	ISE	CDR + 60 days
6	5.1	SMBA-105 Uncompensated momentum contribution of ± 0.3 N-m- sec per axis (TBR).	ISE	CDR + 60 days
7	5.4	SMBA-112 The initial data rate allocations of the instrument shall not exceed 2 Mbps orbital average and not exceed 10 Mbps peak (TBR).	ISE	CDR + 60 days
8	5.4.2	Table 5.4.2-1 Mission End-to-End Pointing Requirements: Pointing Accuracy (TBR), Pointing Knowledge (TBR)	ISE	CDR + 60 days
9	5.6.1.4	Spacecraft ECI to spacecraft Control Frame and/or a time-tagged spacecraft ephemeris (ECEF) with a minimum frequency of 8 Hz (TBR) if required by the instrument to meet performance requirements.	ISE	CDR + 60 days

10	5.6.1	Table 5.6.1-5 Suggested Instrument Calibration Maneuver Requirements (TBS) and SMBA-125	Instrument Provider	CDR + 60 days
11	5.7.2	SMBA-151 Worst case expected launch load of 15 g independently in each axis (TBR).	ISE	CDR + 60 days
12	5.7.3.3.1	SMBA-169 Open loop gain margin no less than 12 dB within $\pm 10\%$ of structural mode frequencies, and no less than 6 dB at other frequencies. (TBR)	ISE	CDR + 60 days
13	6.2	Table 6.2-1 Radiated Susceptibility Levels Due to Factory/Transport, Launch Site, Launch Vehicle, Ascent, and On-Orbit Phases (TBR), values (TBD/TBR), Note [6] (TBR)	ISE	PER
14	6.2	Table 6.2-2 Radiated Susceptibility Levels Due to In-Band Sources in On-Orbit Phases (TBD/TBR/TBS) and values (TBD/TBR/TBS)	ISE Instrument Provider	PER
15	7.3.2	Table 7.3.2-1 Spacecraft Transmitter Frequency Susceptibility Test Levels (TBD/TBR) and values (TBD/TBR)	ISE Instrument Provider	PER
16	7.3.2.3	Table 7.3.2-2 Intentional and Unintentional Radiated Electric Field Emissions Limits at the SMBA Instrument Interface, 18 to 235 GHz (TBR) and values (TBR/TBD/TBS)	ISE Instrument Provider	PER
17	7.3.2.3	Table 7.3.2-3 SMBA Receive Sensitivity (TBR), values (TBR/TBS), and Notes (TBS)	ISE Instrument Provider	PER
18	7.3.2.3	Receive Sensitivity (TBR) degrees from the aperture boresight(s)	ISE	PER
19	7.6.3	SMBA-272 outgas at a rate greater than $4.62\text{E-}7$ gram/second (TBS).	Instrument Provider	PER
20	Appendix A	Verification Cross Reference Matrix (VCRM) [TBS]	Instrument Provider	CDR

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1 INTRODUCTION

1.1 Purpose

This Sounder for Microwave-Based Applications (SMBA) Instrument Performance Requirements Document (PRD) specifies instrument-level functional, mission assurance, safety, design, performance, integration, test, and delivery requirements for the instrument provider.

1.2 Document Scope

This specification establishes the requirements for the SMBA instrument. The scope of the PRD includes all functional, design, performance, integration, test, and delivery requirements for the instrument, but does not include mission-specific accommodation and interface requirements for the instrument on the spacecraft. The instrument provider is responsible for developing and maintaining a hierarchy of lower-tier specifications that derive and allocate requirements to subsystems and components.

1.3 NEON Program and NEON Series-1 Project Overview

Over 2016-2018, the National Environmental Satellite, Data, and Information Service (NESDIS) initiated the National Oceanic and Atmospheric Administration (NOAA) Satellite Observing System Architecture (NSOSA) study to determine the most cost-effective space architectures for NOAA's future weather environmental remote sensing missions. Among the results of the NSOSA study was a confirmation for the continuing and evolving need for global environmental observations from low earth orbit (LEO). In response, NESDIS defined the Near Earth Orbit Network (NEON) program to meet existing and emerging global observational needs from LEO. Organizationally, the NEON program and its projects reside within the NESDIS Office of LEO Observations and the National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (GSFC) LEO Programs Division (LPD). The NEON Series-1 (NS1) Project is within the NEON Program.

The NS1 Project will field the next generation LEO microwave (MW) sounding observational capabilities that support Department of Commerce (DOC) Primary Mission Essential Functions (PMEF), and NOAA strategic objectives associated with terrestrial weather prediction and warning, healthy oceans, and resilient coastal communities and economies. NS1 is the follow-on microwave instrument continuity mission to the Joint Polar Satellite System (JPSS) Advanced Technology Microwave Sounder (ATMS) observations. Satellite-based Microwave temperature and humidity soundings to be provided by the NS1 satellites are the two largest impact sources of observations for forecasts. NS1 will utilize microwave sounding to provide necessary atmospheric state information for numerical weather prediction and other applications. The National Weather Service's weather products need three-dimensional atmospheric state information which is provided by atmospheric sounding in the form of vertical profiles (state) of atmospheric water vapor and temperature.

The NS1 Project scope includes satellites that host low Earth orbit (LEO) Earth-observing microwave sounding instruments, related ground services (i.e., command and control, and data routing, processing, distribution, and archival), and launch services. The NS1 mission consists of flight, launch, and ground segments along with the interfaces necessary for the collection of

environmental data. This includes the periodic data downlink to ground stations and continuous Direct Broadcast (DB) allowing for data product generation and distribution to users. NS1 consists of single instrument satellites operating in a sun synchronous orbit each with a design life of 5-years. It includes development, integration, test, launch, and science operations of the Sounder for Microwave-Based Applications (SMBA) instrument. The SMBA instrument will produce global atmospheric temperature and moisture profiles allowing the NEON Program to provide science continuity with ATMS while introducing the addition of more frequency bands, hyperspectral data collection, and Radio Frequency Interference (RFI) detection capabilities.

NS1 will utilize best practices and processes of the commercial space industry to meet Federal program level requirements while delivering these critical SMBA hyperspectral microwave sounding observation capabilities utilizing small satellites that can be flexibly deployed.

1.4 The Sounder for Microwave-Based Applications

The SMBA instrument is the principal instrument aboard the NEON Series-1 satellites. The instrument (hereafter referred to as the 'SMBA instrument' or simply the 'instrument'), is a cross-track microwave sounder that collects hyperspectral microwave radiometric data that are used to calculate the vertical distribution of temperature (T), moisture (q), and pressure in the Earth's atmosphere.

The instrument measures atmospheric radiances within the microwave range of approximately 20-300 GHz over a swath width of approximately 2,500 km from an altitude of 824 km.

Hyperspectral capability indicates fine spectral resolution, which is beneficial for both science and radio frequency interference (RFI) detection capability. Hyperspectral microwave capability refers to microwave sounding systems with approximately 100 or more channels. The increased number of channels results in a finer spectral resolution than is achieved by traditional microwave sounding techniques over the same bandwidth.

Instrument data is acquired continuously, stored on-board the satellite, and subsequently downlinked to a ground network for capture, preprocessing, and routing to meteorological centers.

The instrument produces the application packets as inputs to the Raw Data Records (RDR). Engineering, instrument telemetry, science, radiometric, and spectral calibration data are also provided to enable the RDRs to be calibrated and geolocated. Together these enable calibration and geolocation. Additionally, diagnostic data may be acquired. The application packets are sent to the National Oceanic and Atmospheric Administration (NOAA) Ground System where they are used to help generate higher level products such as RDRs, Temperature Data Records (TDR), Sensor Data Records (SDR), and Environmental Data Records (EDR).

The instrument system performance requirements are provided in subsequent sections.

1.5 Requirements Definitions

Throughout this PRD:

-
- Contractually imposed mandatory requirements are identified by “shall” statements with requirement identification numbers of the form SMBA-XXXX. Any deviation from these requirements requires approval of the Government Contracting Officer.
 - Preferred good practices are identified by “should” statements. “Should” designates performance, which is requested, but not mandatory. Noncompliance of a “should” statement does not require Government approval but shall be addressed at the earliest applicable project milestone.
 - Permission is identified by “may” or “can” statements.
 - Expected outcome or action is identified by “will” statements.
 - The term “instrument” refers to the SMBA instrument which may be constructed of one or more components that are individually mounted to the spacecraft.
 - The term “component” is used in various ways throughout this document. For structural requirements, the term component is used to describe the instrument whether it be a singular “component” or a suite of interconnected components that mount to the spacecraft. For all other sections, the term “component” is used more broadly to describe lower level parts or an aggregation of parts into a small functional unit (e.g., parts, sensors, switches, relays, small mechanisms, etc.).
 - The term “spacecraft bus” refers to the NEON Series-1 spacecraft without instruments.
 - The term “satellite” refers to the NEON Series-1 spacecraft bus integrated with the flight instrument(s). The term “satellite” is equivalent to the term “observatory” used in other NASA and NEON Program documents.
 - The term “spacecraft” may refer to the spacecraft bus.
 - The term “instrument provider” refers to the SMBA instrument contractor with whom the Government has entered into a contract to execute the SMBA instrument Statement of Work (which is titled “Sounder for Microwave-Based Applications (SMBA) Instrument Statement of Work (SOW) (NS1-SMBA-SOW-0002)”) and satisfy the requirements of this PRD.
 - The term “Government” represents the appropriate NOAA and/or NASA NEON Program or NEON Series-1 Project management office or, where “the Government” is indicated as providing a function for or interface to the SMBA instrument provider, the function or interface may be provided by Government support services contractors.
 - The term “TBR” (to be reviewed) means that the stated information will be reviewed for appropriateness jointly by the instrument provider and the Government and the information may be changed prior to final definition by the Government during the course of the contract.
 - The term “TBD” (to be determined) indicates further research or analysis is needed to determine the information during the course of the contract.
 - The term “TBS” (to be supplied) indicates information that will be supplied/specified during course of instrument’s development.

2 APPLICABLE/REFERENCE DOCUMENTS

2.1 Applicable Documents

The applicable documents listed below are called out within individual requirements specified in the PRD. All or portions of these documents apply directly to the performance required and contain provisions that constitute requirements of this PRD. In the event of a conflict between the language in an applicable document and in this PRD, the PRD will govern.

Table 2.1-1. Applicable Document List

Document Number	Revision	Document Title
NS1-SMBA-xxx-xxxx	*	NEON Series-1 Spacecraft to SMBA Instrument Interface Control Document (TBS)
NS1-SMBA-xxx-xxxx	*	NEON Series-1 Spacecraft to SMBA Instrument Mechanical Interface Control Document (TBS)
470-LEO-ANYS-00018	Rev -	Radiation Environment for the NEON Program
ISO 14644-1	May 1, 1999	Cleanrooms and Associated Control Environments
MIL-STD-461C	C August 4, 1986	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference (EMI)
MIL-STD-461G	G December 11, 2015	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-462	Rev - Notice 2 from May 1, 1970	Measurement of Electromagnetic Interference Characteristics (Notice 2)
NASA-STD-5001	B w/ change 3	Structural Design and Test Factors of Safety for Spaceflight Hardware
GSFC-STD-7000 Section 2.5.2.1.1 Section 2.5.2.1.2 Section 2.5.2.2.3	B	General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects
NASA-STD-8719.14	C	NASA Process for Limiting Orbital Debris
NASA-STD-8719.24	Rev A, Annex Rev B with Change 1	NASA Payload Safety Requirements
NPR 8715.6	B	NASA Procedural Requirements for Limiting Orbital Debris

*See Contract Attachment N, Applicable Documents List, for current revision

2.2 Reference Documents

The reference documents listed below are referred to within the PRD and contain information relating to the work required. The documents do not contain provisions that constitute requirements of this PRD, but rather serve to amplify or clarify the requirements.

Table 2.2-1. Reference Documents List

Document Number	Revision	Document Title
ASTM E595	5/14/2021	Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment
CCSDS 301.0-B-4	November 2010	Consultative Committee for Space Data Systems (CCSDS) Time Code Formats Standard
GSFC-STD-7000	B	General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects
IEST-STD-CC1246	E	Product Cleanliness Levels and Contamination Control Program
472-00228	B	JPSS Flight Project Contamination Control Plan
JSC-CR-06-070	August 2006	Space Vehicle RF Environments
NASA-STD-5017	B	Mechanism Design Requirements
http://outgassing.nasa.gov		Outgassing Data for Selecting Spacecraft Materials
https://www.bipm.org/en/committees/jc/jcgm/publications		Joint Committee for Guides in Metrology. (2012). <i>International vocabulary of metrology: Basic and general concepts and associated terms (VIM)</i> (3rd ed., JCGM 200:2012). Bureau International des Poids et Mesures.
http://www.itl.nist.gov/div898/handbook/		NIST/SEMATECH e-Handbook of Statistical Methods
E. Kim et al., "An Evaluation of NOAA-20 ATMS Instrument Pre-Launch and On-Orbit Performance Characterization"		E. Kim et al., "An Evaluation of NOAA-20 ATMS Instrument Pre-Launch and On-Orbit Performance Characterization," in IEEE Transactions on Geoscience and Remote Sensing, vol. 60, pp. 1-13, 2022, Art no. 5302813, doi: 10.1109/TGRS.2022.3148663.
D0001-M01-S01-001_JPSS_ATBD_ATMS_SDR_B	Version 1	NOAA ATMS SDR ATBD https://www.star.nesdis.noaa.gov/jpss/ATMS.php

3 GENERAL REQUIREMENTS

3.1 On-Orbit Coordinate System

The NEON Series-1 satellite's on-orbit coordinate system uses a right-hand, orthogonal, body-fixed XYZ coordinate system as follows (see Figure 3.1-1): the +Z-axis is downward towards geodetic nadir, the Y-axis is along the orbit normal plane (+Y is opposite the orbital angular momentum), and the X-axis is along the spacecraft velocity vector (+X toward the direction of spacecraft travel).

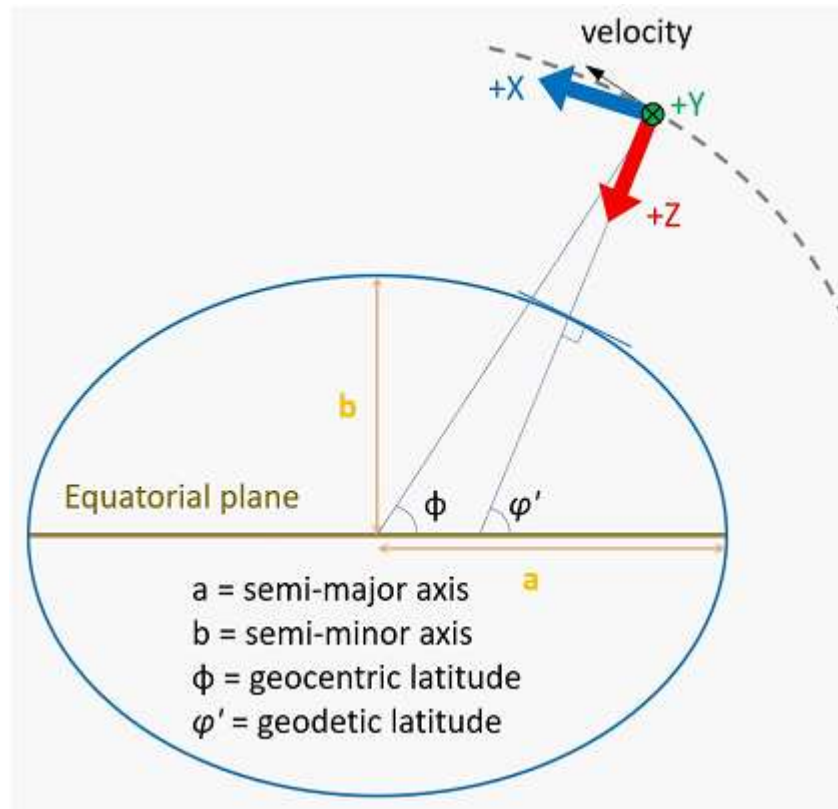


Figure 3.1-1. On-Orbit Coordinate Systems

NEON Series-1 Project uses J2000-referenced orbital elements for attitude knowledge and the WGS84 geodetic reference system for environmental data geolocation.

The instrument should use the international system of units, *Système Internationale* (SI), unless precluded by design or manufacturing heritage. All interface documents should provide units in SI and English. The primary (as-designed) units should be displayed above the converted units with the latter in parentheses.

3.2 Mission Time Convention

SMBA-1	The instrument shall time stamp packets in the secondary header using Coordinated Universal Time (UTC), including the leap-second convention, also known as CCSDS Day Segmented Time Code (CCSDS 301.0-B-4), for all instrument telemetry while the instrument processor is running.
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Note: Telemetry frames that are generated by hardware during ascent or processor down time may format the time stamp using CCSDS Unsegmented Time Code (CCSDS 301.0-B-4).

4 SMBA SCIENCE REQUIREMENTS

In Section 4, all requirements apply over the life of the instrument.

4.1 SMBA Channel Requirements

A frequency band is a portion of the electromagnetic spectrum associated with a letter designation. For typical use in SMBA, refer to the frequencies and associated letter designation in Table 4.1-1.

Table 4.1-1. SMBA Letter Designations for Frequency Bands

Letter Designation	Frequency Range	Band Purpose
K band	20 - 25 GHz	Column Integrated Observations & Quality Control
Ka band	31 - 33 GHz	Window
V band	50 - 62 GHz	Temperature (T)
W band	85 - 92 GHz	Window
F band	108 - 123 GHz	Temperature (T)
D band	164 - 167 GHz	Window
G band	173 - 193 GHz	Humidity (q)
J band	200 - 238 GHz	Window & Ice Cloud Detection

The SMBA instrument design may cover a subset of the frequency range associated with a letter designation. The frequency range is not prescribing precise start and stop frequencies or bandwidths. The instrument frequency plan should take into consideration required heritage channels, additional hyperspectral channels, science benefits, RFI detection methods, downlink throughput, hardware constraints, International Telecommunication Union (ITU) allocations, etc. Science benefits might include, for example, using more hyperspectral bandwidth in a frequency band than utilized by the heritage channels to improve Noise Equivalent Delta Temperature (NEDT) or use more hyperspectral bandwidth to characterize features in the oxygen and water vapor lines.

A channel is the portion of a frequency band defined by a spectral response function (SRF). For the SMBA instrument, SRF is defined as a magnitude of energy versus frequency transfer function, where the input is at the antenna aperture and the output is at the instrument output. The output of the instrument for a channel corresponds to the integrated energy under the SRF. A channel is only within one frequency band and does not span across multiple frequency bands.

Center frequency is defined as the median frequency between the half-power points of a passband.

Bandwidth is defined as the spectral width of a passband, where the width is the frequency range between the half power points that define the start and stop frequencies. The total bandwidth of a channel is the sum of the bandwidths of the passbands comprising the channel.

Center frequency stability is defined as the maximum deviation from the center frequency over the operational life of the instrument.

Hyperspectral capability indicates a fine spectral resolution of the electromagnetic spectrum within a given specified frequency band. Fine spectral resolution enabled by a hyperspectral capability will be beneficial to address science needs and RFI detection needs (Section 4.2). While hyperspectral capabilities will allow finer spectral resolution, the instrument need not downlink all of the collected bandwidth. The downlinked channels should be a balance between data volumes and spectral resolution.

SMBA-2 Hyperspectral capability shall be provided for the frequency bands identified in Table 4.1-2.

Table 4.1-2. Hyperspectral and RFI Detection Capabilities by Band

Frequency Bands	Hyperspectral Capability	RFI Detection Capability	Hyperspectral Resolution (MHz) ^[4]
K	Required ^[1]	Required	≤ 10.0
Ka	Required ^[1]	Required	≤ 10.0
V	Required ^[2]	Required	≤ 1.0 for spectrum within 30 MHz of the absorption peaks at 56.9682 GHz and 57.6125 GHz, with the objective to extend this capability to additional absorption peaks as design permits. ≤ 10.0 elsewhere
W	Required ^[1]	Required	≤ 20.0
F	Required ^[2]	Not Required	≤ 20.0
D	Not Required	Not Required	N/A
G	Required ^[3]	Not Required	≤ 100.0
J	Not Required	Not Required	N/A

Notes:

[1] Capability required for Window Channels continuously in operational mode.

[2] 20 or more channels required in V band and 10 or more channels in F Band for Temperature Sounding continuously in operational mode.

[3] 10 or more channels required in G Band for Moisture Sounding continuously in operational mode.

[4] Hyperspectral Resolution also applies to RFI detection spectral resolution.

SMBA-3 For science continuity, the SMBA instrument shall provide the heritage channels at the center frequencies, bandwidths, and NEDT as identified in Table 4.1-3 with the channel performance of Table 4.1-4.

SMBA-4

In addition to the heritage channels, the instrument shall provide new hyperspectral and/or RFI detection channels within the same band that meet the channel performance of Table 4.1-4.

Note: New channels may use or reuse any frequency spectrum within the same band. New channels may be formed by aggregating the minimum hyperspectral resolutions. Aggregated channel bandwidths do not need to be uniform across the frequency band.

Table 4.1-3. NEON Program Microwave Sounder Heritage Channel Requirements

Center Frequency^[1] (GHz)	Bandwidth (GHz) +/- 5%	NEDT^[2] (K)
23.8	0.27	0.30
31.4	0.18	0.35
50.3	0.18	0.40
52.8	0.4	0.27
53.166	0.140	0.32
53.326	0.140	
53.481	0.17	0.29
53.711	0.17	
53.867	0.142	0.32
54.029	0.142	
54.4	0.4	0.27
54.94	0.4	0.27
55.5	0.33	0.30
57.290344	0.33	0.34
57.073344	0.078	0.50
57.507344	0.078	
56.920144	0.036	0.52
57.016144	0.036	
57.564544	0.036	
57.660544	0.036	
56.946144	0.016	0.78
56.990144	0.016	
57.590544	0.016	
57.634544	0.016	
56.958144	0.008	1.10
56.978144	0.008	
57.602544	0.008	
57.622544	0.008	
56.963644	0.003	1.79
56.972644	0.003	
57.608044	0.003	
57.617044	0.003	
88.2	2.0	0.20

89	4	0.15
114.75	1.0	0.30
115.95	0.8	0.30
116.65	0.6	0.40
117.25	0.6	0.40
117.80	0.5	0.40
118.24	0.38	0.50
118.58	0.30	0.50
165.5	3.0	0.30
176.31	2.0	0.28
190.31	2.0	
178.81	2.0	0.28
187.81	2.0	
180.31	1.0	0.40
186.31	1.0	
181.51	1.0	0.40
185.11	1.0	
182.31	0.5	0.57
184.31	0.5	
229	2.0	0.40

Notes:

[1] In the Center Frequency column, each boxed row is one channel and contains one or more passbands. The passbands in each row are the passbands to be aggregated to form the heritage channel. For example, Table 4.1-3 lists one channel with two passbands centered at 176.31 GHz and 190.31 GHz. Fewer passbands in a channel may be used, provided the Temperature Sensitivity requirements are met for the heritage channel bandwidth.

[2] NEDT values:

- (a) Assume a 300K scene temperature observation.
- (b) Assume no RFI in measurement.
- (c) Apply to the instantaneous field of view (IFOV) as defined in Section 4.4. Instrument provider to show NEDT computed values from the IFOV.

Note: Calibration accuracy, NEDT, and Thermal Noise Contribution to NEDT requirements are found in Sections 4.3 and 4.4, respectively. The NEDT in Table 4.1-3 includes both thermal noise and 1/f noise contributions and therefore are not to be scaled. The Thermal Noise Contribution to NEDT in Table 4.1-4 is to be scaled for integration time and bandwidth.

SMBA-5 The center frequency stability of the channels shall be less than or equal to the values listed in Table 4.1-4.

Table 4.1-4. NEON Program Microwave Sounder Channel Performance Requirements

Frequency Bands	Center Frequency Stability (MHz)	Total Calibration Accuracy (absolute value of E_{acc}) (K)	Thermal Noise Contribution to NEDT ^[1] (K)
K	5	1.0	19.8
Ka	5	1.0	18.9
V for < 56 GHz	0.3	0.5	21.6
V for $\geq$ 56 GHz	0.3	0.5	25.0
W	18	1.0	36.0
F	1	0.5	40.0
D	22	0.5	66.5
G	9	0.5	72.0
J	22	0.5	72.0

Notes:

[1] Thermal Noise Contribution to NEDT values:

(a) Assume a 300K scene temperature observation.

(b) Apply to an integration time of 1 msec and a bandwidth of 1 MHz. To scale the values for the actual integration time and bandwidth, divide the value by square root of (actual integration time in msec*actual bandwidth in MHz).

(c) Assume no RFI in measurement.

(d) Apply to the instantaneous field of view (IFOV) as defined in Section 4.4. Instrument provider to show NEDT computed values from the IFOV.

SMBA-6 In frequency bands requiring hyperspectral capability, the hyperspectral resolution shall be as specified in Table 4.1-2.
Note: Heritage channels (Table 4.1-3) have bandwidths greater than the minimum hyperspectral resolution (Table 4.1-2) and will be formed by aggregating the minimum hyperspectral resolutions. Aggregated channel bandwidths do not need to be uniform across the frequency band.

SMBA-7 For all hyperspectral and RFI detection bands, the number of channels, number of passbands, center frequency of each passband, bandwidth of each passband, and frequency resolution shall be reprogrammable on orbit.

4.2 Radio Frequency Interference Detection Capability

The global expansion of other ITU services into bands adjacent to and overlapping with SMBA bands poses a threat to the quality of the radiometric data and, therefore, to the accuracy of numerical weather prediction (NWP) forecasts and other products. Accurate RFI detection will allow NWP models and science community users to identify contaminated observations and act accordingly.

Table 4.1-2 requires on-board RFI Detection Capability for frequency bands that might experience RFI. Given the importance of these bands for science performance and continuity, the hyperspectral capability enhances the RFI Detection Capability of the SMBA instrument.

The on-board RFI Detection Capability should incorporate flexible, repeatable, and effective methods to identify and characterize the continually changing man-made sources of interference over the SMBA mission life.

SMBA-8 Detection of RFI by man-made signals that impact operational data products shall be reported for all spectral bands identified in Table 4.1-2. Note: Man-made signals may be continuous, pulsed, narrowband, broadband, etc.

RFI detection should occur at the appropriate processing level to allow assessment of the RFI detection algorithm and to maximize the preservation of science data.

SMBA-9 The instrument shall be capable of on-board RFI detection simultaneously with science data collection.

SMBA-10 The instrument shall be capable of concurrently executing multiple RFI detection algorithms in operational mode and diagnostic mode.

SMBA-11 In operational and diagnostic modes, the instrument shall be capable of collecting, storing, and relaying to the spacecraft raw data used for RFI detection. Note: Raw data can be used on the ground for assessment of RFI detection algorithms. Raw data refers to internal instrument data along the processing chain prior to the formation of channels. Raw data may include data in time domain and/or frequency domain.

SMBA-12 SMBA RFI detection shall have the capability to enable or disable detection functions through on-board scripts and ground commanding. Note: Examples of RFI detection functions include selecting frequency bands, selecting geolocations, and selecting type of raw data to downlink.

SMBA-13 When RFI is detected, an RFI detection packet shall be generated.

A SMBA RFI detection packet provides the information to compute metrics on RFI detection algorithm performance. Ground processing will produce science data quality flags and metrics based on the RFI detection packet.

SMBA-14 An RFI detection packet shall identify the algorithm that detected the RFI, threshold exceedance, frequency range corrupted, channel(s) affected, and time stamp. Note: Spacecraft provides time stamp and ephemeris data.

RFI detection is performed by processing the signal mathematically and comparing against a threshold. Given the evolving RFI environment, reprogrammability of the RFI detection capability will allow SMBA to adapt.

SMBA-15 On orbit, the SMBA instrument shall have the capability to reprogram at a minimum both new RFI detection algorithms and thresholds through ground command.

4.3 System Calibration Uncertainty

The SMBA definitions and requirements associated with calibration uncertainty are intended to be in accordance with the National Institute of Standards and Technology (NIST), Bureau International des Poids et Mesures (BIPM), and E. Kim et al., "An Evaluation of NOAA-20 Advanced Technology Microwave Sounder (ATMS) Instrument Pre-Launch and On-Orbit Performance Characterization." If discrepancies exist among definitions in these documents, the usage in this PRD takes precedence.

Uncertainty is defined as the total uncertainty comprised of the accuracy contributions and the precision contributions, as conceptually shown in Equation 1 below.

Accuracy is defined as the systematic portions of uncertainty that are constant or vary in a predictable manner, also known as bias. Bias is the statistical mean of the uncertainty and can be signed positive or negative.

Precision is defined as random portions of uncertainty that vary in an unpredictable manner. Random error is one-sigma standard deviation of the uncertainty.

Calibration uncertainty of SMBA is defined as the difference between the brightness temperature inferred from the microwave radiometer (referred to the antenna collecting aperture) and the true brightness temperature.

Figure 4.3-1 provides a notional structure on how the calibration uncertainty sources contribute to the System Calibration Uncertainty. Calibration uncertainty contributions may include, but are not limited to, those shown in Figure 4.3-1.

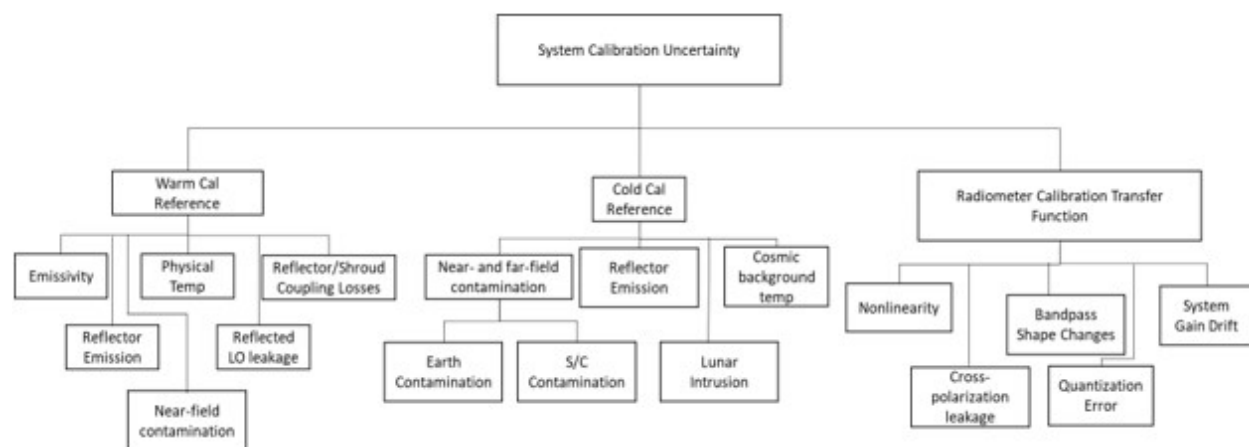


Figure 4.3-1. Notional Calibration Uncertainty Contributions

Notes:

-
- [1] A calibration reference is defined as an element that provides a signal that corresponds to a known brightness temperature. It can be a physical thing (like a blackbody; ex: on-board target or cold space). A noise diode is also an example of a reference.
- [2] Traditional radiometers have used two-point calibration methodology: cold space for the cold reference temperature and an on-board blackbody for the warm reference temperature. The instrument design is not constrained to this methodology or these calibration reference types. Other types of calibration references exist such as noise diodes and cold Field Effect Transistors (FET).
- [3] Proposed calibration references that are NIST-traceable must include a description of the traceability plan that enables them to be a standard, along with an estimate of the (absolute) accuracy.

Values for contributions in Figure 4.3-1 are characterized on the ground, on-orbit, and/or by analysis.

In a typical total power radiometer using a two-point calibration scheme, the system calibration uncertainty, E_{unc} , is conceptually described by Equation 1 below—i.e., it includes contributions from accuracy uncertainty and precision uncertainty. Different radiometer architectures and calibration schemes may necessitate different computations.

Equation 1: $E_{unc} = f(E_{acc}, E_{prec})$

Equation 2: $E_{acc} = \delta E_{hot} + (1 - \delta)E_{cold} + E_{sys}$

Equation 3: $E_{prec} = \sqrt{E_{prec,hot}^2 + E_{prec,cold}^2 + E_{prec,sys}^2}$

Equation 4: $\delta = (TB_{scene} - TB_{cold}) / (TB_{hot} - TB_{cold})$

E_{acc} : total calibration accuracy

E_{hot} : calibration accuracy related to hot reference

E_{cold} : calibration accuracy related to cold reference

E_{sys} : calibration accuracy in radiometer calibration transfer function

E_{prec} : total calibration precision; the 1 sigma root sum square (RSS) of the contributing individual precisions

TB_{hot} : measured brightness temperature of the hot reference

TB_{cold} : measured brightness temperature of the cold reference

TB_{scene} : measured brightness temperature of the observed scene

Total calibration precision, E_{prec} , excludes precision uncertainty from NEDT. Therefore, requirements in Section 4.3, System Calibration Uncertainty, exclude uncertainty from NEDT. Requirements related to the precision associated with NEDT (both white noise and 1/f noise) are considered separately in Section 4.4, Temperature Sensitivity, and Section 4.7, Noise Characteristics.

Requirements in Section 4.3, Calibration Uncertainty, apply on-orbit, after bias corrections, unless otherwise noted.

SMBA-16	For each channel, the absolute value of the total calibration accuracy, E_{acc} , shall be less than the values listed in Table 4.1-4.
SMBA-17	The orbital variation of accuracy for any single orbit shall be less than 0.2 K.
SMBA-18	The lifetime stability of orbital-mean accuracy shall be less than 0.2 K.
SMBA-19	For a given channel, the maximum calibration accuracy difference across the scan shall be less than 0.3 K. Note: Scan angle dependent bias characterizes the consistency of calibration accuracy within a scan.
SMBA-20	The magnitude of antenna temperature accuracy of the warm calibration reference shall be less than 0.20 K.
SMBA-21	The emissivity of any on-board and calibration test equipment (CTE) blackbody calibration references shall be greater than or equal to 0.99990 within an uncertainty less than 0.00003.
SMBA-22	The physical temperature spatial gradient across any warm load calibration reference shall be less than 0.2 K for any warm load view duration.
SMBA-23	The physical temperature measurement uncertainty of any single temperature sensor for on-board blackbody calibration references shall be less than or equal to +/- 0.1 K.
SMBA-24	The magnitude of antenna temperature accuracy of the cold space calibration reference shall be less than 0.5 K for K, Ka, and W bands, and 0.2 K for all other bands.

4.4 Radiometric Sensitivity

Radiometric sensitivity, NEDT, is defined as the standard deviation of the radiometer output in kelvins (K) when the antenna is viewing a 300 K uniform and stable target. The NEDT is computed as the root sum square of the thermal noise and 1/f noise contributions per Equation 5 below. Thermal noise, also known as white noise, is noise with a uniform power spectral density. 1/f noise, also known as pink noise or flicker noise, is noise with a power spectral density with a negative slope up to -1.

$$\text{Equation 5: } NEDT = [(NEDT_{thermal})^2 + (NEDT_{1/f})^2]^{1/2}$$

NEDT computations apply to the instantaneous field of view (IFOV) and are computed in accordance with D0001-M01-S01-001, NOAA ATMS SDR Algorithm Theoretical Basis

Document (ATBD), Section 4. The SMBA PRD has separate requirements for the thermal noise contribution to NEDT (to allow for scaling with integration time and bandwidth) and the 1/f noise contribution (specified as a ratio of 1/f noise to the total noise).

IFOV is the portion of the Earth's surface enclosed by the projected -3dB antenna contour at an instant of time (i.e., instantaneous footprint). An IFOV does not include smearing due to scanning or satellite orbital movement, nor any aggregating or spatial resampling of different footprints.

IFOVs are combined to form beam positions along scan. The integration time is the time interval between viewing the beam centers (boresight points) of successive beam positions along scan.

SMBA-25	The thermal noise contribution to the total radiometric sensitivity of the channels shall be less than or equal to the values listed in Table 4.1-4, when scaled for bandwidth and integration time.
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4.5 Channel Dynamic Range

SMBA-26	The dynamic range of the instrument shall envelope the brightness temperature range of Earth scenes, calibration references, and be sufficient to meet the RFI requirements in Section 4.2.
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SMBA-27	The output of each channel sample period shall have an uncertainty less than or equal 0.01% of full scale because of any previous signals in that channel or crosstalk from other channels.
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4.6 Channel Linearity

Nonlinearity is the departure from the expected response of an ideal linear radiometer. Channel nonlinearity is determined from a second-order or higher fit to the calibration accuracy data obtained during thermal-vacuum calibration testing. Residual nonlinearity is the residual error after applying a nonlinearity correction to the individual calibration accuracy points.

SMBA-28	As input to the computation of residual nonlinearity, the nonlinearity shall be computed for each channel.
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SMBA-29	The residual nonlinearity, after calibration corrections to any measurement within the dynamic range, shall be less than or equal to 0.1 K 1-sigma for all channels.
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4.7 Noise Characteristics

The 1/f noise is computed in accordance with D0001-M01-S01-001 NOAA ATMS SDR ATBD Section 4 Equation (11).

The 1/f noise ratio, $R_{1/f}$, is the power ratio of 1/f to the total noise is calculated as:

Equation 6: $R_{1/f} = \frac{(NEDT_{1/f})^2}{NEDT^2}$

SMBA-30 The 1/f noise ratio shall be less than or equal to 0.10.

The inter-channel noise correlation is measured by correlating the noise signal in one channel to the concurrent noise signal in another channel while viewing the same Earth scene or calibration reference, resulting in a correlation coefficient. The inter-channel noise correlation requirement only applies to channels that do not have overlapping SRF passbands.

SMBA-31 The inter-channel noise correlation coefficient shall be less than or equal to 0.15.
Note: A channel correlates perfectly with itself and has an inter-channel noise correlation coefficient of 1.00.

4.8 Beam Characteristics

Each frequency band of the instrument is considered to form an antenna beam. In the following sections, if only one beam is discussed, it refers to any and all beams.

The antenna beam electrical boresight is defined as the direction of maximum gain of the antenna.

The term "Beam Position" means the position of a "Beam-Center," which is defined as the position of the antenna beam electrical boresight at the mid-point of the integration time.

Beamwidth is defined as the angular width between the half-power points relative to the maximum gain of the antenna in the electric field plane that contains the boresight vector.

Antenna main beam is defined as the angular region around the boresight within 2.5 times the beamwidth of the antenna. (Note: Not +/-2.5 times.)

Antenna main beam efficiency is defined as the ratio of all energy received within the main beam, i.e., including cross-polarized energy, to the total of all energy received by the antenna.

SMBA-32 The beamwidth in any plane containing the electrical boresight axis shall be within plus or minus 10% of the nominal beamwidth.

SMBA-33 Antenna main beam efficiency shall be no less than 95.0% at all frequency bands and all beam positions.

4.8.1 Beam Polarization

SMBA quasi-horizontal and -vertical polarizations are defined by the following two equations:

Equation 7: Quasi Horizontal Polarization (QH) = $TV \sin^2 \Theta + TH \cos^2 \Theta$

Equation 8: Quasi Vertical Polarization (QV) = $TV \cos^2 \Theta + TH \sin^2 \Theta$

Note that from Equation 7 and Equation 8, the QV and QH reduce to TV (with electric field vector perpendicular to the orbital velocity direction) and TH (with electric field parallel to the orbital velocity direction), respectively, at nadir when theta is zero degree. Here theta is the scan-angle (from nadir). TV and TH are the brightness temperatures of the vertically polarized and horizontally polarized components, respectively, following the conventional definition. The vertical polarization is the component whose electric field is lying in the plane of incidence, and the horizontal polarization is the component whose electrical field is perpendicular to the plane of incidence. In other words, the SMBA instrument receives a linear combination of the pure vertical and horizontal components. The total electric field vector of the received wave (either QV, or QH) is lying in a plane perpendicular to the propagation direction and making a polarization angle Φ_p (in degrees), with respect to a reference (zero degree) line L, which is the intersection of the tangent (at beam center) plane and the plane of incidence. The total electric field vector rotates (i.e., the polarization angle changes) as scan angle theta changes. For Quasi Vertical, $\Phi_p = \Theta$. For Quasi Horizontal Polarization, $\Phi_p = 90 - \Theta$.

SMBA-34	Each channel shall have minimum of one linear polarization.
SMBA-35	Polarization twist angle shall be less than 0.1 degree. Note: Polarization twist angle is defined as the mechanical angular misalignment between the feed horn polarization orientation and optical polarization reference.
SMBA-36	Each frequency band shall have a cross-polarization leakage less than -20dB for K, Ka, and W bands, and less than -15dB for all other frequency bands.

4.8.2 Beam Pointing

From an atmospheric sounder, NWP modelers and other users need a series of measurements over a range of altitudes to construct an atmospheric profile of temperature or moisture (Atmosphere Vertical Temperature Profile (AVTP), Atmosphere Vertical Moisture Profile (AVMP)) where a profile is a snapshot of the thermodynamic or moisture state along a line from the top to the bottom of the atmosphere.

There are important physical connections between the state at one altitude and the state just above and below that altitude; likewise, the measurement at one altitude overlaps with the measurement just above and below that altitude. The states are connected along a line from the top to the bottom of the atmosphere. Therefore, a sounder will make measurements within the projected beam along a line from the sensor to the Earth's surface. Different frequency bands and channels yield measurements at different altitudes.

In order for those measurements at different altitudes to all fall long the same line, the beams for all bands will point along the same line of observation from the sensor to the Earth's surface

(spatial simultaneity) and at the same time since scanning at orbital velocity will move to a new observation line (a new profile) within milliseconds (temporal simultaneity).

Note that the spatial simultaneity requirement applies along the entire line from the top to the bottom of the atmosphere. Just requiring the footprints of all the beams to be coincident only constrains the beams at the Earth's surface. Both endpoints of the line (beams) should be as coincident as possible.

SMBA-37 For any given beam position, all antenna beam electrical boresight axes of the instrument shall point in the same direction and intersect the same location on the Earth's surface, subject to the pointing accuracy requirements, within 1/10 of a footprint integration time.

4.8.3 Channel Rejection

SMBA-38 Each channel shall have a gain that is no greater than 40 dB below the band-center value for all frequencies outside of a band ± 0.70 times the specified half-power bandwidths.

SMBA-39 Isolation between frequency bands shall be greater than 70 dB for all frequencies outside of a band.

All channels that use upper and lower mixer sideband signals should employ stopbands to reduce local oscillator noise to levels compatible with meeting the sensitivity and the calibration accuracy requirements.

4.8.4 Gain

Passband gain is defined as the overall gain of the instrument from the antenna aperture to the instrument output, averaged over the passband bandwidth.

SMBA-40 For any channel with more than one passband, the gains of all passbands within that channel shall be within ± 1 dB.

SMBA-41 Channels shall have peak-to-peak gain variation within each passband no greater than 1.0 dB (TBR) for K, Ka, and V bands, 1.5 dB (TBR) for W, F, D, G, and J bands.
Note: This requirement applies over the 80% of the passband bandwidth centered on the center frequency of the passband.

4.8.5 Scan Characteristics

SMBA-42 The instrument shall provide contiguous or overlapping spatial sampling (3dB contours of the antenna patterns) both along-track and cross-track when operated in the mission orbit. Note: Nyquist spatial sampling is an acceptable option, as well.

SMBA-43	The instrument shall provide global coverage greater than or equal to 95% when operated in the mission orbit.
SMBA-44	The instrument IFOV shall have a spatial horizontal resolution at nadir less than or equal to 32 km (T) and 16 km (q) when operated in the mission orbit.
SMBA-45	The instrument shall integrate while scanning (i.e., not step scan).
SMBA-46	The instrument shall scan the antenna beam electrical boresight cross-track to the satellite motion.
SMBA-47	The total scan period shall be TBS seconds $\pm$ TBS msec.
SMBA-48	Instrument shall synchronize its scan start time with a timing signal it receives from the spacecraft command and telemetry (C&T) bus.
SMBA-49	The time offset from the rising edge of the spacecraft timing signal to the start of beam position number one of the instrument scan shall have an accuracy of ± 1 msec.
SMBA-50	The scan start timing offset from the spacecraft synch pulse shall be settable over a range of zero to the scan period.

4.9 Spectral Response Function

The SRFs are created via a combination of measurement and analysis. For a channel having multiple passbands, the SRF of each passband is unique and must be independently ascertained.

Measuring the SRF creates a one-to-one mapping between the end-to-end gain values (from the antenna aperture through the instrument output) and the RF frequencies of radiation relevant to each channel.

SRFs are used to generate: (1) Band absorption coefficients for radiative transfer models used by environmental retrievals algorithms and NWP models. NWP models continue to require increasingly accurate SRF measurements. (2) Band conversion coefficients in the SDR algorithm that calculate warm/cold target radiance from their physical temperatures and Earth spectral brightness temperature from Earth radiance.

SMBA-51	The magnitude (gain) knowledge of a SRF shall be within 0.1 dB for K, Ka, and V bands and 0.3 dB for W, D, G, F, and J bands, where the channel response is greater than -40 dB relative to channel maximum response.
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SMBA-52	The magnitude (gain) knowledge of the SRF shall be within 0.5dB where the channel response is lower than -40dB relative to channel maximum response.
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SMBA-53	The number of data points in SRF measurements shall be greater than or equal to 300 data points for each passband, including data points on the skirts and adjacent stopbands.
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4.10 Calibration

SMBA-54	Instrument shall perform in-flight calibration for all channels.
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SMBA-55	The instrument shall be designed such that the ground calibration needs to be performed only once for each delivered instrument.
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SMBA-56	No ground calibration shall be required after storage before launch.
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SMBA-57	All channels shall use an end-to-end calibration method, including the full optical chain.
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SMBA-58	The cold calibration reference shall be the cosmic microwave background radiation, also referred to as cold space.
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SMBA-59	During cold calibration, the instrument shall view the cold calibration reference directly without the use of additional optical elements.
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Note: Requirements SMBA-60 through SMBA-99 are reserved for future instrument science requirements pertaining to specific implementation approaches.

5 NEON SERIES-1 INSTRUMENT REQUIREMENTS

The following paragraphs contain requirements extracted from the NEON Series-1 Project-level requirements documentation and are directly traceable to NOAA's Mission Requirements.

5.1 Instrument Accommodations

- | | |
|-----------------|--|
| SMBA-100 | The instrument shall comply with the accommodation requirements, in accordance with the requirements delineated herein and the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx), including launch vehicle mechanical, electrical, environmental requirements, and constraints. |
| SMBA-101 | The instrument shall define and maintain static and dynamic clearances to the respective instrument envelope(s), define and maintain the instrument operational, calibration, and thermal field-of-views, and define integration and test target accommodations in the Spacecraft-to-SMBA Instrument MICD (NS1-SMBA-xxx-xxxx). |
| SMBA-102 | The instrument mass, including blankets and instrument harnessing, shall not exceed 150 kg. Mass properties, including center of mass and moments of inertias, will be defined in the Spacecraft-to-SMBA MICD. |
| SMBA-103 | The instrument shall not exceed an orbit average power load of 295 W in any instrument operational mode. The instrument power allocation per mode and peak power will be defined in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx). |
| SMBA-104 | The specific thermal interface needs (inclusive of margin) of the instrument shall be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx). The allowable heat transfer to the spacecraft during science operations should not exceed 175 W (TBR) at an interface temperature of +15 °C (TBR). |
| SMBA-105 | The instrument shall not exceed an uncompensated momentum contribution of ± 0.3 N-m-sec per axis (TBR). The specific level of on-orbit disturbances induced to the spacecraft across the instrument-to-spacecraft mounting interface shall be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx). |

5.2 Mission Orbit Requirements

SMBA-106 The instrument shall operate for the mission lifetime in a polar Sun-synchronous orbit with the following characteristics:
 Nominal Altitude: 824 km +/- 17 km
 Nominal Inclination: 98.71 degrees +/- 0.10 degrees
 Ground Track Repeatability Accuracy: +/- 20 km at the equator
 Ground Track Repeat Cycle: 16 days

SMBA-107 The instrument shall operate for the mission lifetime in the polar Sun-synchronous orbit defined in SMBA-106 when deployed/maintained orbit in either of the following local time of the ascending nodes (LTANs):

 Nominal Ascending Equator Crossing Time: 1330 (local time) +/- 10 minutes
 or
 Nominal Ascending Equator Crossing Time: 1730 (local time) +/- 10 minutes

5.3 Mission Lifetime, Commissioning, and Storage Requirements

SMBA-108 The instrument shall be designed for a mission lifetime of 5 years from completion of on-orbit commissioning.

SMBA-109 The instrument shall be capable of a 13-year ground storage period that includes routine maintenance and monitoring activities to ensure no loss of functional capabilities.

SMBA-110 The instrument shall be capable of being placed in a ground storage state that requires no intervention by personnel for up to 12-month periods.

5.4 Mission Data Requirements

The spacecraft command and data handling (C&DH) subsystem collects instrument health and status telemetry, instrument memory dump, RFI detection data, and mission data packets through the command and data interface. The C&DH subsystem multiplexes these packets and attaches appropriate protocol for downlink and storage. The need for following functions will be determined jointly by the instrument and spacecraft providers and defined in the applicable instrument ICD:

- Spacecraft-to-Instrument transfers consisting of real-time ground commands, stored commands, memory loads, state of health indications, and auxiliary data (e.g., time code).
- Instrument-to-Spacecraft transfers consisting of, for example, mission data (including auxiliary data received from the spacecraft that is time correlated to the instrument data), RFI detection data, instrument health and status telemetry, indication of instrument mode

transitions, instrument diagnostic data, memory dumps, and Survival Mode temperatures (not part of the data bus).

The instrument average data rate is defined to be the total data transmitted for one orbit divided by the orbit period. The instrument peak data rate is defined as the maximum data volume divided by the collection period.

SMBA-111	Spacecraft-to-Instrument command & data transfers and Instrument-to-Spacecraft data transfers shall be through ethernet. Note: The intent of this requirement is for a standard ethernet protocol, not a custom protocol, for compatibility with multiple spacecraft providers.
SMBA-112	The initial data rate allocations of the instrument shall not exceed 2 Mbps orbital average and not exceed 10 Mbps peak (TBR). The instrument average and peak data rates per instrument mode will be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).
SMBA-113	All data that comprises the RDR shall be transferred to the spacecraft uncompressed or using lossless compression.
SMBA-114	Receipt of individual instrument commands shall be verifiable via instrument health and status telemetry.

5.4.1 Mission Data Geolocation Requirements

The instrument pointing accuracy is the difference between the actual and the intended antenna boresight direction. This includes the as-built static error and the effects of all structural, thermal, and environmental sources that induce changes in the boresight alignment.

SMBA-115	The known geolocation of the instrument boresight, referenced to the center of the effective field of view for any channel, shall be within 1.50 km (radial, 3 sigma) of the true location of that field of view at nadir on the WGS84 reference ellipsoid, at any time during nominal operations.
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5.4.2 Instrument Pointing, Accuracy, and Stability Requirements

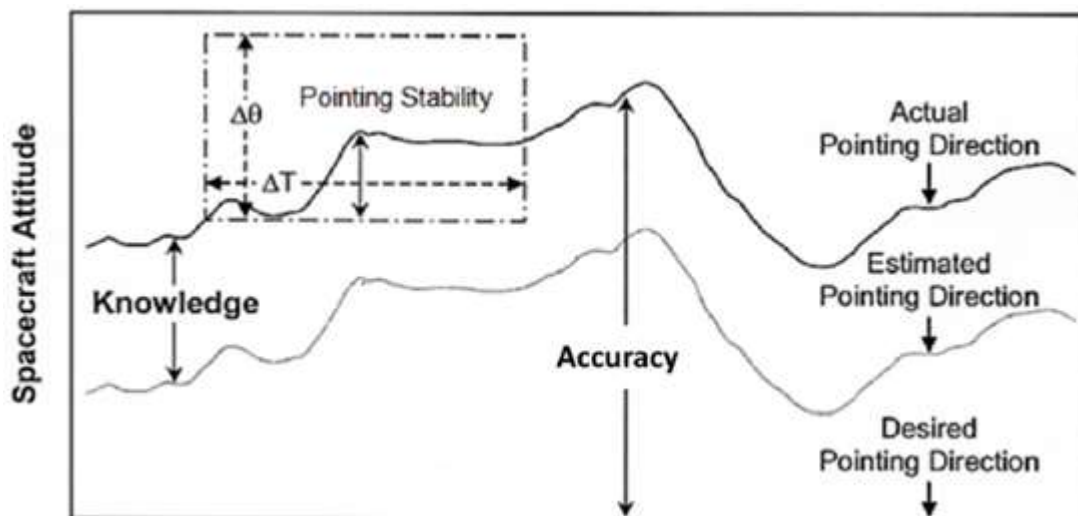


Figure 5.4.2-1. Pointing Terminology

Pointing terminology definitions are illustrated in Figure 5.4.2-1 and are defined as follows for each spacecraft axis:

- Pointing Knowledge – The angle between the actual pointing direction and the estimated pointing direction.
- Pointing Accuracy – The angle between the actual pointing direction and the desired pointing direction.
- Pointing Stability – The peak-to-peak variation in the actual pointing direction over a time interval.
- Accuracy and Knowledge terms consist of three error categories:
 - i. Bias: Errors that remain constant over the duration of the mission.
 - ii. Drift: A long term variation of pointing direction.
 - iii. Jitter: A short term variation of the pointing direction.
- Static Errors – Errors that stay constant over mission time. Bias errors are static errors.
- Dynamic Errors – Errors that vary over time. Drift and Jitter errors are dynamic errors.
- Pointing Stability is due to dynamic error sources Drift and Jitter.
- 3-Sigma – A set of values is considered to meet a 3-sigma requirement if no fewer than 99.73% of the values are within the specified limits of the requirement.

All contributions to pointing uncertainties relating the spacecraft attitude measurements made in the spacecraft attitude reference (AR) frame to the alignment of the instrument interface (mounting surface datums) are allocated to the spacecraft.

All contributions to pointing uncertainties relating the instrument interface (mounting surface datums) to the instrument boresight are allocated to the instrument.

Contributions to pointing uncertainties resulting from clearances across the mounting interface (pin/fastener to hole tolerance) at the instrument interface (mounting surface datums) are allocated to the instrument.

SMBA-116 End-to-End pointing accuracy, pointing knowledge, and pointing stability shall meet the requirements specified in Table 5.4.2-1 per axis.

Table 5.4.2-1. Mission End-to-End Pointing Requirements^[1]

End-to-End Pointing Requirement	Requirement (@ antenna boresight)
Pointing Accuracy	± 2400 arcsec 3σ (TBR)
Pointing Knowledge	± 265 arcsec 3σ (TBR)
Pointing Stability	N/A
Spacecraft Contributions	GN&C Requirement (@ AR)
Pointing Accuracy	± 900 arcsec 3σ
Pointing Knowledge	± 112 arcsec 3σ

Notes:

- [1] The instrument provider should combine all static errors and dynamic errors separately then add the static and dynamic errors in reporting pointing accuracy and knowledge budgets for each axis, using the method identified in Figure 5.4.2-2. The static term for ground-to-on-orbit thermal shift should be broken into 2 pieces. The model predict, without a Model Uncertainty Factor (MUF), should be accommodated in the Pointing Accuracy budget and the uncertainty, with a MUF = 1.6, should remain in the Pointing Knowledge budget. Thermal distortion dynamic error terms should be treated together as a worst-case summation. A MUF = 1.6 should be used for all thermal distortion errors. The instrument provider is free to use a more conservative error combining method.

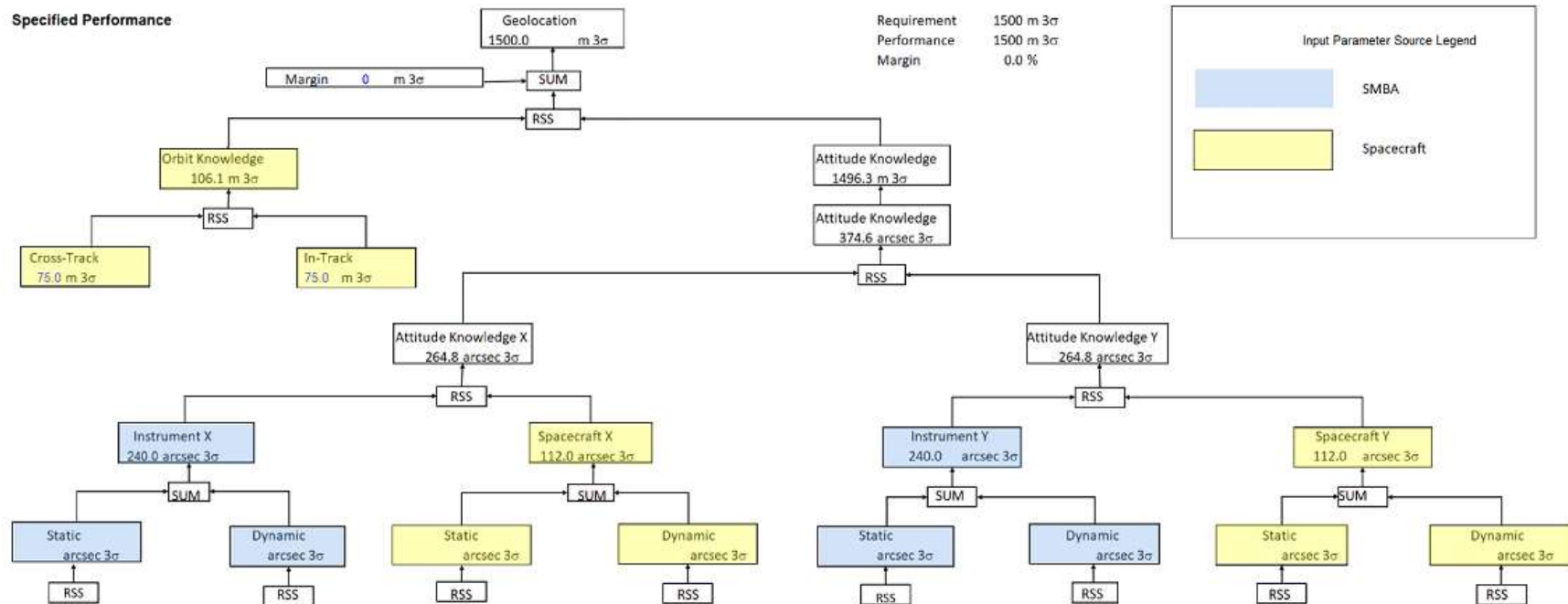


Figure 5.4.2-2. Analysis of Geolocation Knowledge Uncertainty at Nadir

5.4.3 Instrument Timing Requirements

The spacecraft will provide the instrument with a time-of-day (TOD) pulse and TOD packet that is accurate to +/- 1 millisecond of UTC to enable the instrument to synchronize a time reference for instrument data.

SMBA-117	The instrument shall use the TOD pulse and TOD packet to synchronize a time reference for instrument data with an accuracy of +/- 500 microseconds of the spacecraft time. Using this synchronization, any instrument packet time tag will be within +/- 1.5 milliseconds of UTC.
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In event of brief loss of TOD pulses, both the instrument and spacecraft should continue in the current mode, including instrument processing of secondary header time stamp, until return of TOD pulses.

5.5 NEON Series-1 Project Mission Phases

The following section delineates requirements applied to the mission phases. The mission phases of the NEON Series-1 Project are as follows:

- Satellite Assembly, Integration and Test, and Ground Storage
- Pre-launch/Launch Site Processing
- Launch and Orbit Insertion Phase
- Satellite Activation, Checkout, and Commissioning Phase
- Mission Operations
- End-of-Life Decommissioning

5.5.1 Satellite Assembly, Integration and Test, and Ground Storage Phase

SMBA-118	The instrument shall be capable of remaining integrated to the spacecraft while undergoing testing at periodic intervals (e.g., 12-month periods) during assembly, integration and test (I&T), and ground storage.
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SMBA-119	During assembly, cleaning, system I&T, and launch site processing, the instrument shall be capable of being processed in an ISO 14644-1 Class 8 cleanroom environment.
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NOTE: All instruments are expected to be covered and purged in a gaseous nitrogen whenever exposed to an environment worse than Class 8.

5.5.2 Pre-Launch/Launch Site Processing Phase

During Pre-Launch/Launch Site Processing phase, the satellite is performing functional checkout of satellite (spacecraft and instrument) subsystems, integrating to the launch vehicle, and performing final mission readiness activities to support launch.

5.5.3 Launch and Orbit Insertion Phase

During Launch/Orbit Insertion phase, the satellite is performing functional checkout of spacecraft subsystems and maneuvering the satellite into the final mission orbit. During this phase, it is expected that the instrument is OFF, and its environment is being monitored/maintained by spacecraft-provided survival services.

5.5.4 Satellite Activation, Checkout, and Commissioning Phase

During the first 45 days after launch, the satellite will go through activation and checkout of all spacecraft subsystems and the instrument. This will include turn-ons, calibrations, outgassing, systems operations verifications, etc. The commissioning will culminate with a successful Project Post-Commissioning Review.

Note: While full commissioning is dependent on instrument contributions, the spacecraft is expected not to hinder these instrument contributions.

5.5.5 Mission Operations Phase

SMBA-120	The instrument shall maintain an operational availability of greater than 99.5% over any 30-day period for the mission lifetime. This includes the total time spent in planned instrument outages that prevent acquisition/transmission of operational data. Note: Collision avoidance maneuvers, orbital maintenance maneuvers, and any on-orbit failures including any automatic fault recovery and/or faults that require an undetermined amount of time to detect, diagnosis, and generate corrective action are explicitly excluded from this calculation.
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5.5.6 End-of-Life Decommissioning Phase

SMBA-121	The instrument shall comply with NPR 8715.6, NASA Procedural Requirements for Limiting Orbital Debris, and the companion document NASA-STD-8719.14, Revision C, Process for Limiting Orbital Debris. The instrument should limit orbital debris generation as well as limit the number and size of debris fragments that survive atmospheric re-entry.
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5.6 Instrument Functional Requirements

5.6.1 Typical Satellite Modes

Figure 5.6.1-1 depicts the satellite mode and sub-mode configurations and mode transitions from pre-launch through on-orbit operations and decommissioning.

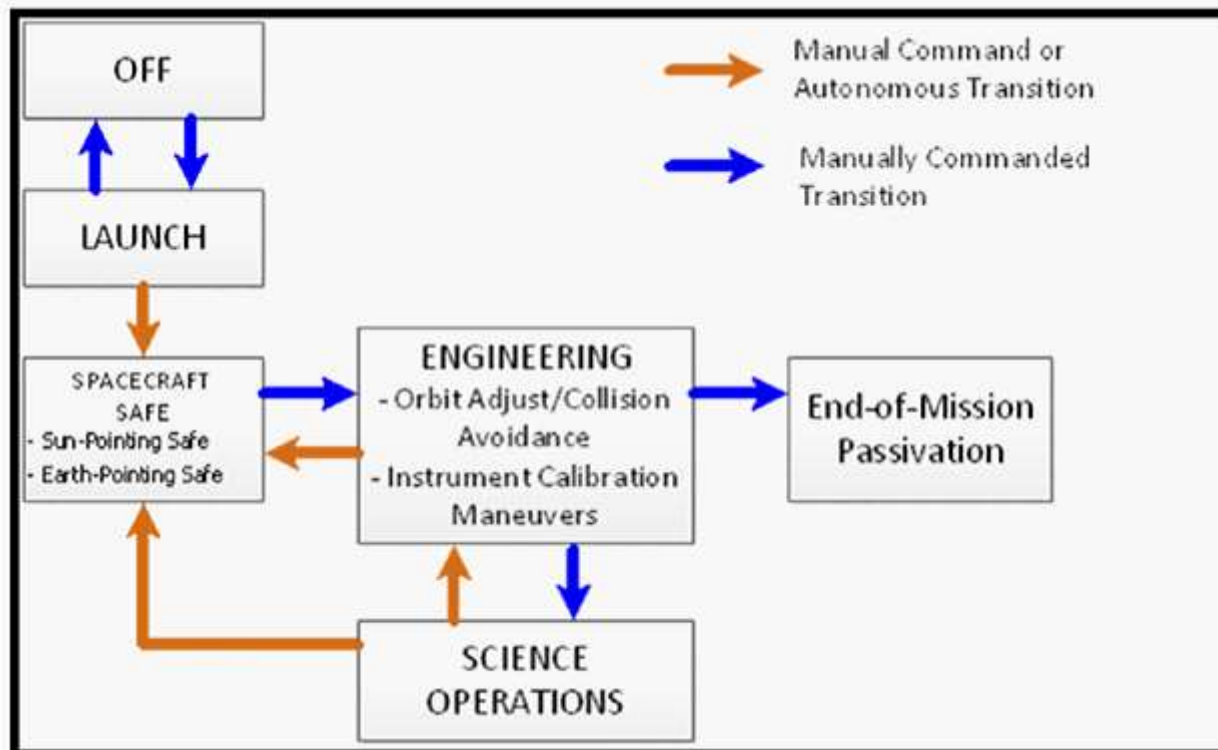


Figure 5.6.1-1. Typical Satellite Mode Transitions

The spacecraft should transition between modes by manual command and/or autonomously (as notionally depicted in Figure 5.6.1-1).

5.6.1.1 Spacecraft OFF Mode

The spacecraft will be able to be remotely configured into OFF Mode during ground operations. OFF Mode is defined as having all power removed including internal battery power.

5.6.1.2 Launch Mode Functional Requirements

The spacecraft will be capable of launching with the instrument in Instrument OFF Mode. Following orbit insertion and initial spacecraft deployments, the spacecraft will transition the instrument into Instrument Survival Mode.

5.6.1.3 Spacecraft Safe Modes

In the event of certain anomalous conditions, as determined by the spacecraft, the spacecraft will place itself and the instruments into a Safe Mode. Power system anomalies and attitude system anomalies are examples of conditions which could result in the entry into the Spacecraft Safe Mode. This mode could involve changing the nominal mission attitude of the spacecraft. This is an emergency mode and, as such, is a power conservation mode. As part of the process of entering Spacecraft Safe Mode, the spacecraft will initiate the corresponding Instrument Safe Mode. For severe emergencies, the spacecraft may remove operational power to the instruments after ensuring the instrument survival heater services are enabled.

The spacecraft will provide a tiered approach to Spacecraft Safe Mode commensurate with the anomaly:

- Earth-Pointing Safe Mode (as described in Section 5.6.1.3.1)
- Sun-Pointing Safe Mode (as described in Section 5.6.1.3.2)

The spacecraft should be capable of acquiring the appropriate safe hold attitude from any arbitrary attitude while maintaining a safe environment for the spacecraft and instruments and without violation of instrument solar constraints. The spacecraft should limit exposure of the instruments to solar input onto sensitive surfaces, to the extent that sensitive surfaces and exposure limits are defined in the individual Spacecraft-to-instrument ICDs.

SMBA-122	The instrument shall be designed to meet all performance requirements in operational mode following the execution of either safing event described in sections 5.6.1.3.1 and 5.6.1.3.2.
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5.6.1.3.1 Earth-Pointing Safe Mode

The spacecraft will provide a coarsely controlled Earth-Pointing Safe Mode capability that maintains a safe environment for the spacecraft and instruments for an indefinite period of time as the preferred mode for selected anomalous conditions. The Earth-Pointing Safe Mode attitude reference is defined as orienting the spacecraft body frame such that the spacecraft +Z-axis is aligned with geodetic or geocentric nadir and the +X-axis is aligned with the projection of the instantaneous orbital velocity vector onto the plane perpendicular to geodetic or geocentric nadir. See illustration in Figure 5.6.1-2.

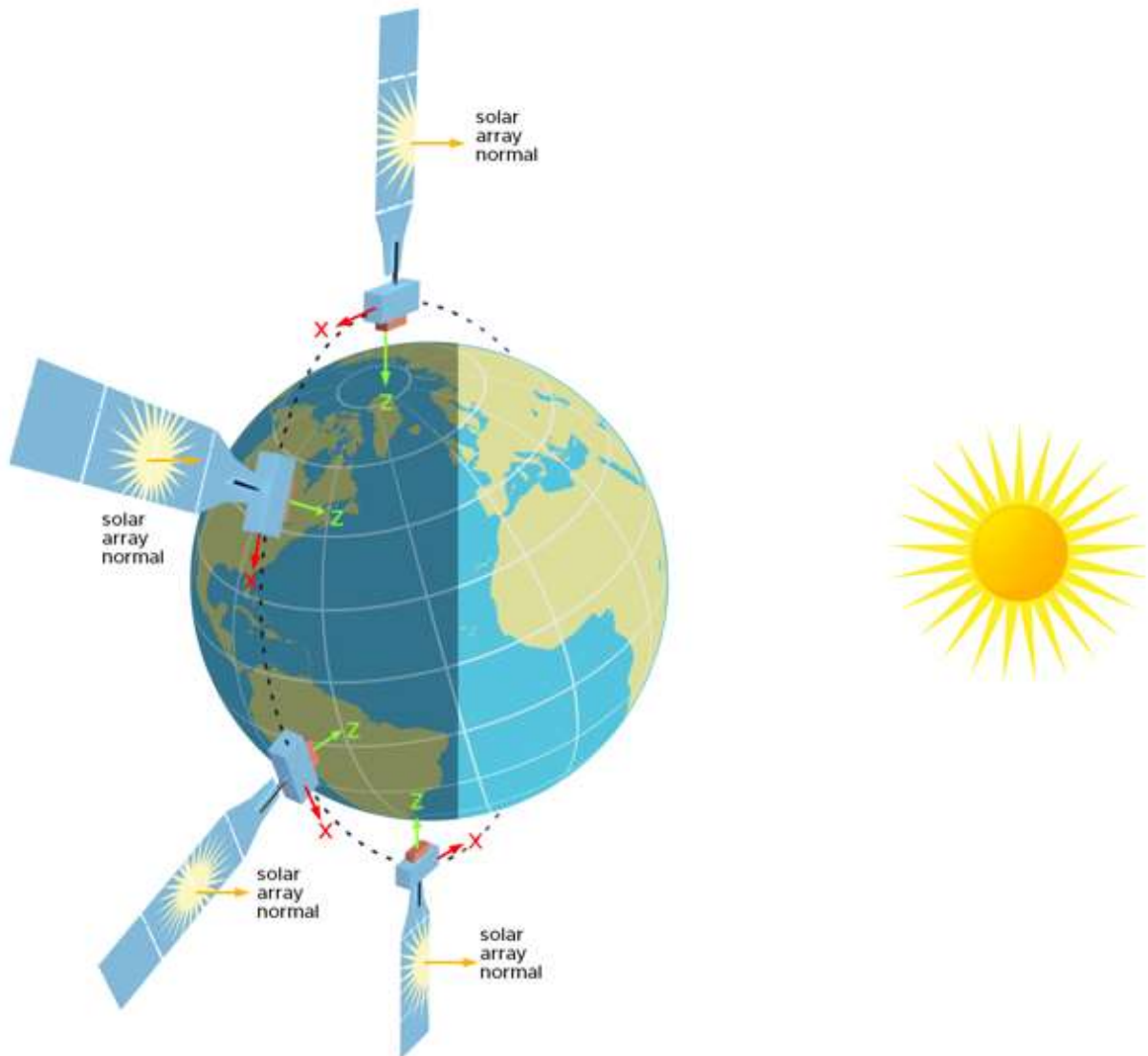


Figure 5.6.1-2. Typical Spacecraft Configuration for Earth-Pointing Safe Mode

5.6.1.3.2 Sun-Pointing Safe Mode

The spacecraft will provide a solar-inertial Sun-Pointing Safe Mode capability (see illustration in Figure 5.6.1-3) to point the solar array to the Sun while maintaining a safe environment for the spacecraft and instrument for an indefinite period of time.

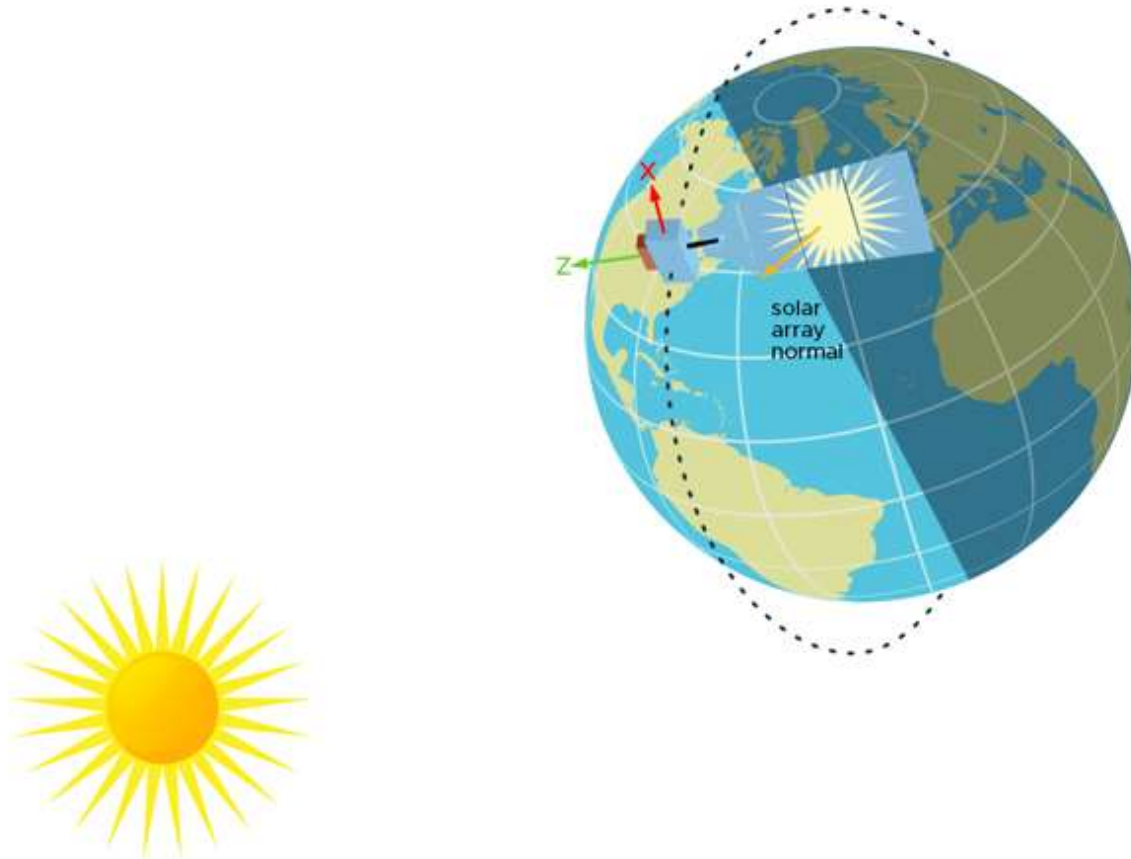


Figure 5.6.1-3. Typical Spacecraft Configuration for Sun-Pointing Safe Mode

5.6.1.4 Spacecraft Engineering Mode

Engineering Mode is the collection of modes where the spacecraft initially performs check-out, instrument outgassing, and other commissioning activities, as well as performs routine maintenance activities such as orbit adjust maneuvers and instrument sensor calibration maneuvers. While the attitude control requirements may be the same as in Science Operations Mode, time spent in Engineering Mode does not get counted towards satisfaction of the mission availability requirements with the exception of instrument calibration maneuvers.

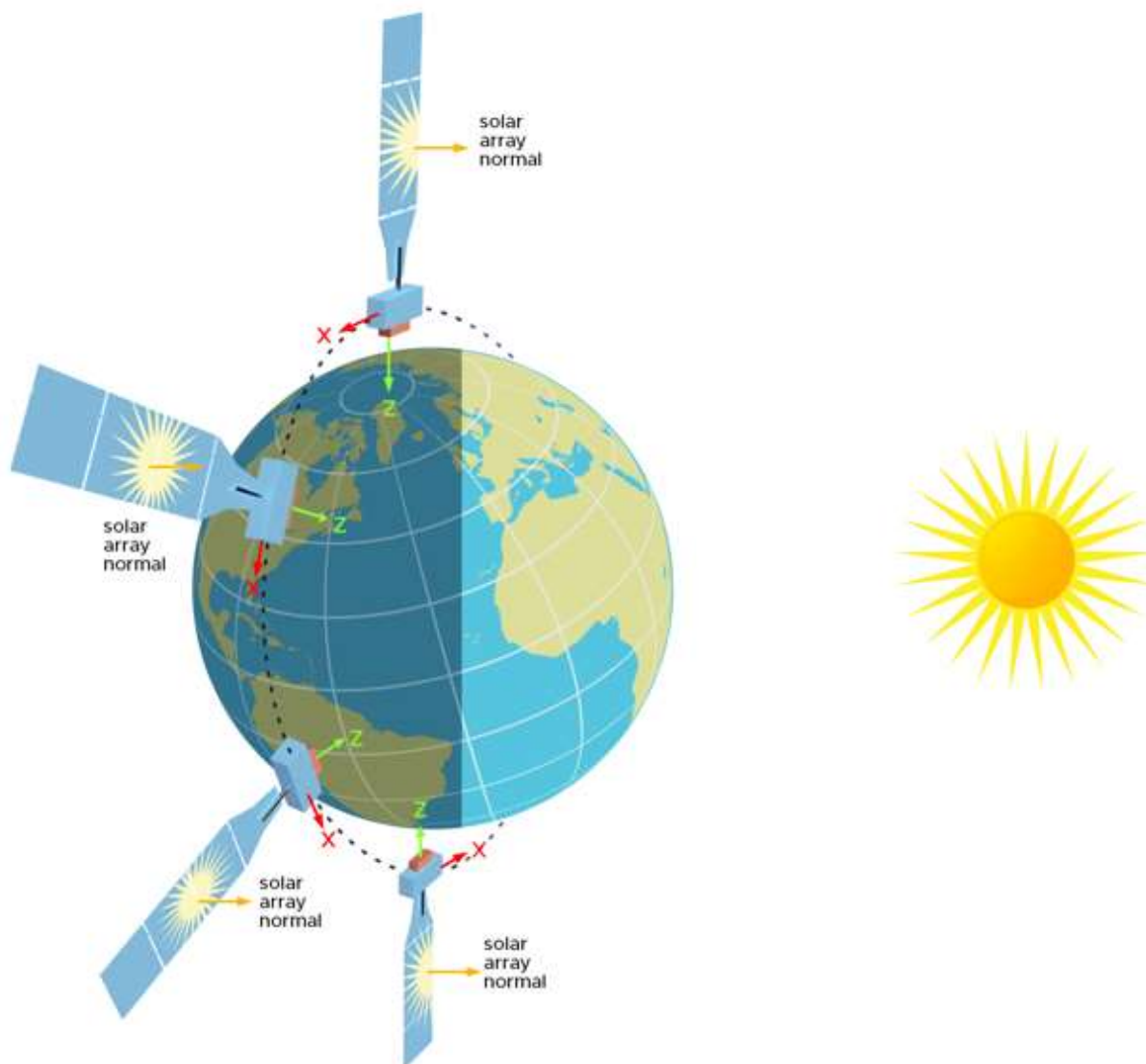


Figure 5.6.1-4. Typical Spacecraft Configuration for Engineering Mode and Science Operations Mode

During Engineering Mode and Science Operations Mode, the spacecraft will orient the spacecraft body frame such that the spacecraft +Z-axis is aligned with geodetic nadir and the +X-axis is aligned with the projection of the instantaneous orbital velocity vector onto the plane perpendicular to geodetic nadir. See illustration in Figure 5.6.1-4.

During Engineering Mode and Science Operations Mode, the spacecraft will be capable of providing time-tagged spacecraft attitude estimates (Earth Centered Inertial (ECI)) to spacecraft Control Frame) and/or a time-tagged spacecraft ephemeris (Earth Centered Earth Fixed (ECEF)) with a minimum frequency of 8 Hz (TBR) if required by the instrument to meet performance requirements.

SMBA-123	The instrument shall provide the application packets necessary to produce the RDRs while in either Engineering mode or Science Operational mode.
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5.6.1.4.1 Orbit Adjust/Collision Avoidance

SMBA-124	The instrument shall be capable of remaining in Operational mode without damage while the spacecraft performs orbit correction.
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5.6.1.4.2 Instrument Calibration Maneuvers

While in Spacecraft Engineering Mode, the spacecraft performs the calibration maneuvers upon ground command and returns to nominal satellite operations (including Earth-pointing orientation) upon completion.

SMBA-125	The instrument shall have the capability of performing the calibration maneuvers listed in Table 5.6.1-5 (TBS) upon ground command, non-simultaneously, and returning to nominal satellite operations (including Earth-pointing orientation) upon completion.
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Table 5.6.1-5. Suggested Instrument Calibration Maneuver Requirements (TBS)

Maneuver	Description	Duration	Constraints	Rationale
SMBA Negative Roll	Roll in the <u>negative</u> S/C roll angle direction such that the Earth limb is within the SMBA Earth Sector Field of View (FOV). Dwell for minimum of 5 minutes. Roll in the <u>positive</u> S/C roll angle direction such that SMBA nadir position is aligned nadir Occurrence: - Once during LEO&A - Once prior to end of mission (preferred)	Dwell at each Earth limb for minimum of 5 minutes with Earth limb in view. More time preferred.	Keep Sun out of SMBA science field of regard. Perform maneuver over open ocean and clear sky. Significant weather events, such as hurricanes, heavy rains, etc., along with Moon views, must be avoided in the scan FOV. SMBA to be in nominal scan mode throughout the maneuver.	Measure: - Magnitude of crosstrack scan bias - Magnitude of sidelobe contamination - Radiometric equivalency of Earth/Space-view - Reflector emissivity
SMBA Positive Roll	Roll in the <u>positive</u> S/C roll angle direction such that the Earth limb is within the SMBA Earth Sector FOV. Dwell for minimum of 5 minutes. Roll in the <u>negative</u> S/C roll angle direction such that SMBA nadir position is aligned nadir. Occurrence: - Once during LEO&A - Once prior to end of mission (preferred)	Dwell at each Earth limb for minimum of 5 minutes with Earth limb in view. More time preferred.	Same as Negative Roll	Same as Negative Roll

SMBA 360-degree Roll (preferred, replaces positive/negative rolls)	Roll in the negative S/C roll angle direction such that the Earth limb is within the SMBA Earth Sector FOV. Dwell for minimum of 5 minutes. Continue Roll in the negative S/C roll angle direction such that the Earth limb is within the SMBA Earth Sector FOV. Dwell for minimum of 5 minutes. Continue Roll in the negative S/C roll angle direction such that SMBA nadir position is aligned nadir. Occurrence: • Once during LEO&A • Once prior to end of mission (preferred)	Dwell at each Earth limb for minimum of 5 minutes with Earth limb in view. More time preferred.	Keep Sun out of SMBA science field of regard. Perform maneuver over open ocean and clear sky. Significant weather events, such as hurricanes, heavy rains, etc., along with Moon views, must be avoided in the scan FOV. SMBA to be in nominal scan mode throughout the maneuver.	Measure: • Magnitude of crosstrack scan bias • Magnitude of sidelobe contamination • Radiometric equivalency of Earth/Space-view • Reflector emissivity The 360-degree rolls are more useful to help characterize the instrument.
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SMBA Back-flip (360-degree Pitch)	Pitch so that the entire SMBA science field of regard is viewing cold space (i.e., SMBA nadir towards zenith). Collect minimum of 5 minutes around zenith. The Moon should be visible within the SMBA Earth Sector FOV during the zenith scans. Perform RF Interference checks (transmitter ON-OFF cycles) before and after zenith scans for a minimum of 6 times (3 times before and 3 times after zenith scans). Continue to pitch such that SMBA nadir position is aligned nadir. Occurrence: - Once during LEO&A - Once prior to end of mission 10 days, to capture all phases of the Moon, for 2D lunar scans (preferred)	RFI checks - Start with Transmitter OFF - Transmitter ON (20 sec) - Transmitter OFF (10 sec) - Repeat - Perform pre- and post- zenith scans Collect data a minimum of 5 minutes around zenith with SMBA in nominal scan mode.	Keep Sun out of SMBA science field of regard. Perform maneuver over open ocean. Capture the Moon in SMBA FOV during the zenith scans (i.e., in the center of the smallest beamwidth). Perform the maneuver at a constant pitch rate (0.15 deg/sec to 0.18 deg/sec preferred). SMBA to be in nominal scan mode throughout the maneuver.	Primary: Obtain knowledge of non-uniformities/biases across Earth view sector and estimate flat reflector emissivity; perform on-orbit evaluation of beam-pointing; and, check for RFI from on-board transmitters. Secondary: Amplitude calibration, sidelobe characterization. 2D lunar scans: - Characterize reflector emissivity and scan biases - Capture observations at different phases of the Moon for lunar calibration and antenna beam pointing error characterization
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Note: Calibration maneuvers listed are typical for on-orbit commissioning. Instrument provider should propose maneuvers necessary for successful calibration and science operations.

5.6.1.5 Spacecraft Science Operations Mode

Science Operations Mode is the predominant operating mode during the mission. All time spent in the Science Operations Mode counts towards the satisfaction of the mission availability requirements defined in Section 5.5.5. The spacecraft will provide a three-axis stabilized nadir pointing attitude control system capable of meeting the geolocation requirements of the science instruments defined in Section 5.4.1.

The spacecraft should take no more than 10 minutes to return from Engineering Mode or Earth-Pointing Safe Mode to Science Operations Mode.

The spacecraft should take no more than 30 minutes to return from Sun-Pointing Safe Mode to attitude control with the assumption that the instrument interface temperature conditions are capable of Science Operations Mode.

5.6.1.6 End-of-Mission Passivation Mode

At the end of the mission, the instrument will be powered off prior to the start of the disposal maneuvers.

5.6.2 Typical Instrument Modes

This section defines requirements associated with the five typical individual operating modes of the instrument. The typical five main instrument operating modes are:

- Instrument OFF Mode
- Instrument Survival Mode
- Instrument Safe Mode
- Instrument Engineering Mode
- Instrument Operational Mode

While the modes can be dependent on the spacecraft, when the spacecraft is in Operational (Science) Mode, the instrument can be in any of the above modes.

SMBA-126	The instrument shall implement the following modes and functionalities, as appropriate, with the mode characterizations shown in Table 5.6.2-1 and mode transitions shown in Figure 5.6.2-1.
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Table 5.6.2-1. Instrument Mode Characterization

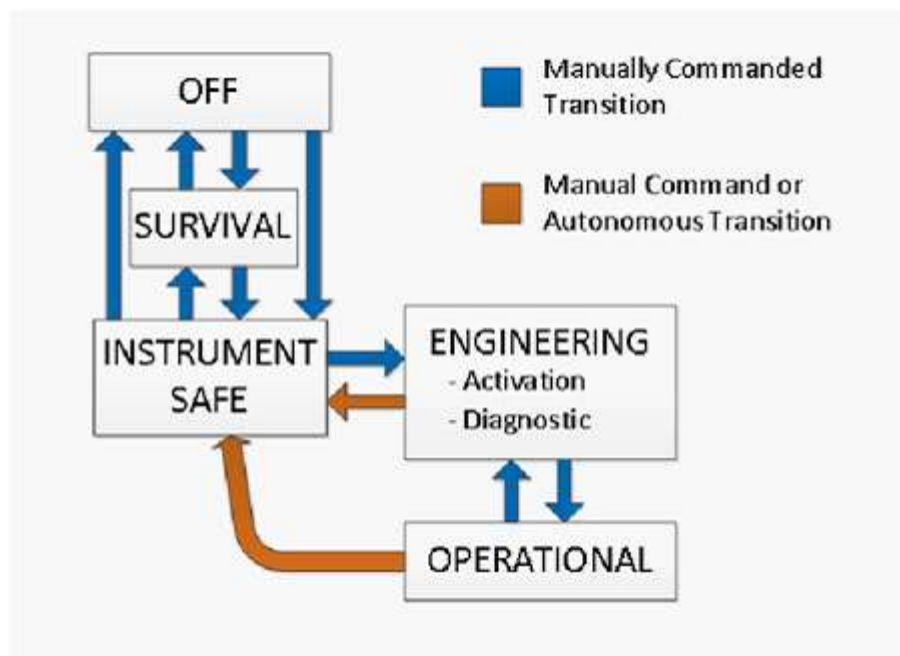
Instrument Mode	Survival Power	Operational Power	CMD & TLM	Meet Section 4 Science Performance
OFF	OFF	OFF	No	No
Survival	Enabled	OFF	No	No
Instrument Safe	Enabled ^[1]	ON	Yes	No
Engineering - Activation - Diagnostic	Enabled ^[1]	ON	Yes	No ^[2]
Operational	Enabled ^[1]	ON	Yes	Yes

Notes:

Yes = capability pertains to that mode/functionality

[1] Instrument Survival heater service may or may not be enabled, outside of Survival Mode.

[2] Science performance in Diagnostic mode is not precluded, but is not required.

**Figure 5.6.2-1. Typical Instrument Mode Transition Diagram**

SMBA-127 The instrument shall be command-able into any of the instrument modes, with transitions as illustrated in Figure 5.6.2-1.

SMBA-128 Although preferred instrument-provided sequences may be defined, the instrument shall cause no damage to itself or the spacecraft when transitioning from any mode to any other mode.

5.6.2.1 Instrument OFF Mode

In the OFF Mode, no operational power is supplied to the instrument. In this mode, the spacecraft controls the instrument via the pulse command lines, rather than by the C&T bus. The spacecraft also monitors relay status indicators via the discrete bi-level telemetry interfaces.

Instrument OFF Mode may be used for ground storage & transportation, launch, and spacecraft power crisis situations.

Note: In the OFF Mode, the instrument will receive no power from the spacecraft via the operational or the survival power service.

5.6.2.2 Instrument Survival Mode

Survival Mode is entered when a critical power shortage has been identified by the spacecraft. Survival Mode is a low power mode from which the Observatory can eventually recover to full operational status. Only those functions required for Observatory safety, diagnostics, and recovery will be powered. Instrument operational power will be OFF and instrument survival power will be enabled. In Survival Mode, the spacecraft will be responsible for sampling critical instrument temperatures via the instrument passive analog temperature sensors (normal instrument telemetry is not available with operational power OFF). Initiation of Survival Mode by the spacecraft should not require commands to turn on the instrument survival heaters.

SMBA-129	In Survival Mode, the instrument shall be able to survive indefinitely while the spacecraft is in either safe mode attitude defined in Section 5.6.1.3.
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5.6.2.3 Instrument Safe Mode

In the event of certain on-orbit spacecraft bus subsystem anomalies, the instrument must be properly configured to protect against prolonged exposure to abnormal environments such as those caused by abnormal pointing of the spacecraft. When possible, the spacecraft will send pre-defined reconfiguration commands to the instrument specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx), including moving the scanning reflector to a stowed position. In addition, the instrument must put itself into a Safe Mode upon detection of an internal anomaly which could be detrimental to the instrument or any other part of the Observatory.

SMBA-130	For instances of an instrument-detected on-orbit failure, where failure to take prompt corrective action could result in damage to the instrument, the instrument shall autonomously place itself into a Safe Mode. Autonomous transition to Safe Mode may be inhibited by the on-orbit upload of safety limits for mission-critical parameters.
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SMBA-131	The instrument shall enter Safe Mode upon receipt of a spacecraft command to enter Safe Mode.
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SMBA-132	The instrument shall autonomously configure to its Safe Mode within 45 seconds after entry into Safe Mode is initiated.
SMBA-133	The instrument shall provide a mode status flag in housekeeping telemetry, to indicate when it is in Safe Mode.
SMBA-134	Upon initiation of Safe Mode, the instrument shall configure itself such that no damage will occur if the next action from the spacecraft is to turn OFF the instrument, even though that may not be the normal flow.

5.6.2.4 Instrument Engineering Modes

Instrument Engineering Modes are the collection of modes where the instrument initially performs the post-launch check-out and commissioning activities such as activation, outgassing, and calibrations, as well as performs routine maintenance activities such as software updates, outgassing, and sensor calibrations. While the science performance may be the same as in Operational Mode, time spent in Engineering Modes does not get counted towards satisfaction of the mission availability requirements defined in Section 5.5.5 with exception of science calibration maneuvers.

5.6.2.4.1 Instrument Activation Mode

Activation Mode refers to instrument turn-on, and subsequent instrument component warm up, or cool down, to the operating temperatures. Activation Mode terminates when all instrument temperatures, biases, and currents have stabilized within specified operational limits.

SMBA-135	During power up, the instrument shall enable the command and telemetry functions and provide for orderly turn-on (channel enable, scan motor turn-on, etc.) based on commands to the instrument.
SMBA-136	Following power-up, the instrument shall be functional without requiring a software or data upload.

5.6.2.4.2 Instrument Diagnostic Mode

Diagnostic Mode refers to instrument engineering activities (diagnostic procedures) to evaluate science performance, update software tables/configurations, and/or collect status/health telemetry.

SMBA-137	To support early orbit checkout, and for anomaly resolution, the instrument contractor shall provide the capability to selectively disable any on-orbit processing operation that combines or compresses raw data in any manner.
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SMBA-138 Instrument Diagnostic mode shall support housekeeping, telemetry, and software updates, as well as troubleshooting activities including capability of different sampling of telemetry (rates and memory locations).

In Diagnostic Mode, the instrument can point the scan reflector to any commanded scan reflector position. The commanded position is maintained until a new pointing or scanning command is received.

SMBA-139 In Diagnostic Mode, when commanded, the instrument shall scan and position the antenna beam electrical boresight to enable measurement of scene radiance at specified locations.

SMBA-140 In Diagnostic Mode, the instrument shall have the capability to provide continuous sampling of science data.

SMBA-141 In Diagnostic Mode, the instrument shall have the capability to scan at a constant velocity.

During ground testing, the instrument should have the capability to use RFI test cases developed collaboratively between NASA and the chosen instrument provider. This capability allows for RFI detection algorithm performance to be assessed. RFI test case data can be imported via test port, simulated in the lab environment, and/or commanded via built-in test. The instrument should provide insight along the processing stream from end-to-end for validating RFI detection algorithm performance and radiometric performance.

The instrument should have the capability to perform autonomous and commanded built-in tests. Autonomous built-in tests are performed in the background to monitor instrument health and safety in order to support fault identification, in both operational and diagnostic modes. Commanded built-in tests are performed only when commanded in the diagnostic mode. Table 5.6.2-2 provides suggestions for built-in tests.

Table 5.6.2-2. Built-in Tests

(This table can be revised to reflect the specifics of the instrument chosen)

Test	Description	Built-in Test type
Safety	Monitor key voltages and temperatures against safety limits	Autonomous
ADC	Monitor reference voltage injected into multiplexer/ADC	Autonomous
Memory	Memory test performed as part of boot-up	Commanded
Dwell test	Sample a specific housekeeping channel at maximum acquisition rate and provide data back to the ground	Commanded
Point & stare	Continuous sampling while pointing scanner to a selected beam position	Commanded
Constant Velocity Scan	Continuous sampling while scanning at constant velocity	Commanded

5.6.2.5 Instrument Operational Mode

Instrument Operational Mode is the predominant operating mode during the mission. All time spent in this mode counts towards the satisfaction of the mission availability requirements defined in Section 5.5.5.

SMBA-142	In the Instrument Operational Mode, the instrument shall provide the data to produce the RDRs, as well as health and status telemetry, RFI detection, and housekeeping data.
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5.6.3 Contingency Management

The instrument should be capable of indefinitely maintaining its health and safety without ground support.

SMBA-143	The instrument shall perform autonomous fault detection, isolation, and correction for mission-critical functions where faults could cause damage, loss of control, or loss of science.
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SMBA-144	The instrument shall employ telemetry monitoring (TMONS) or similar capability to aid in the detection and reporting of out-of-limit conditions or other abnormal affects that could be caused by external sources of RFI within 1 orbit of the occurrence.
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SMBA-145	An unresolvable instrument subsystem fault shall result in autonomous transition to Instrument Safe Mode.
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5.7 Instrument Design Requirements

The following sections delineate the design requirements and guidelines of the various instrument technical disciplines taken from applicable NASA and industry design standards.

5.7.1 Mechanical Alignment Requirements

A permanently affixed optical alignment target should be used to reference antenna pattern measurements and provide the spacecraft integrator a known relative indicator of boresight alignment.

SMBA-146	The alignment of the instrument beam electrical boresight axes shall be measured with respect to the primary optical alignment target.
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SMBA-147	The alignment of the optical alignment targets with respect to the antenna subsystem mounting surfaces shall change by less than or equal to ± 0.025 degrees as a result of any testing.
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SMBA-148 The instrument alignment reference shall be viewable from two orthogonal directions when integrated to the satellite.

5.7.2 Structural Requirements

The instrument will survive the critical loading conditions that exist within the envelope of handling, launch environment, and mission requirements. The detailed requirements are provided in the following section.

SMBA-149 The Instrument, configured for launch, shall have a fixed-base fundamental resonant mode frequency of greater than 100 Hz.

SMBA-150 Instrument structure(s) shall be designed to have positive margin of safety for all failure modes to the worst simultaneous combination of design limit loads, applied temperature, and other accompanying environmental phenomena for each design condition without experiencing yielding or detrimental deformation.

Limit load is defined as the highest expected load, including environmental effects. Limit loads due to the launch environment or other acceleration environment will be applied through the center-of-mass of the components, as configured for the appropriate instrument mode, to design the component mounting interfaces.

SMBA-151 Instrument structure(s) shall be designed to have positive margin of safety for all failure modes to the worst case expected launch load of 15 g independently in each axis (TBR).

The factors of safety shown in Table 5.7.2-1, or those defined in the notes to the table, as applicable to the type of calculation listed in the table, will be applied to limit loads to calculate structural margins.

Table 5.7.2-1. Flight Hardware Design/Analysis Factors of Safety Applied to Limit Loads
[1] [2]

Type	Static	Sine	Random/Acoustic ^[4]
Metallic Yield	1.25 ^[3]	1.25	1.6
Metallic Ultimate	1.4 ^[3]	1.4	1.8
Stability Ultimate	1.4	1.4	1.8
Beryllium Yield	1.4	1.4	1.8
Beryllium Ultimate	1.6	1.6	2.0
Composite Ultimate	1.5	1.5	1.9
Bonded Inserts/Joints Ultimate	1.5	1.5	1.9

Notes:

[1] Factors of safety for pressurized systems to be compliant with NASA-STD-8719.24.

[2] Factors of safety for glass and structural glass bonds specified in NASA-STD-5001, Tables 4 and 5.

- [3] If qualified by analysis only, positive margin will be shown for factors of safety of 2.0 on yield and 2.6 on ultimate.
- [4] Factors shown should be applied to statistically derived peak response based on root mean square level. As a minimum, the peak response will be calculated as a 3-sigma value.

5.7.3 Mechanism Design Requirements

The following design requirements apply to the mechanisms used in the design of the instrument. NASA-STD-5017, Mechanism Design Requirements, provides suggested best practices for mechanism designs.

5.7.3.1 Torque/Force Margins

SMBA-152 The minimum torque at the motor shaft shall never be less than $7.06\text{E-}3$ N-m (1 oz-in).

Torque margin is a measure of the excess capability of a mechanism to impart, maintain, or prevent motion, or to provide required acceleration. Torque margin is intended to ensure that a mechanism retains reserve torque that can be applied in the event of an unforeseen effect that increases resistive torque or reduces motive torque within the mechanism, similar to the factor of safety used in a structural margin calculation.

Torque margin is not applicable to mechanisms that must provide a specific torque value within a narrow tolerance range to perform their basic function, such as a spring holding a relief valve closed or an ejection mechanism requiring a specific (not minimum) separation velocity.

Driving Torques: T_{avail} is the minimum available torque generated by the driving or holding component (e.g., spring, motor, pyrotechnics, solenoids, heat-actuated devices, pneumatic or hydraulic systems, brake). The minimum available torque from an energized motor is typically understood to include “magnetic losses” such as hysteresis and eddy current drag torques, “mechanical losses” such as bearing, brush, and windage drag torques, and “torque ripple” caused by detent (cogging) torque, asymmetry in motor design and construction, commutation strategy, etc.

T_k are the individual maximum known resisting torques that are not strongly influenced by environmental conditions and cycles.

For static and dynamic torque margins, T_v terms are the individual maximum “variable” resisting torques whose values may change with environmental conditions and cycles (e.g., friction torque, viscous drag torques, wire harness torque due to flexing or set). For holding torque margins, T_v is used to denote the individual maximum disturbing torques regardless of their degree of variability.

T_f are the individual maximum “fixed” resisting torques that are not strongly influenced by environmental conditions and cycles (e.g., vehicle maneuver-induced torques, return spring torques, unbalanced pressure loads limited by relief mechanisms).

T_a is the torque required to achieve a specified acceleration of the driven component.

Resistive Torques: Sum of Torque known = sum of the fixed torques or forces that are known and quantifiable, such as accelerated inertias ($T=I_a$) and not influenced by friction, temperature, life, etc. A constant FS is applied to the calculated torque required. Sum of Torque variable = sum of the torques or forces that may vary over environmental conditions and life such as static or dynamic friction, alignment effects, latching forces, wire harness loads, damper drag, and variations in lubricant viscosity, including degradation or depletion of lubricant over life.

There are many forms of torque margin equations in use in various standards. Each can be reformulated to appear like the others; the only difference among them is in the magnitude and nature of the different safety factors applied to the terms. This form was chosen due to its relative simplicity, its ability to handle several torque margin calculations with a single equation, and its suitability for application of maturity-based safety factors.

- SMBA-153** Sufficient torque or force margins shall be provided to assure reliable mechanism operation over the life of the mission.
- Margins to be determined using worst-case conditions and include all flight drive electronics effects and limitations.
 - Torque margin to be verified using measured data acquired from unit qualification for a new design. Subsequent units to be verified using acceptance level operational test data.
 - The Torque Margin (TM) to be greater than zero and be calculated using the following formula:
 - $TM = \{T_{avail} / (FS_k T_k + FS_v T_v + FS_a T_a)\} - 1$
 - The minimum available driving torque for the mechanism to be determined based on the factor of safety (FS) defined in SMBA-154.

- SMBA-154** The Mechanism Design Factor of Safety shall be 1.15 for Acceleration Torques (FS_a), 1.35 for Known Torques (FS_k), and 2.5 (FS_v) for Variable Torques.
Note: Reference NASA-STD-5017B for further clarification.

5.7.3.2 Binding/Jamming/Seizing

- SMBA-155** Mechanisms shall either be designed to preclude installation in an incorrect orientation or else be clearly labeled in a manner that indicates proper installation orientation and prevents improper installation.

Designs should include provisions to prevent binding, jamming, or seizing.

Examples of possible provisions include dual rotating surfaces or other mechanical redundancies; robust strength margins such that self-generated internal particles are precluded; shrouding and debris shielding; proper dimensioning and tolerancing; thermal analysis; proper selection of materials and lubrication design to prevent friction welding, galling, etc.; and testing to determine detrimental contamination levels.

5.7.3.2.1 Clearances

SMBA-156	Static and dynamic clearances between the mechanism and any other structure, component, thermal covering, and field of view shall be established and maintained, during all phases of the mission at the highest level of assembly. Note: NASA-STD-5017B, Section 4.2, provides guidance on best practices to maintain static and dynamics clearances.
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Maintaining clearances within and around mechanisms is necessary both to maintain proper mechanism function and to prevent the mechanism from causing problems with other systems. Establishing clearances allows one to both design the mechanism to maintain the clearances and then verify the existence of those clearances in the design and on the flight hardware.

Established clearances should account for the following:

- Manufacturing, assembly, and alignment tolerances.
- Temperature- and temperature-gradient-induced deformations or instabilities.
- Deflections due to vibration.
- Deflections due to external loads, including gravity effects.
- Deflections due to operational loads.
- Deflections due to changes in pressurization, including thermal blanket billowing.
- Motion of cable harnesses, tubing, and sensor wiring.
- Environments arising from transportation.
- The full range of adjustability of the mechanism parts.

5.7.3.2.2 Tolerancing

SMBA-157	Dimensional tolerances on all moving parts and intentional interference-fit parts shall be established and documented via a tolerance stack-up/clearance analysis to ensure that proper functional performance is maintained under all natural and induced environmental conditions and configurations.
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Natural and induced environmental conditions and configurations include but are not limited to: (1) thermally induced in-plane distortions to include, differential thermal growth and shrinkage, (2) deflections due to internal and externally applied loads; and (3) mechanical adjustment (rigging).

5.7.3.2.3 Lubrication

5.7.3.2.3.1 Lubricant Compatibility

SMBA-158 Lubricants used in the mechanism shall be compatible with the following: (1) interfacing materials (including components and fluids), (2) other lubricants used in the mechanism, (3) all natural and induced environments encountered by the mechanism, (4) outgassing/creep requirements (e.g., for nearby optical surfaces), if applicable, and (5) hygroscopic requirements, if applicable.

5.7.3.2.3.2 Lubricant Life

SMBA-159 The selection of lubricant for use in the mechanism shall be based upon development tests of the lubricant that demonstrate its ability to provide adequate lubrication under all specified operating conditions over the design lifetime.

5.7.3.2.3.3 Bearing Lubrication

SMBA-160 The selection of lubricant for use in critical moving mechanical assemblies shall be based upon development tests or flight heritage of the lubricant that demonstrate its ability to provide adequate lubrication under all specified operating conditions over the design lifetime.

SMBA-161 An analysis shall be performed for any liquid lubricant application, subject to depletion, to show that there is an adequate amount of lubricant in the system (not including degradation) for the duration of the mechanism's operational life with a factor greater than 10.

If solid lubrications are used, specific written procedures should control the method of application, subsequent handling, and ground testing of the components and assemblies to avoid exposure to moisture or humidity.

5.7.3.3 Scan Motor

SMBA-162 The torque capability of the scan drive motor shall be sufficient to perform its proper functions when the frictional torque is three times the predicted worst case.

SMBA-163 The predicted worst-case frictional torque shall include worst-case combinations of environment (particularly temperature extremes), tolerances, lubricant contamination and/or evaporation, and other possible degradation at the end of the specified operational period.

5.7.3.3.1 Scan Control Capability

SMBA-164	The scan position encoder shall provide 360-degree position readout.
SMBA-165	The beam scan angle of every beam position shall be digitized to an accuracy of no fewer than 16 bits.
SMBA-166	Filament type optical encoders shall not be used.
SMBA-167	The scan drive control loop shall have stiffness adequate to achieve specified pointing performance in the presence of at least three times the predicted worst-case frictional torque.
SMBA-168	The scan drive control loop shall have an open loop phase margin no less than 25 degrees.
SMBA-169	The scan drive control loop shall have an open loop gain margin no less than 12 dB within $\pm 10\%$ of structural mode frequencies, and no less than 6 dB at other frequencies. (TBR)
SMBA-170	The scan drive subsystem shall be analyzed and tested to ensure that scan control characteristics meet the overall system pointing error budget.

5.7.3.3.2 Contingency Capability

SMBA-171	The scan drive motor shall meet all performance requirements if in a locked condition, powered and commanded to start for up to 30 minutes.
SMBA-172	The scan drive mechanism shall meet all performance requirements in operational mode following any available shutdown mode or sequence.

5.7.3.4 Mechanism Performance and Strength Analysis

SMBA-173	Mechanism components and linkages shall have sufficient strength to tolerate an actuation force/torque stall condition at any point of travel and still maintain a positive margin of safety with the ultimate factor of safety applied.
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SMBA-174	Bearings shall have analysis demonstrating acceptable material, mounting, preload, performance, and structural integrity, accounting for the following conditions: (1) maximum combined axial, radial, and moment loads sustained during ground handling, launch, on-orbit, or other operational modes, (2) system stiffness requirements, (3) effects of temperature, temperature gradients, fits, tolerances and initial preload on torque, stiffness, and life, (4) lubrication, (5) wear, (6) smoothness of operation, (7) friction torque, considering breakaway and running, in the installed state, (8) reliability and life, and (9) effects of alignments, fits, tolerances, thermal, and load-induced distortions on preload, stress, and bearing shoulder height requirements.
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5.7.4 Electrical Systems Requirements

The spacecraft will provide electrical power generation, power management, power distribution, energy storage, load shedding control, battery state-of-charge management, and protection against over-voltage, under-voltage, and over-current during the mission lifetime in all mission modes.

The instrument should be designed to operate over a range of 26 volts direct current (VDC) to 34 VDC at the load interface.

SMBA-175	The instrument shall be designed to survive voltages from +0V up to +40 VDC at any instrument operational or survival power interface.
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5.7.5 Electromagnetic Interference and Electromagnetic Compatibility Requirements

Electromagnetic Compatibility (EMC) is necessary in each of the following areas:

- self-compatibility of an instrument to itself;
- compatibility between an instrument and all internal sources and receivers on the satellite (other instruments and the spacecraft bus); and
- compatibility between an instrument and all external sources and receivers "visible" on the ground, on the launch vehicle, and on-orbit.

While operating in the configurations necessary for transport, test, launch, separation, and on-orbit mission activities, the spacecraft bus, instruments, ground support equipment, and test equipment will operate together without performance degradation due to Electromagnetic Interference (EMI) from each other, or the external environment, and without interfering with equipment in the external environment.

SMBA-176	The instrument shall be designed to limit conducted and radiated emissions as specified by the EMI/EMC emission test limits defined in Section 7.3.
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SMBA-177	The instrument shall be designed to meet performance specifications when subjected to the conducted and radiated susceptibility tests defined in Section 7.3 as well as the On-Orbit RF environment in Section 6.2.
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All connectors that are not used for flight should have flight-quality EMI-preventative covers that are captive when installed for launch.

5.7.6 Thermal Control Requirements

The instrument provider will work with the spacecraft vendor to define and document the thermal interface requirements in the ICD.

SMBA-178	The instrument shall be thermally safe for continuous operation in all spacecraft and instrument modes.
SMBA-179	The instrument shall be capable of maintaining instrument subsystems within their operational design temperature range when the spacecraft is in Science Operations Mode or Engineering Mode.
SMBA-180	The instrument shall be capable of maintaining instrument subsystems within their design survival temperature range when the instrument is in Safe or Survival Mode.

5.7.7 Instrument Flight Software Requirements

SMBA-181	Flight processors providing computing resources shall be sized for worst-case utilization not to exceed the capacity shown in Table 5.7.7-1 (measured as a percentage of total available resource capacity).
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Table 5.7.7-1. Flight Processor Resource Utilization Limits

Test Method	Measured
RAM	70%
PROM	100%
EEPROM	70%
CPU	70%

SMBA-182	On power-up or upon hard reset, the on-board computer shall execute flight software after completion of the power-on-self-test and bootstrap process.
SMBA-183	The on-board computer shall be hard reset on execution of a discrete hardware command, or on execution of a flight software-initiated instruction or sequence of instructions.

A soft reset allows the instrument to re-establish spacecraft bus communications without loss of mission constants data, instrument error history, and where possible, loss of current mission processing parameters and data values.

SMBA-184	Flight software shall preserve the contents of selected memory regions (e.g., event log, parameter data tables) following a soft reset.
SMBA-185	The flight software shall be deterministic in terms of scheduling and prioritization of critical processing tasks to ensure their timely completion.
SMBA-186	The instrument shall have the capability to update flight software through ground commanding.

6 SPACE ENVIRONMENT REQUIREMENTS

Specified below are natural environment characteristics in the presence of which the instrument will meet all other performance requirements.

6.1 General Radiation

The instrument will be compatible with the natural radiation environments for its operational orbit. To envelope the needs over the potential operational periods for the instrument, the document defines the radiation environment for the evaluation of degradation due to total ionizing and non-ionizing dose and of single event effects for the worst case sixty-three (63) month period for the NEON Program's missions spanning from January 2028 to December 2043.

SMBA-187 The instrument shall be capable of meeting all performance requirements at the total ionizing and non-ionizing dose levels and single event effect conditions as defined in Radiation Environment for the NEON Program (470-LEO-ANYS-00018).

Total Ionizing Dose (TID): For nominal 2.5 mm (~100 mils) of equivalent aluminum shielding, the expected dose is 35.4 krad-Si (95% confidence).

Total Non-Ionizing Dose (TNID): For nominal 2.5 mm (~100 mils) of equivalent aluminum shielding, the expected displacement damage is equivalent to 1.34×10^{11} p/cm² at 63 MeV in Silicon (95% confidence).

Single Event Effects (SEE): Spectra includes entire periodic table of galactic cosmic ray and solar heavy ions, as well as protons for nominal and peak conditions such as solar flare/coronal mass ejections.

The 95% confidence level should be used as TID/TNID requirement for SEE parts.

6.2 EMI/EMC - RF Environment

SMBA-188 The instrument shall operate without performance degradation while exposed to the field defined in the Radiated Susceptibility Table 6.2-1 "On-Orbit External Performance" and "On-Orbit Steady State Performance" environments while in the intended operational mode.

SMBA-189 The instrument shall automatically recover from and operate without performance degradation after experiencing the field while in the intended operational mode for the "Launch Site & On-Orbit Operate-Through" phases of operation defined in Table 6.2-1.

SMBA-190	Instrument components shall survive after being exposed to the field while in the intended mode for the "Launch Site Survival" environments defined in the RS103 Table 6.2-1. Note: For this requirement, "survive" is defined as enduring the exposure levels and returning to nominal performance without degradation after exposure. Instrument components are not expected to meet functional or performance requirements during exposure.
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Note: See Section 7.3 for RS103 Test Method details.

Table 6.2-1. Radiated Susceptibility Electric Field Levels Due to Launch Site (Factory/Transport, PPF, Launch Vehicle, Ascent) and On-Orbit Phases (TBR)

Frequency	Launch Site Survival (V/m) ^[2,3]	Launch Site & On-Orbit Operate-Through (V/m) ^[2]	On-Orbit External Performance (V/m) ^[5]	On-Orbit Steady State Performance (V/m) ^[1]
14 kHz - 2 MHz	20	20	-	-
2 MHz - 30 MHz	20	20	20	20
30 MHz - 200 MHz	20	20	20	20
200 MHz - 420 MHz	20	20	20	20
420 MHz - 500 MHz	100	50	30	20
500 MHz - 1 GHz	30	30	20	20
1 GHz - 1.20 GHz	100	40	30	20
1.20 GHz - 1.40 GHz	100	50	30	20
1.40 GHz - 1.71 GHz	100	40	30	30
1.71 GHz - 2.2 GHz	100	40	30	20
2.2 GHz - 2.3 GHz	200	72	30	36
2.3 GHz - 2.5 GHz	200	72	45	20
2.5 GHz - 5.25 GHz	200	72	40	20
5.25 GHz - 5.925 GHz	200	120	60	20
5.925 GHz - 7.25 GHz	100	40	40	20
7.25 GHz - 8.5 GHz	100	45	40	45
8.5 GHz - 11.0 GHz	200	80	60	20
11.0 GHz - 12.8 GHz	100	20	20	20
12.8 GHz - 14 GHz	100	20	20	20
14 GHz - 18 GHz	20	20	20	20
18 GHz - 20 GHz	20	20	20	20
20 GHz - 25 GHz ^[4]	20	20	[7]	[7]
25 GHz - 25.5 GHz ^[4]	20	20	0.5 TBR	0.5, 20 ^[6] TBR
25.5 GHz - 27.0 GHz	20, 100 ^[6]	20, 50 ^[6]	1.0	1.0, 50 ^[6]
27.0 GHz - 31 GHz ^[4]	20	20	1.0 TBR	0.5, 20 ^[6] TBR
31 GHz - 33 GHz ^[4]	20	20	[7]	[7]
33 GHz - 40 GHz	20	20	6.0 TBR	2.0 TBR
40 GHz - 50 GHz ^[4]	20	20	TBD	TBD
50 GHz - 62 GHz ^[4]	20	20	[7]	[7]
62 GHz - 85 GHz ^[4]	20	20	TBD	TBD
85 GHz - 92 GHz ^[4]	20	20	[7]	[7]
92 GHz - 108 GHz ^[4]	20	20	10	2
108 GHz - 123 GHz ^[4]	20	20	[7]	[7]
123 GHz - 164 GHz ^[4]	20	20	TBD	TBD
164 GHz - 167 GHz ^[4]	20	20	[7]	[7]
167 GHz - 173 GHz ^[4]	20	20	TBD	TBD
173 GHz - 193 GHz ^[4]	20	20	[7]	[7]
193 GHz - 200 GHz ^[4]	20	20	TBD	TBD
200 GHz - 235 GHz ^[4]	20	20	[7]	[7]
235 GHz - 238 GHz ^[4]	20	20	[7]	[7]

Notes:

- [1] The On-Orbit Steady State values up to 20 GHz are baselined at 20 V/m per MIL-STD-461G Space Platforms. Values that exceed 20 V/m are due to the local transmitters and include ~6dB EMI susceptibility margin. For 20 GHz to 40 GHz levels are reduced from MIL-STD-461G to accommodate adjacent SMBA in-band frequencies. Specific environments may be lower. See Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).

- [2] The large values (100-200 V/m) do not apply across the entire frequency ranges shown, but at discrete frequencies within those bands. Consult NASA NEON for further information. The 'Launch Site Survival' environments may be reduced by any combination of procedures, facility shielding, or shipping container shielding.
- [3] The following survival test ranges are to be analyzed for compliance but not tested on the NEON Flight Units: "Launch Site Survival" ≥ 100 V/m.
- [4] Frequencies may be analyzed for compliance in lieu of testing.
- [5] On-Orbit External Modulation - 100 μ s pulses at 1 kHz for frequencies above 1 GHz.
- [6] Second higher level applies at angles greater than 15 (TBR) degrees from the SMBA boresight.
- [7] Levels are as specified by Table 6.2-2.

Table 6.2-2. Radiated Susceptibility Levels Due to In-Band Sources in On-Orbit Phases (TBD/TBR/TBS)

Frequency (GHz)		Sensitivity (dB μ V/m)	
From	To	Boresight (On-Orbit External)	>15 degrees (On-Orbit Steady State)
20.00	21.20	105 TBR	128 TBR
21.20	21.40	TBS	TBS
21.40	22.21	TBD	TBD
22.21	22.50	TBS	TBS
22.50	23.60	TBD	TBD
23.60	24.00	2 TBR	45 TBR
24.00	24.80	42 TBR	85 TBR
24.80	25.00	64 TBR	106 TBR
31.00	31.30	43 TBR	92 TBR
31.30	31.80	3 TBR	52 TBR
31.80	32.40	43 TBR	92 TBR
32.40	33.00	82 TBR	132 TBR
50.00	50.20	TBD	TBD
50.20	50.40	0 TBR	56 TBR
50.40	52.60	TBD	TBD
52.60	59.30	TBS	TBS
59.30	62.00	TBD	TBD
85.00	86.00	TBD	TBD
86.00	87.20	TBS	TBS
87.20	89.20	15 TBR	72 TBR
89.20	92.00	TBS	TBS
108.00	109.50	TBD	TBD
109.50	111.80	TBS	TBS
111.80	114.25	TBD	TBD

114.25	122.25	TBS	TBS
122.25	123.00	TBD	TBD
164.00	167.00	17 TBR	85 TBR
173.00	174.80	TBD	TBD
174.80	175.31	TBS	TBS
175.31	191.31	12 TBR	81 TBR
191.31	191.80	TBS	TBS
191.80	193.00	TBD	TBD
200.00	209.00	TBS	TBS
209.00	226.00	TBD	TBD
226.00	231.50	TBS	TBS
231.50	235.00	TBD	TBD
235.00	238.00	TBS	TBS

7 VERIFICATION REQUIREMENTS

The following paragraphs delineate the functional performance tests and demonstrations to be conducted on the instrument and its subsystems.

7.1 Mission Requirements Verification

A comprehensive verification program as specified will be conducted to ensure that the instrument, instrument components, instrument subsystems, interface with the spacecraft, and ground support systems meet the specified requirements. Certain testing will be conducted at the instrument provider facilities, at the spacecraft provider facilities, during instrument level test, instrument integration onto the spacecraft, as well as at the integrated satellite level.

The baseline instrument verification testing sequence, both at the instrument provider and spacecraft provider locations, is defined in NEON Series-1 SMBA Instrument SOW (NS1-SMBA-SOW-0002).

All instrument qualification, protoflight, and acceptance testing should be done in accordance with GSFC-STD-7000B as specified and tailored in this document. In event of conflicts, the requirements of this specification will take precedence.

All instrument qualification and protoflight tests will be conducted with hardware of the final design that has passed the in-process production screens.

Test levels including any requests for tailoring or notching will be submitted to NASA for review prior to testing.

7.2 Electrical Functional Test Requirements

The following paragraphs delineate the functional performance tests and demonstrations to be conducted on the NEON Series-1 satellite and its components, subsystems, and instruments.

7.2.1 Electrical Interface Testing

SMBA-191	As a part of the integration of a component or subsystem into the next higher level of assembly, electrical tests shall be performed to verify the interface configuration (power, grounds, commands, telemetry, signals, timing, etc.).
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During the initial integration of the instrument on the spacecraft, turn-on and turn-off transients will be measured and characterized to ensure that equipment inrush current due to turn-on transients will neither adversely affect the power bus nor overstress its circuit protection device. Refer to the NEON Series-1 Spacecraft-to-SMBA instrument ICD (NS1-SMBA-xxx-xxxx) for the instrument inrush current limits.

SMBA-192	All instrument hardware shall be verified by test or inspection for the proper polarity, orientation, and position of all components (sensors, switches, and mechanisms).
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SMBA-193	For deployment circuits, the Instrument shall demonstrate that electrical current margins on explosive or non-explosive initiator circuits are at least 20dB for the no-fire case and 6 dB for the all-fire case under all applicable environmental conditions.
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7.2.2 Comprehensive Performance Tests (CPT)

Refer to the NS1-SMBA-SOW-0002 for performing the CPTs.

7.2.3 Limited Performance Tests (LPT) and Aliveness Tests

Refer to the NS1-SMBA-SOW-0002 for performing the LPTs.

7.2.4 NEON Series-1 Ground System Compatibility Tests (GCT)

Refer to the NS1-SMBA-SOW-0002 for performing the GCTs.

7.2.5 Flight Hardware Operating Time (Burn-In) and Failure Free Performance

SMBA-194	One thousand (1000) hours of operating/power-on time shall be accumulated on all flight electronic hardware (including all redundant hardware) prior to launch.
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SMBA-195	Of the 1000 hours operating/power-on time accumulated on flight hardware prior to launch, at least 200 hours shall be in vacuum split evenly between hot and cold dwells.
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SMBA-196	Of the 1000 hours operating/power-on time accumulated on flight hardware prior to launch, the last 350 hours shall be failure-free.
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7.3 EMC/EMI Test Requirements

Where not otherwise specified by this document, electromagnetic verification testing should be conducted in accordance with MIL-STD-461G.

SMBA-197	The instrument shall be tested in accordance with Department of Defense Interface Standard - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, MIL-STD-461, Rev G, as tailored by this Section and Section 6.2.
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SMBA-198 EMI/EMC tests CE101, CMCE (Power Cables), CE102, CS101, CS114 (DM), RE102, and RS103 shall be conducted for all instruments, as applicable based on test specifications, as shown in Table 7.3-1.

SMBA-199 EMI/EMC tests CMCE (Signal Cables), CS06, CS114CM, CS115, RE101, and RS101 shall be conducted for first flight, new, and re-designed instruments, as applicable based on test specifications, as shown in Table 7.3-1.

Table 7.3-1. Instrument and Subsystem Level Test Applicability Matrix

EMI/EMC Test Method	Test Method Reference	Flight Unit Applicability ^[1]	Test Description	Test Limit or Level	Test Tailoring, Deviations, and Notes
CS101	MIL-STD-461G	ALL	CS, power leads, 30 Hz to 150 kHz	Figures 7.3.4-1 & 7.3.4-2	Section 7.3.4.1
CS06	GSFC-STD-7000B & MIL-STD-462/461C	1st Flight Unit	CS, power leads, transients	GSFC-STD-7000B Figure 2.5-18	Section 7.3.4.2
CS114(CM)	MIL-STD-461G	1st Flight Unit	CS, bulk cable injection, 10 kHz to 200 MHz	MIL-STD-461G FIGURE CS114-1, Curve 3	Section 7.3.4.3
CS114(DM)	GSFC-STD7000B	ALL	CS, power lead injection, 150 kHz to 50 MHz	Figure 7.3.4-3	Section 7.3.4.4
CS115	MIL-STD-461G	1st Flight Unit	CS, Bulk Cable injection, Impulse excitation (cables only, not leads)	Power Cables: MIL-STD-461G FIGURE CS115-1, 5A max Signal Cables: 1A min	Section 7.3.4.5
CE101	MIL-STD-461G	ALL	CE, Power Leads, 30 Hz to 150 kHz	MIL-STD-461G Figure A-10, Curve 2	Section 7.3.3.1
CMCE (Primary Power Cables)	GSFC-STD-7000B	ALL	Common Mode Conducted Emissions, 10 kHz to 30 MHz on Primary Power Cables	Figure 7.3.3-1	Section 7.3.3.2
CMCE (All Cables)	GSFC-STD-7000B	1st Flight Unit	Common Mode Conducted Emissions, 10 kHz to 200 MHz on all cables	Figure 7.3.3-1	Section 7.3.3.2
CE102	MIL-STD-461G	ALL	CE, Power Leads, 10 kHz to 10 MHz	MIL-STD-461G FIGURE CE102-1, BASIC Curve	Section 7.3.3.3
RS101	MIL-STD-461G	1st Flight Unit	RS, Magnetic Field, 30 Hz to 100 kHz	MIL-STD-461G FIGURE RS101-1	Section 7.3.2.4
RS103	MIL-STD-461G	ALL	RS, Electrical Field, 14 kHz to 40 GHz	See Section 6.2	Section 7.3.2.1 & 7.3.2.2
RE101	MIL-STD-461G	1st Flight Unit	RE, Magnetic Field, 30 Hz to 100 kHz	MIL-STD-461G FIGURE RE101-2	Section 7.3.1.3
RE102	MIL-STD-461G	ALL	RE, Electrical Field, 14 kHz to 18 GHz	Tables 7.3.1-1	Section 7.3.1.1

Notes:

[1] 1st Flight Unit also refers to new and re-designed units per SMBA-199.

SMBA-200 For all MIL-STD-461G EMI/EMC testing, the line impedance stabilization network (LISN) in MIL-STD-461G shall be tailored, as necessary, to be more representative of the spacecraft power bus impedance. When not otherwise specified, the 5 microhenry LISN specified in Section A.4.3.6, Figures A-2 & A-3, of MIL-STD-461G, along with a minimum of 5000uF at the LISN power source input(s) (line-to-line) will be utilized in place of the standard 50 microhenry LISN. The return sides LISN input should be grounded to chassis ground to establish a single point ground if not established elsewhere.

Flight-like cables (constructed with expected flight wire groupings, twisting, shielding, and shield terminations) should be used for all EMC testing.

All connectors that are not used for flight should have flight-quality EMI-preventative covers that are captive when installed for launch. For system-level testing, connectors that are not used for flight will be capped with EMI-preventative covers.

7.3.1 Radiated Emissions (RE)

7.3.1.1 Radiated Emissions, Electric Field, RE102

The purpose of this requirement is to verify that electric field emissions from the satellite and its associated cabling do not interfere with on-board RF receivers or launch vehicle receivers. The specific requirements for the test, including test methods, are detailed in requirements below:

SMBA-201 The instrument shall meet the radiated emissions limits specified in Table 7.3.1-1.

SMBA-202 The basic test method of MIL-STD-461G RE102 shall apply for the Satellite Radiated (E-Field) Emissions Test.

SMBA-203 The Instrument RE102 test shall be conducted in the worst case instrument modes.

SMBA-204 For the receiver notches specified in Table 7.3.1-1, the RE102 EMI antennas shall be positioned such that emissions are measured in the direction of the spacecraft receive antennas as depicted in the MICD or ICD.

SMBA-205 The frequency range for the RE102 test shall be 2 MHz to 18 GHz for the Nominal Operational Mode and 14 kHz to 18 GHz for the Launch Mode.

SMBA-206 Unintentional radiated emission measurements in the specified Table 7.3.1-1 receiver bands (notches) shall be made in accordance with MIL-STD-461G RE102 with the EMI receiver preceded by a low-noise preamplifier at the measurement antenna such that the test system noise figure is less than or equal to 3 dB.

SMBA-207 The RE102 bandwidth limits shall be as specified in Table 7.3.1-1.

Table 7.3.1-1. Unintentional Radiated Electric Field Emissions, 14kHz to 18 GHz

Frequency (MHz)	Limit Level (dB μ V/m)	Minimum 6dB Resolution Bandwidth (Hz)	Receiver Protected
0.014 ^[1]	58	1K	
0.018 ^[1]	56	1K	
0.15 ^[1]	56	1K	
0.15 ^[1]	56	10K	
2	56	10K	
30	56	10K	
30	56	100K	
400	56	100K	LV
400	56, 28 ^[1]	100K	LV
500	56, 28 ^[1]	100K	LV
500	56	100K	LV
1000	56	100K	
1000	56	1M	
1100	56	1M	
1100	42, 33 ^[1]	1M, 100K ^[1]	GPS-L2
1350	42, 33 ^[1]	1M, 100K ^[1]	GPS-L2
1350	56	1M	
1475	56	1M	
1475	43, 33 ^[1]	1M, 100K ^[1]	GPS-L1
1556	43, 33 ^[1]	1M, 100K ^[1]	GPS-L1
1556	25	10K	GPS-L1
1562	25	10K	GPS-L1
1562	14	1K	GPS-L1
1589	14	1K	GPS-L1
1589	25	10K	GPS-L1
1592	25	10K	GPS-L1
1592	43, 33 ^[1]	1M, 100K ^[1]	GPS-L1
1675	43, 33 ^[1]	1M, 100K ^[1]	GPS-L1
1675	56	1M	
2000	56	1M	
2000	45	1M	
2015	45	1M	
2015	25	10K	
2025	25	10K	SC-CMD
2025	15	1K	SC-CMD
2110	15	1K	SC-CMD
2110	25	10K	SC-CMD

2120	25	10K	
2120	45	1M	
2150	45	1M	
2150	50	1M	
5000	50	1M	
5000-18000	+20 dB per decade	1M	
18000	61	1M	

Notes:

[1] Applies only to Launch Mode if powered for launch.

7.3.1.2 RE, Magnetic Field, RE101

SMBA-208 Radiated AC magnetic field emissions from the instrument shall not exceed the levels shown in MIL-STD-461G Figure RE101-2 at a distance of 7 cm per MIL-STD-461G, RE101 from 30 Hz to 100 kHz.
Note: Test requirement applies to new or redesigned hardware; otherwise previous test artifacts may be substituted in lieu of testing.

7.3.2 Radiated Susceptibility (RS)

7.3.2.1 RS, Electric Field, Orbit and Launch Levels

The purpose of this requirement is to verify the ability of the instruments, spacecraft hardware, and associated cabling to withstand electric fields generated from intentional RF transmitters during all phases of the project, including on-orbit, on the launch vehicle, at the launch site, and all test environments. The specific requirements for the test, including test methods, are detailed in requirements below:

SMBA-209 The Instrument shall meet the radiated susceptibility requirements defined in Section 6.2 when subjected to the radiated electric field tests as defined in Sections 7.3.2.1 and 7.3.2.2.

SMBA-210 For the Satellite Orbit and Launch Mode Radiated (E-Field) Susceptibility Tests, the MIL-STD-461G RS103 test method shall apply.

SMBA-211 The frequency range for the RS103 tests shall be 14 kHz to 18 GHz for the Launch Mode (if powered) and 2 MHz to 40 GHz for the on-orbit Operational Mode. Note: Testing above 18 GHz may be limited to frequencies identified in Table 6.2-1.

SMBA-212 The dwell time for each frequency step shall be tailored from the MIL-STD-461G default of 3 seconds to one scan period or integer multiple of the scan period at the instrument locations.

SMBA-213	For the On-Orbit External levels for frequency ranges 200 MHz to 40 GHz (as identified in Table 6.2-1), at least one test antenna position shall radiate within the Field of View (aperture) of the Instrument.
SMBA-214	For the Satellite RS103 Test on-orbit configuration, the levels in Table 6.2-1 columns 2, 3 and 4 shall apply.
SMBA-215	For the Satellite RS103 Test Launch Configuration, if powered, the levels in Table 6.2-1 columns 1 and 2 shall apply.
SMBA-216	For all Radiated Susceptibility Electric Field Tests, sufficient filtering shall be employed on EMI test equipment to prevent broadband noise and harmonics from interfering with the SMBA receiver bands (see Table 7.3.1-4, SMBA Receive Sensitivity).
SMBA-217	For all Radiated Susceptibility Electric Field Tests above 200 MHz, analysis shall show that the output power applied at the test antenna produces the required levels at the Instrument and ensures that measured levels are not the result of peak resonances or reflections.

7.3.2.2 RS, Electric Field, Satellite Transmitter Frequencies

The purpose of this requirement is to verify the ability of the satellite to withstand electric fields generated from intentional RF transmitters during the on-orbit phase of the mission. The specific requirements for the test, including test methods, are detailed in requirements below:

SMBA-218	The MIL-STD-461G RS103 test method shall apply for the satellite transmitter frequency radiated susceptibility test as defined in Section 7.3.2.2.
SMBA-219	Minimum scan rates and step sizes shall be reduced from the MIL-STD-461G per Table 7.3.2-1.
SMBA-220	Minimum dwell times for stepped scans shall be tailored from the MIL-STD-461G requirement of 3 seconds per Table 7.3.2-1 to one scan period or integer multiple of the scan period.
SMBA-221	The susceptibility scans shall be performed using the modulation format of the specific transmitter(s) that will be producing the fields on the satellite; however, if this is not known or test equipment limitations prevent such modulation, then modulation will be pulse modulated (on/off ratio of 40 dB minimum) at a 1 kHz rate with a 50% duty cycle per MIL-STD-461G.

SMBA-222 Satellite transmitter frequency RS103 test parameters and exposure levels for the instrument shall be as specified in Table 7.3.2-1.

Table 7.3.2-1. Spacecraft Transmitter Frequency Susceptibility Test Levels (TBD/TBR)

Transmitter Name	Start Frequency (MHz)	Stop Frequency (MHz)	Analog Scan, Maximum Rate (MHz/sec)	Stepped Scan, Maximum Step Size (*f0)	Minimum Dwell Time per step (Sec.)	Level (V/m) ^[1]
S-Band (Center Freq)	TBD	TBD	NA	NA	64	36
S-Band (Full BW)	TBR 2200	TBR 2290	0.0166	0.00001 (1.0E-05)	3 ^[2]	36
X-Band, (Center)	TBD	TBD	NA	NA	64	45
X-Band, (Full BW)	TBR 7750	TBR 7900	0.166	0.00006 (6.0E-05)	3 ^[2]	45
Ka-Band, (Center)	TBD	TBD	NA	NA	64	50, 1.0 ^[3]
Ka-Band, (Full BW)	TBR 25500	TBR 27000	2.0	0.0002 (2.0E-04)	3 ^[2]	50, 1.0 ^[3]

Notes:

[1] The limits above are generic levels and include a ~6dB margin. These levels may be tailored to expected distances from the specified emitting antenna, however margin must be maintained to account for multipath effects.

[2] MIL-STD-461G default is 3 seconds, Exceptions – test should use SMBA scan period(s) (see SMBA Instrument to Spacecraft ICD (NS1-SMBA-xxx-xxxx)).

[3] Reduced level within critical stayout zone of SMBA boresight.

7.3.2.3 RS, SMBA Instrument Compatibility

SMBA-223 The instrument provider shall confirm performance and evaluate the ability to meet the radiated emissions requirements shown in Tables 7.3.2-2 and 7.3.2-3 at the instrument apertures.

Spacecraft components and other instruments will meet the radiated emissions requirements shown in Table 7.3.2-2. The SMBA instrument will not be susceptible below these levels at angles > 15.5 (TBR) degrees from the aperture boresight(s).

Table 7.3.2-2. Intentional and Unintentional Radiated Electric Field Emissions Limits at the SMBA Instrument Interface, 18 to 235 GHz (TBR)

(This table will be revised to reflect the specifics of the instrument chosen)

Frequency Range From (MHz)	Frequency Range To (MHz)	Emission Limits (dBμV/m) [1]	Bandwidth	Susceptible Receiver or Intentional Transmitter
18000	31800	69 - 74	1 MHz	General Requirement
21200	21400	TBS	TBS	SMBA K-Band
22210	22500	TBS	TBS	SMBA K-Band
23567.25	24032.75	45	100 Hz	Heritage Channel
25500	27000	154	300 MHz	SC Ka Transmitter

31241.5	31558.5	52	1000 Hz	Heritage Channel
31558.5	31800	TBS	TBS	SMBA Ka Band
31800	40000	74 - 76	1 MHz	General Requirement
50141.5	50458.5	56	180 MHz	Heritage Channel
50200	50400	TBS	TBS	SMBA V-Band
52465	53135	61	400 MHz	Heritage Channel
53096	53236	TBS	140 MHz	Heritage Channel
53256	53396	TBS	140 MHz	Heritage Channel
53316	53596	60	170 MHz	Heritage Channel
53596	53876	60	170 MHz	Heritage Channel
53796	53938	TBS	142 MHz	Heritage Channel
53958	54100	TBS	142 MHz	Heritage Channel
54065	55782.25	60	400 MHz	Heritage Channels
56889.22	57691.46	47	330 MHz	Heritage Channels
52600	59300	TBS	TBS	SMBA V-Band
86000	92000	TBS	TBS	SMBA W-Band
86500	89900	71	2000 MHz	Heritage Channel
87000	91000	TBS	4000 MHz	Heritage Channel
100000	102000	TBS	TBS	TBD
105000	108000	TBS	TBS	TBD
108000	122250	TBS	TBS	SMBA F-Band
114250	115250	TBS	1000 MHz	Heritage Channel
115550	116350	TBS	800 MHz	Heritage Channel
116350	116950	TBS	600 MHz	Heritage Channel
116950	117550	TBS	600 MHz	Heritage Channel
117550	118050	TBS	500 MHz	Heritage Channel
118050	118430	TBS	380 MHz	Heritage Channel
118430	118730	TBS	300 MHz	Heritage Channel
118730	122250	TBS	TBS	SMBA F-Band
148500	151500	TBS	TBS	TBD
162950	168050	84	3000 MHz	Heritage Channel
164000	167000	TBS	TBS	SMBA D-Band
174610	178010	86	2000 MHz	Heritage Channel
174800	191800	TBS	TBS	SMBA G-Band
177110	180510	87	2000 MHz	Heritage Channel
179460	181160	84	1000 MHz	Heritage Channel
180660	182360	84	1000 MHz	Heritage Channel
181885	182735	81	500 MHz	Heritage Channel
183885	184735	81	500 MHz	Heritage Channel
184260	185960	84	1000 MHz	Heritage Channel
185460	187160	84	1000 MHz	Heritage Channel
186110	189510	87	2000 MHz	Heritage Channel
188610	192010	87	2000 MHz	Heritage Channel
200000	209000	TBS	TBS	SMBA J-Band
217000	231500	TBS	TBS	SMBA J-Band
228000	230000	TBS	TBS	Heritage Channel
230000	231500	TBS	TBS	SMBA J-Band

Notes:

- [1] General requirement is 69 dB μ V/m at 18 GHz and increases +20 dB/decade to 31.8 GHz, then 74 dB μ V/m increasing +20 dB/decade. Limits apply at the respective channel Bandwidths at the location of the SMBA apertures. Levels above the general requirement are intentional spacecraft emission limits. This table assumes the emitter is located at an angle outside the critical stayout zones identified in the ICD. Less stringent limits may apply at angles farther from these areas. Refer to Table 7.3.2-3. No RF energy may be directed to within critical stay out zones referenced from the SMBA scan planes.

Table 7.3.2-3. SMBA Receive Sensitivity (TBR)**(This table will be revised to reflect the specifics of the instrument chosen)**

Frequency Band	Center Frequency (MHz)	Max. Bandwidth (MHz)	Frequency Range		EMI Sensitivity of SMBA Passbands [4]			
			From (MHz)	To (MHz)	dBm	dBμV/m Bore-sight ≤15.5° [1][5]	dBμV/m ≤40° [2] [5]	dBμV/m >40° [3] [5]
K	TBS	TBS	20000	25000	TBS	TBS	TBS	TBS
K	23800	270	21300	22000	-60	72.79	115.51	129.01
			22000	22800	-69	63.79	106.51	120.01
			22800	23567.25	-91	42.22	84.94	98.44
			23567.25	24032.75	-131	2.45	45.17	58.67
			24032.75	24800	-91	42.67	85.39	98.89
			24800	25900	-70	64.07	106.79	120.29
			25900	26300	-27	107.07	149.79	163.29
			26300	26550	-17	117	160	173
			26550	26850	-17	117	160	173
			26850	28900	-17	118	160	174
Ka	TBS	TBS	31000	33000	TBS	TBS	TBS	TBS
Ka	31400	180	28900	30400	-53	82.76	132.17	161.17
			30400	31241.5	-93	43.1	92.51	121.51
			31241.5	31558.5	-133	3.26	52.67	81.67
			31558.5	32400	-93	43.42	92.83	121.83
			32400	33900	-54	82.73	132.14	161.14
V	TBS	TBS	50000	62000	TBS	TBS	TBS	TBS
V	50300	180	47641.5	49300	-53	79.53	136.58	149.58
			49300	50141.5	-93	39.75	96.8	109.8
			50141.5	50458.5	-133	-0.15	56.9	69.9
			50458.5	51300	-93	39.95	97	110
			51300	52800	-53	80.15	137.2	150.2
V	52800	400	50300	51800	-49	83.98	141.03	154.03
			51800	52465	-89	44.16	101.21	114.21
			52465	53135	-129	4.27	61.32	74.32
			53135	53800	-89	44.38	101.43	114.43
			53800	55300	-49	84.56	141.61	154.61
V	53246 ± 80	140	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
V	53596 ± 115	170	51096	52596	-50	83.11	140.16	153.16
			52596	53336	-90	43.3	100.35	113.35
			53336	53596	-130	3.45	60.5	73.5
			53596	53856	-130	3.44	60.49	73.49
			53856	54596	-90	43.51	100.56	113.56
			54596	56096	-50	83.68	140.73	153.73
V	53948 ± 81	142	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
V	54400	400	51900	53400	-49	84.25	141.3	154.3
			53400	54065	-89	44.42	101.47	114.47
			54065	54735	-129	4.53	61.58	74.58
			54735	55400	-89	44.64	101.69	114.69
			55400	56900	-49	84.81	141.86	154.86
V	54940	400	52440	53940	-49	84.34	141.39	154.39
			53940	54605	-89	44.51	101.56	114.56
			54605	55275	-129	4.62	61.67	74.67
			55275	55940	-89	44.72	101.77	114.77
			55940	57440	-49	84.89	141.94	154.94
V	55500	330	53000	54500	-50	83.43	140.48	153.48
			54500	55217.75	-90	43.6	100.65	113.65
			55217.75	55782.25	-130	3.71	60.76	73.76
			55782.25	56500	-90	43.81	100.86	113.86

			56500	58000	-50	83.98	141.03	154.03
V	57290.344	330	54790.34	56290.34	-50	83.71	140.76	153.76
			56290.34	57017.59	-90	43.88	100.93	113.93
			57017.59	57563.09	-130	3.98	61.03	74.03
			57563.09	58290.34	-90	44.08	101.13	114.13
			58290.34	59790.34	-50	84.24	141.29	154.29
V	57290.344 ± 217	78	54790.34	56290.34	-53	80.71	137.76	150.76
			56290.34	57008.4	-93	40.88	97.93	110.93
			57008.4	57138.19	-133	0.95	58	71
			57138.19	57442.49	-93	40.98	98.03	111.03
			57442.49	57572.19	-133	1.01	58.06	71.06
			57572.19	58290.34	-93	41.08	98.13	111.13
			58290.34	59790.34	-53	81.24	138.29	151.29
V	57290.344 ± 322.2 ± 48	36	54790.34	56290.34	-53	80.71	137.76	150.76
			56290.34	56889.22	-93	40.87	97.92	110.92
			56889.22	56951.02	-133	0.93	57.98	70.98
			56951.02	56985.22	-93	40.93	97.98	110.98
			56985.22	57047.02	-133	0.94	57.99	70.99
			57047.02	57533.66	-93	40.98	98.03	111.03
			57533.66	57595.46	-133	1.02	58.07	71.07
			57595.46	57629.66	-93	41.03	98.08	111.08
			57629.66	57691.46	-133	1.04	58.09	71.09
			57691.46	58290.34	-93	41.09	98.14	111.14
			58290.34	59790.34	-53	81.24	138.29	151.29
V	57290.344 ± 322.2 ± 22	16	54790.34	56290.34	-57	76.71	133.76	146.76
			56290.34	56931.72	-97	36.88	93.93	106.93
			56931.72	56960.52	-137	-3.07	53.98	66.98
			56960.52	56975.72	-97	36.93	93.98	106.98
			56975.72	57004.52	-137	-3.06	53.99	66.99
			57004.52	57576.16	-97	36.98	94.03	107.03
			57576.16	57604.96	-137	-2.97	54.08	67.08
			57604.96	57620.16	-97	37.03	94.08	107.08
			57620.16	57648.96	-137	-2.97	54.08	67.08
			57648.96	58290.34	-97	37.08	94.13	107.13
			58290.34	59790.34	-57	77.24	134.29	147.29
V	57290.344 ± 322.2 ± 10	8	54790.34	56290.34	-60	73.71	130.76	143.76
			56290.34	56951.02	-100	33.88	90.93	103.93
			56951.02	56965.22	-140	-6.07	50.98	63.98
			56965.22	56971.02	-100	33.93	90.98	103.98
			56971.02	56985.22	-140	-6.07	50.98	63.98
			56985.22	57595.46	-100	33.98	91.03	104.03
			57595.46	57609.66	-140	-5.97	51.08	64.08
			57609.66	57615.46	-100	34.03	91.08	104.08
			57615.46	57629.66	-140	-5.97	51.08	64.08
			57629.66	58290.34	-100	34.08	91.13	104.13
			58290.34	59790.34	-60	74.24	131.29	144.29
V	57290.344 ± 322.2 ± 4.5	3	54790.34	56290.34	-64	69.71	126.76	139.76
			56290.34	56960.65	-104	29.88	86.93	
			56960.65	56966.6	-144	-10.07	46.98	59.98
			56966.6	56969.65	-104	29.93	86.98	99.98
			56969.65	56975.6	-144	-10.07	46.98	59.98
			56975.6	57605.09	-104	29.98	87.03	100.03
			57605.09	57611.04	-144	-9.97	47.08	60.08
			57611.04	57614.09	-104	30.03	87.08	100.08
			57614.09	57620.04	-144	-9.97	47.08	60.08
			57620.04	58290.34	-104	30.08	87.13	100.13
			58290.34	59790.34	-64	70.24	127.29	140.29
W	TBS	TBS	85000	92000	TBS	TBS	TBS	TBS
W	88200	2000	83250	84750	-42	94.93	151.47	163.47
			84750	86500	-82	55.04	111.58	123.58
			86500	89900	-122	15.35	71.89	83.89
			89900	91650	-82	55.65	112.19	124.19
			91650	93150	-42	95.75	152.29	164.29
W	89000	4000	TBS	TBS	TBS	TBS	TBS	TBS

			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	TBS	TBS	108000	123000	TBS	TBS	TBS	TBS
F	114750	1000	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	115950	800	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	116950 ± 300	600	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	117800	500	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	118240	380	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
F	118580	300	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
D	TBS	TBS	164000	167000	TBS	TBS	TBS	TBS
D	165500	3000	158000	160475	-40	97.42	164.63	176.63
			160475	162950	-80	57.48	124.69	136.69
			162950	168050	-120	17.6	84.81	96.81
			168050	170525	-80	57.71	124.92	136.92
			170525	173000	-40	97.76	164.97	176.97
G	TBS	TBS	173000	193000	TBS	TBS	TBS	TBS
G	183310±7000	2000	173810	174210	-39	98.46	166.86	179.36
			174210	174610	-79	58.51	126.91	139.41
			174610	178010	-119	18.53	86.93	99.43
			178010	188610	-79	58.88	127.28	139.78
			188610	192010	-119	19.22	87.62	100.12
			192010	192410	-79	59.24	127.64	140.14
			192410	192810	-39	99.29	167.69	180.19
G	183310±4500	2000	176310	176710	-39	98.58	166.98	179.48
			176710	177110	-79	58.64	127.04	139.54
			177110	180510	-119	18.65	87.05	99.55
			180510	186110	-79	58.88	127.28	139.78
			186110	189510	-119	19.11	87.51	100.01
			189510	189910	-79	59.12	127.52	140.02
			189910	190310	-39	99.17	167.57	180.07
G	183310±3000	1000	177810	178635	-42	95.66	164.06	
			178635	179460	-82	55.71	124.11	136.61
			179460	181160	-122	15.73	84.13	96.63
			181160	185460	-82	55.88	124.28	136.78
			185460	187160	-122	16.04	84.44	96.94
			187160	187985	-82	56.06	124.46	136.96
			187985	188810	-42	96.11	164.51	177.01
G	183310±1800	1000	179010	179835	-42	95.71	164.11	176.61
			179835	180660	-82	55.77	124.17	136.67
			180660	182360	-122	15.78	84.18	96.68
			182360	184260	-82	55.88	124.28	136.78
			184260	185960	-122	15.98	84.38	96.88
			185960	186785	-82	56	124.4	136.9

			186785	187610	-42	96.05	164.45	176.95
G	183310±1000	500	179810	180847.5	-45	92.75	161.15	173.65
			180847.5	181885	-85	52.8	121.2	133.7
			181885	182735	-125	12.82	81.22	93.72
			182735	183885	-85	52.88	121.28	133.78
			183885	184735	-125	12.95	81.35	93.85
			184735	185772.5	-85	52.96	121.36	133.86
			185772.5	186810	-45	93.01	161.41	173.91
J	TBS	TBS	200000	238000	TBS	TBS	TBS	TBS
J	229000	2000	TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS
			TBS	TBS	TBS	TBS	TBS	TBS

Notes:

- [1] For susceptibility to a source within 15.5° (TBS) of the SMBA antenna boresight, the "Boresight ≤15.5°" column should be used for worst-case analysis.
- [2] For susceptibility to a source between 15.5° (TBS) and 40° (TBS) of the SMBA antenna boresight, the "≤40°" column should be used for worst-case analysis.
- [3] For susceptibility to a source at a 40°(TBS) or greater angle from the SMBA antenna boresight, the ">40°" column should be used for worst-case analysis.
- [4] Limits apply at the Max Bandwidth of the respective SMBA channel.
- [5] See Spacecraft-to-SMBA Instrument MICD (NS1-SMBA-xxx-xxxx) for EMI Boresight Stayout Zones.

7.3.2.4 RS, Magnetic Field, RS101

SMBA-224 The instrument shall operate within specified performance, when subjected to the magnetic field levels shown in MIL-STD-461G, Figure RS101-1 (Navy parameters), from 30 Hz to 100 kHz per the RS101 test method.
Note: Test requirement applies to new or redesigned hardware; otherwise previous test artifacts may be substituted in lieu of testing.

7.3.3 Conducted Emissions Tests

7.3.3.1 CE101

SMBA-225 The instrument shall limit conducted current emissions to the CE101 current level shown in MIL-STD-461G, Section A.5.4, Figure A-10, Curve 2, when tested from 30 Hz to 150 kHz per the MIL-STD-461G Section 5.4 CE101 test method.
Note 1: The CE101 differential mode low-frequency plateau limit from 30 Hz to 2 kHz is determined by adding a factor of 20*log (load current in Amperes) to a base of 100 dBμA.
Note 2: A capacitor will be inserted (line-to-line) on the power supply side of the LISNs specified in SMBA-200.

7.3.3.2 CMCE

SMBA-226

Common Mode Conducted Emissions (CMCE) from Instrument power and signal cables shall not exceed the limits specified in Figure 7.3.3-1 when tested from 10 kHz to 200 MHz per GSFC-STD-7000B Section 2.5.2.1.2 except that the 5 microHenry LISN specified SMBA-200 may be used with the capacitor network on the power supply side of the LISN. Subsequent flight Instruments need only to perform CMCE on power cables from 10 kHz to 30 MHz.

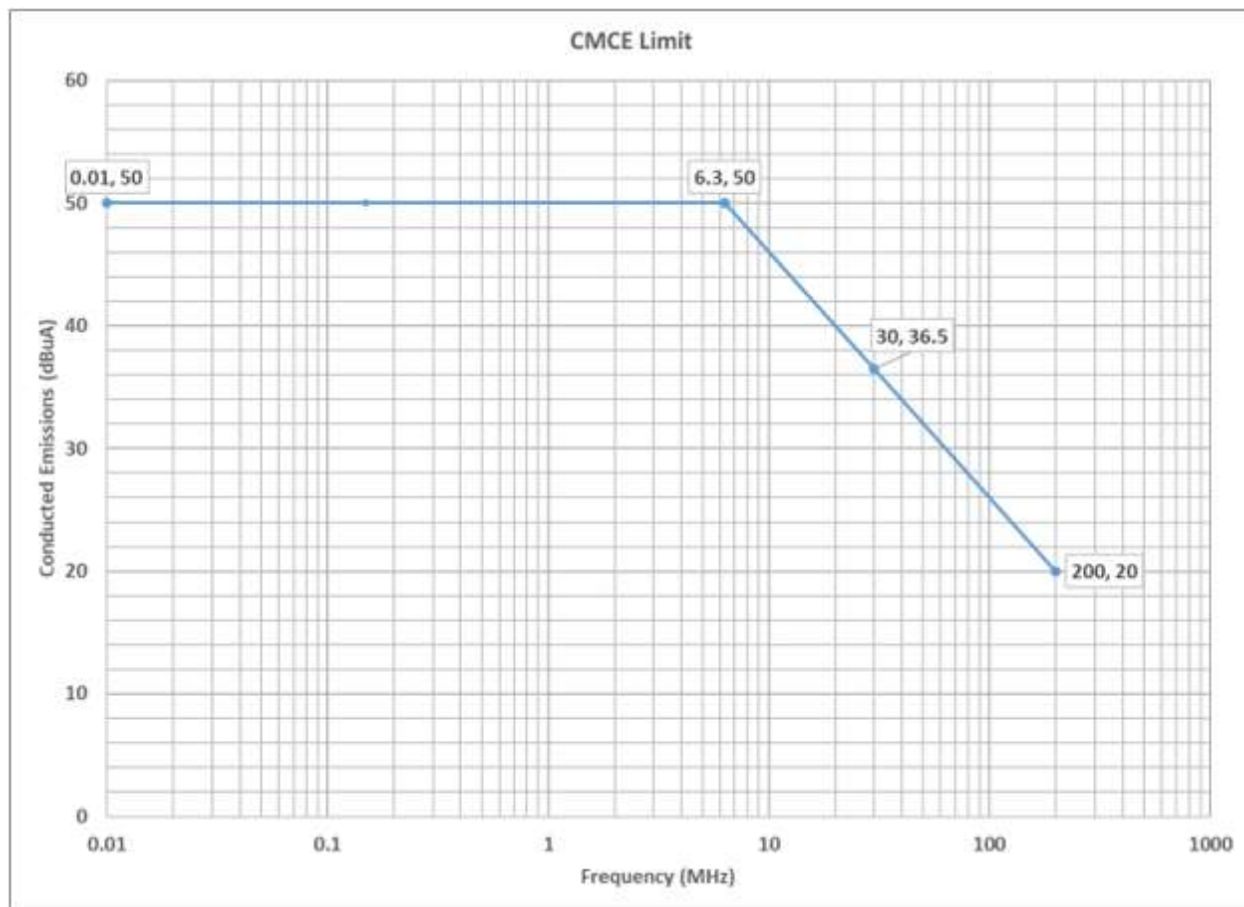


Figure 7.3.3-1. Common Mode Conducted Emissions (CMCE) Limit

7.3.3.3 CE 102

SMBA-227

The instrument shall limit conducted emissions from 10 kHz to 10 MHz to the Basic Curve voltage level shown in MIL-STD-461G, Figure CE102-1, and tested per the CE102 Test method using LISNs specified in SMBA-200.

7.3.4 Conducted Susceptibility Tests

7.3.4.1 CS101

SMBA-228

The instrument shall operate within specified performance, when power leads are subjected to the voltage levels shown in Figure 7.3.4-1, from 30 Hz to 150 kHz, per the MIL-STD-461G CS101 test method. The requirement is also met when the power source is adjusted to dissipate the power level shown in Figure 7.3.4-2 in a 0.5 ohm load and the instrument is not susceptible.

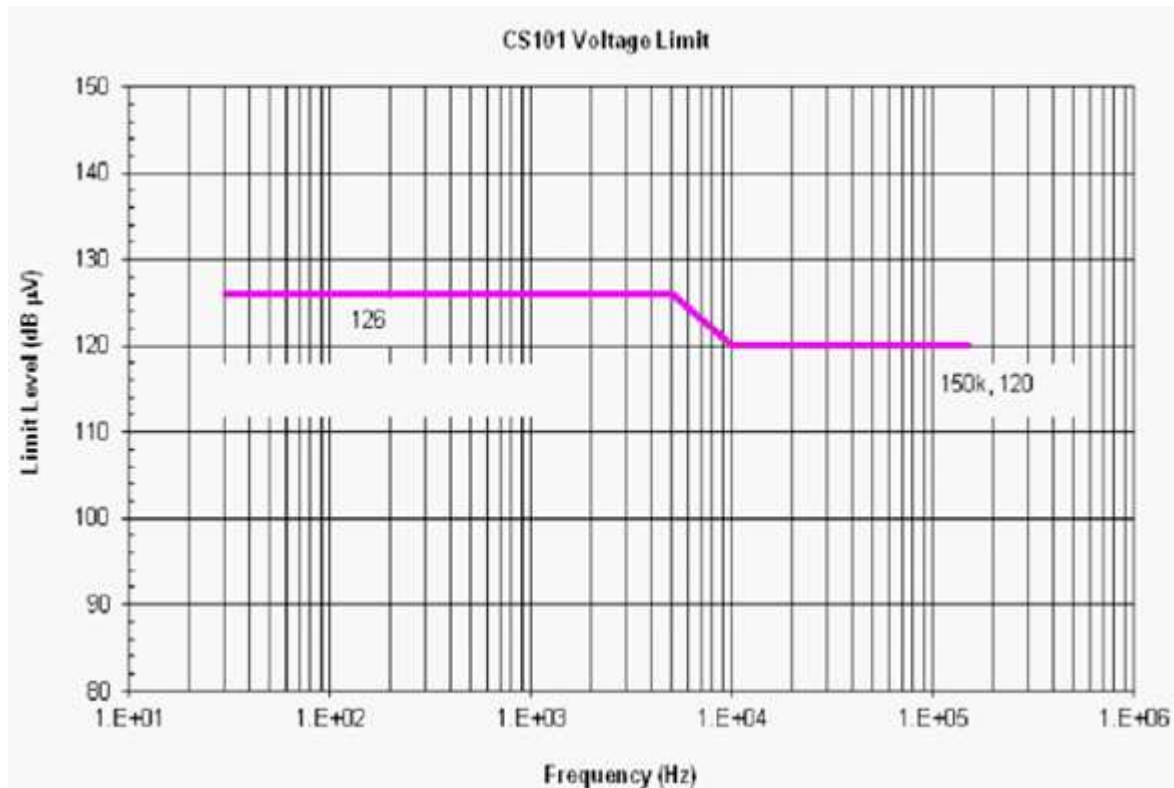


Figure 7.3.4-1. Instrument Power Lead Conducted Susceptibility (CS101) Voltage Limit

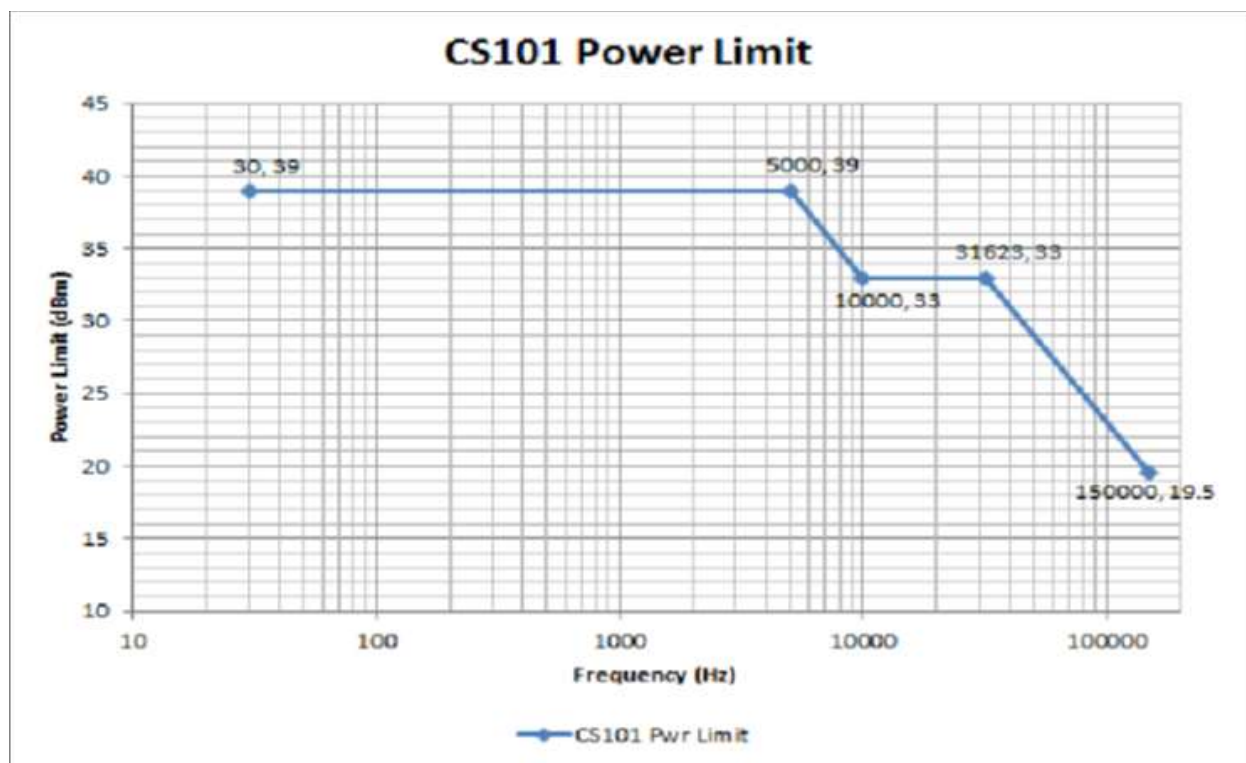


Figure 7.3.4-2. Instrument Power Lead Conducted Susceptibility (CS101) Power Limit

7.3.4.2 CS06

SMBA-229

The instrument shall operate without malfunction, loss of capability, change of operation state/mode, memory changes, or need for outside intervention when subjected to the MIL-STD-462 test CS06 (negative transient only) per GSFC-STD-7000B section 2.5.2.2.3, Figure 2.5-18, with a negative spike tolerance of +5.0V/-0V.

SMBA-230

For the (full level) CS06 test, each spike shall be superimposed on the powerline voltage (+30 V) having the waveform shown in GSFC-STD-7000B, Section 2.5.2.2.3, Figure 2.5-18, and applied to the power input leads once per second, for a total test period of 1 minute in duration. Note: The in-line 5 microHenry LISNs specified in SMBA-200 along with any necessary in line inductance in series or capacitance across the LISN outputs may be used in place of the GSFC-STD-7000B specified capacitor network.

7.3.4.3 CS114CM

SMBA-231

The Instrument shall operate within specified performance, when spacecraft interface power and signal cables are subjected to the calibrated limits shown in Curve 3 of MIL-STD-461G Figure CS114-1, from 10 kHz to 200 MHz per the CS114 test method (Common-Mode).

7.3.4.4 CS114DM

SMBA-232

The Instrument shall operate within specified performance, when the high-side power leads are subjected to an injection probe drive level which has been precalibrated to the level shown in Figure 7.3.4-3, from 150 kHz to 50 MHz. The test method is per MIL-STD-461G CS114 (high power leads with returns and grounds removed) except that a capacitor (10uF minimum) is placed across the supply and return terminals of the LISN outputs and the current limit specified in Figure 7.3.4-3 is used.

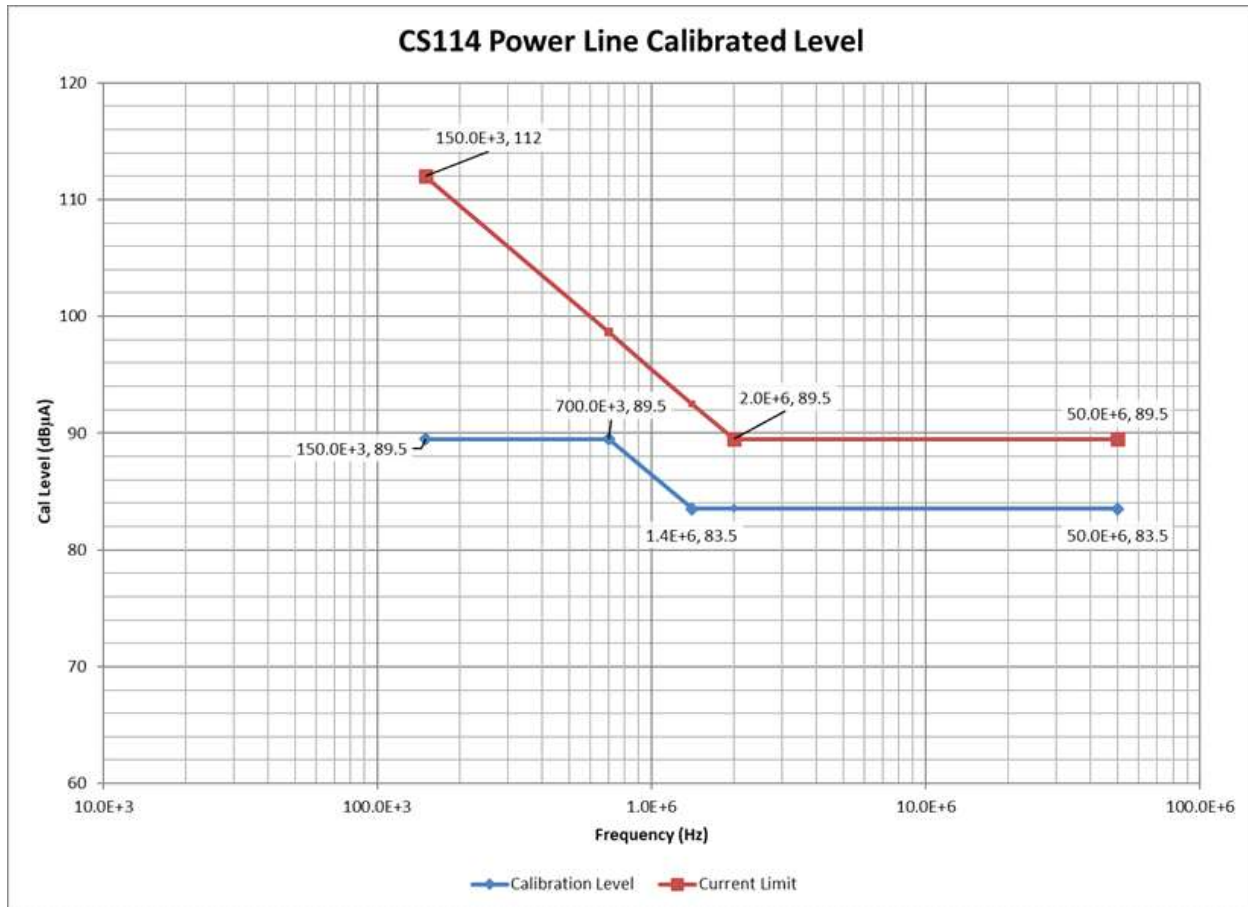


Figure 7.3.4-3. Conducted Susceptibility, Power Line Injection (CS114), Calibrated Level and Current Limit

7.3.4.5 CS115

- SMBA-233** The instrument shall operate without malfunction, loss of capability, change of operation state/mode, memory changes or need for outside intervention when spacecraft interface power cables are subjected to a pre-calibrated signal having rise and fall times, pulse width, and amplitude as specified in MIL-STD-461G, Figure CS115-1, at a 30 Hz rate for one minute per the CS115 test method. Signal cables are to be subjected to the same test except the amplitude may be reduced to 1A min.
Note: Only Bulk Cable Common Mode Injection is applicable.

7.4 Structural and Mechanical Test Requirements

The instrument will experience no unintentional change of state when subjected to the specified dynamic environments specified herein.

- SMBA-234** Structural and mechanical tests and analyses defined in Table 7.4-1 shall be conducted to demonstrate that the flight hardware is qualified for the expected mission environments and that the design of the hardware complies with the specified verification requirements such as factors of safety, interface compatibility, structural reliability, workmanship, and associated elements of system safety.

Table 7.4-1. Structural and Mechanical Verification Test Requirements

Requirement	Payload/ Spacecraft	Subsystem/ Instrument	Unit (Component) Including Instrument Units (Components)
<u>Structural Loads</u>			
Modal Survey	*	T ²	*
Design Qualification	*	A, T/A ¹	*
Structural Reliability Primary & Secondary Structure	*	(A,T) ¹	*
<u>Vibroacoustics</u>			
Acoustics	T	T ²	T ²
Random Vibration	T ²	T ²	T
Sine Vibration	T ³ , T ⁴	T ³	T ³
Mechanical Shock	T	T ⁵	T ⁵
Mechanical Function	A, T	A, T	-
Pressure Profile	-	A, T ²	A
Mass Properties	A/T	A, T ²	*

Notes:

* = May be performed at payload or component level of assembly if appropriate.

A = Analysis required.

T = Test required.

A/T = Analysis and/or Test.

A, T/A¹ = Analysis and Test or Analysis only, using no-test factors of safety of 2.0 on yield and 2.6 on ultimate (with NASA concurrence).(A,T)¹ = Combination of fracture analysis and proof tests on selected elements, with special attention given to beryllium, composites, bonded joints, and weldments.T² = Test to be performed unless assessment justifies deletion.T³ = Test performed to simulate low frequency transient vibration and any sustained periodic mission environment, or to satisfy other requirement such as strength qualification.T⁴ = Test performed for ELV payloads, if practicable, to simulate transient and any sustained periodic vibration mission environment.T⁵ = Test required for self-induced shocks but may be performed at payload level of assembly for externally induced shocks.

7.4.1 Structural Loads Qualification

SMBA-235 The strength qualification test shall be accompanied by a stress analysis that demonstrates a positive margin for the design/analysis factors listed in Table 5.7.2-1.

SMBA-236 Test factors for prototype, protoflight, and acceptance shall follow Table 7.4.1-1.

Table 7.4.1-1. Test Factors and Durations

Test	Prototype Qualification	Protoflight Qualification	Acceptance
Structural Loads ^[1] Level Duration Centrifuge/Static Load ^[4] Sine Burst	1.25 x Limit Load 1 minute 5 cycles @ full level per axis	1.25 x Limit Load 30 seconds 5 cycles @ full level per axis	1.0 x Limit Load 30 seconds 5 cycles @ full level per axis
Acoustics Level ^[2] Duration	Limit Level + 3dB 2 minutes	Limit Level + 3dB 1 minute	Limit Level 1 minute
Random Vibration Level ^[4] Duration	Limit Level + 3dB 2 minutes/axis	Limit Level + 3dB 1 minute/axis	Limit Level 1 minute/axis
Sine Vibration ^[3] Level Sweep Rate	1.25 x Limit Level 2 oct/min	1.25 x Limit Level 4 oct/min	Limit Level 4 oct/min
Mechanical Shock Actual Device Simulated	2 actuations 1.4 x Limit Level 2 x Each Axis	2 actuations 1.4 x Limit Level 1 x Each Axis	1 actuations Limit Level 1 x Each Axis

Notes:

- [1] If qualified by analysis only, positive margins must be shown for factors of safety of 2.0 on yield and 2.6 on ultimate. Beryllium and composite materials cannot be qualified by analysis alone.
Note: Test levels for weldments, beryllium, bonded and composite structure, including metal matrix, are 1.25 x Limit Level for both qualification and acceptance testing.
- [2] As a minimum, the test level shall be equal to or greater than the workmanship level.
- [3] The sweep direction should be evaluated and chosen to minimize the risk of damage to the hardware. If a sine sweep is used to satisfy the loads or other requirements, rather than to simulate an oscillatory mission environment, a faster sweep rate may be considered, e.g., 6-8 oct/min to reduce the potential for over stress.
- [4] Shorter durations may be used in static testing if necessary to protect the hardware from damage due to facility limitations. If a shorter duration is used then the dwell time at load shall be sufficient to demonstrate that the target loading condition has been achieved within the specified tolerances, all test measurements have been recorded, and the structure is stable under the applied loading condition.

7.4.2 Acoustic Testing

SMBA-237 An acoustic test shall be performed on instrument and components unless an assessment of the hardware indicates that they are not susceptible to the expected acoustic environment, or responses are enveloped by random vibration testing. Acoustic testing deferral to higher levels of assembly requires Government approval.

SMBA-238 The instrument shall be designed to withstand the maximum expected launch vehicle flight acoustic environment during ascent as specified in Table 7.4.2-1 plus 3 dB.

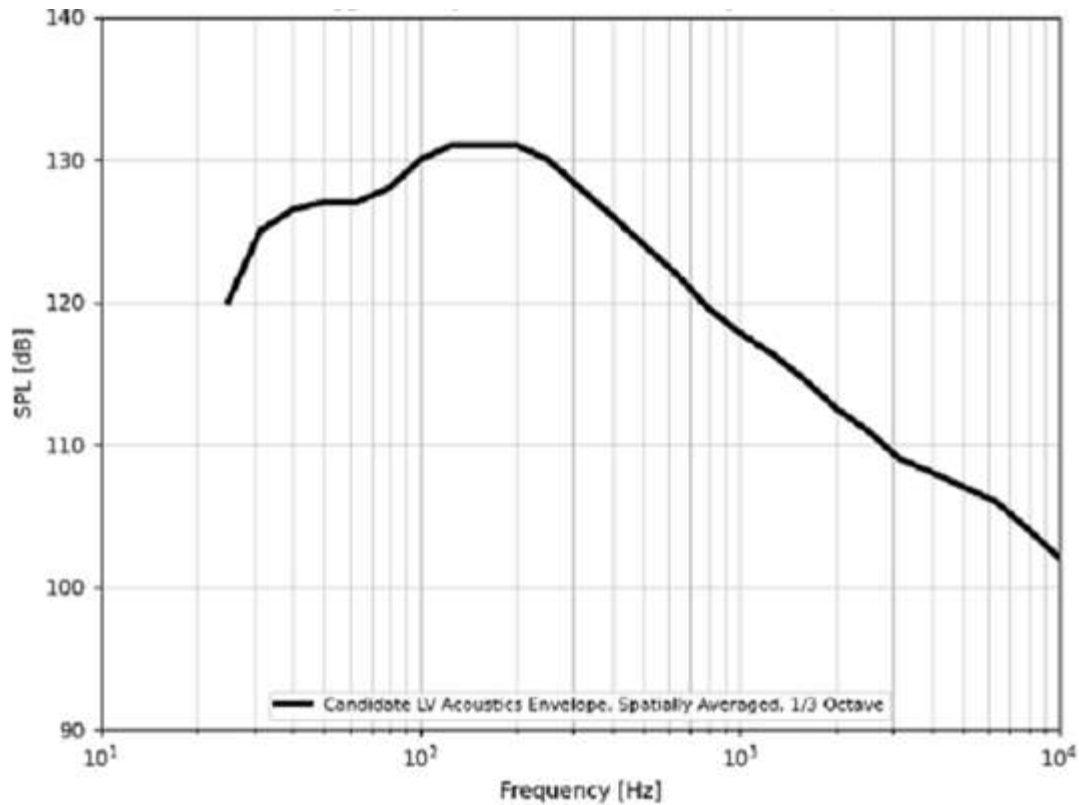


Figure 7.4.2-1. Payload Acoustics Maximum Predicted Environment (MPE) Design Envelope

Table 7.4.2-1. Payload Acoustics MPE Design Envelope

Candidate LV Payload Acoustics Design Envelope (P95/50 MPE)	
Frequency [Hz]	SPL [dB]
25	120
31.5	125
40	126.5
50	127
63	127
80	128
100	130
125	131
160	131
200	131
250	130
315	128
400	126
500	124
630	122
800	119.5
1000	117.8
1250	116.4
1600	114.5
2000	112.5
2500	111
3150	109
4000	108
5000	107
6300	106
8000	104
10000	102
OASPL ^[1]	140

Notes:

[1] Overall Sound Pressure Level

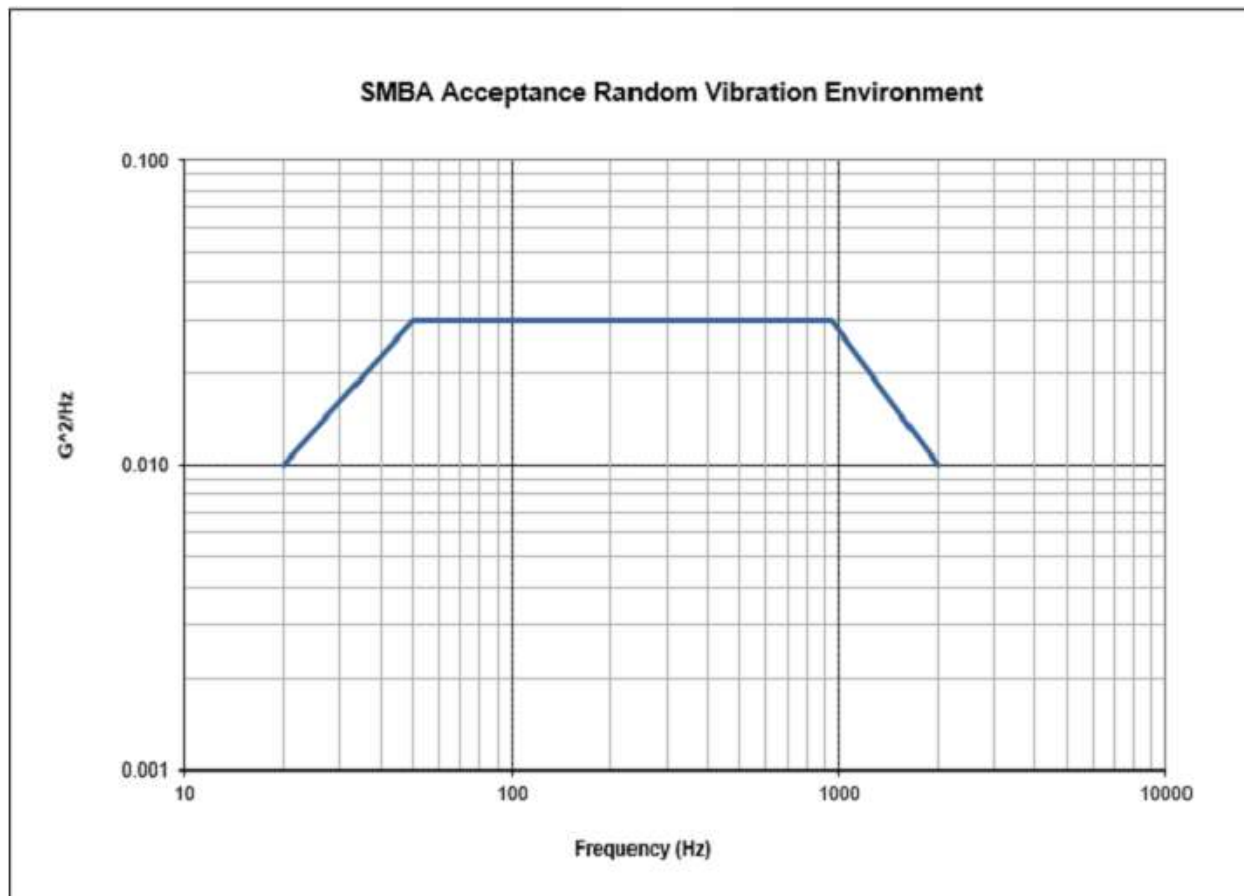
7.4.3 Random Vibration Testing

SMBA-239 All instrument flight units shall be subjected to random vibration testing in three axes.

SMBA-240 The Instrument shall be designed to withstand the maximum expected launch vehicle flight random vibration environment during ascent as specified in Table 7.4.3-1 plus 3 dB.

Table 7.4.3-1. SMBA Acceptance Random Vibration Environment

SMBA Acceptance Random	
FREQ (Hz)	ASD LEVEL (g^2/Hz)
20	0.010
50	0.03
950	0.03
2000	0.010
Overall	6.74 g_{rms}

**Figure 7.4.3-1. SMBA Acceptance Random Vibration Environment**

SMBA-241 The acceptance test levels for any separately mounted component shall be greater than minimum workmanship levels specified in Table 7.4.3-2.

7.4.3.1 Random Vibration After Rework

SMBA-242 As a minimum, reworked components shall be subjected to a single axis workmanship random vibration test to the minimum workmanship test levels specified in Table 7.4.3-2 and Figure 7.4.3-2.

The determination of axis is to be made based on the direction necessary to provide the highest excitation of the reworked area. Testing may be required in more than one axis if a single axis test cannot be shown to adequately test all of the reworked area. If the amount of rework or disassembly required is significant, then 3-axis testing to acceptance levels may be necessary if they are higher than workmanship levels.

Table 7.4.3-2. Component Minimum Workmanship Random Vibration Test Levels

Frequency (Hz)	ASD Level (g^2/Hz)
20	0.01
20-80	+3 dB/oct
80-500	0.04
500-2000	-3 dB/oct
2000	0.01
Overall	6.8 g_{rms}

The plateau acceleration spectral density (ASD) level may be reduced for components weighing between 45.4 and 182 kg, or 100 and 400 pounds according to the component weight (W) up to a maximum of 6 dB as follows:

	Weight in kg	Weight in lb
dB reduction	$=10 \log (W/45.4)$	$=10 \log (W/100)$
ASD(plateau) level	$=0.04 \cdot (45.4/W)$	$=0.04 \cdot (100/W)$

SMBA-243 The sloped portions of the spectrum shall be maintained at plus and minus 3 dB/oct. Therefore, the lower and upper break points, or frequencies at the ends of the plateau become:
 $FL = 80 (45.4/W) [\text{kg}]$ FL = frequency break point low end of plateau
 $= 80 (100/W) [\text{lb}]$
 $FH = 500 (W/45.4) [\text{kg}]$ FH = frequency break point high end of plateau
 $= 500 (W/100) [\text{lb}]$

SMBA-244 The test spectrum shall not go below $0.01 \text{ g}^2/\text{Hz}$. For components whose weight is greater than 182 kg or 400 pounds, the workmanship test spectrum is $0.01 \text{ g}^2/\text{Hz}$ from 20 to 2000 Hz with an overall level of $4.4 \text{ g}_{\text{rms}}$.

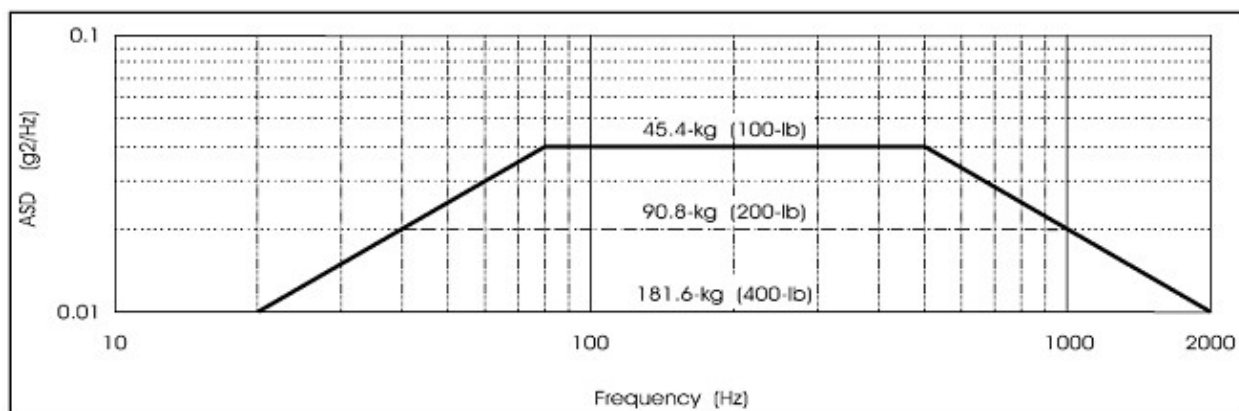


Figure 7.4.3-2. Minimum Workmanship Random Vibration Test Levels

7.4.4 Sine Vibration Testing

SMBA-245 All instrument flight units shall be subjected to sine sweep testing in three axes.

SMBA-246 Sine vibration testing shall be conducted with the instrument in the launch configuration. (If unit is powered during launch, the unit should be powered during testing.)

SMBA-247 There shall be one sweep from 5 Hz to 50 Hz for each axis.
Note: This is the typical frequency range of the sine test, but the range may be extended depending on the specific launch vehicle and the frequency content of the coupled loads analysis.

SMBA-248 The test sweep rate shall be 4 octaves/min for acceptance and protoflight and 2 octaves/min for qualification.

SMBA-249 The instrument shall be designed to withstand 1.25 times the maximum expected launch vehicle sine vibration environment values as specified in Table 7.4.4-1 in all three axes.

Table 7.4.4-1. SMBA Acceptance Sine Environment (All Axes)

Frequency (Hz)	Acceleration (g)
5 to 100	10

Notes:

- [1] Input levels may be notched to limit subsystem/component CG response to the design limit loads.
- [2] Low-frequency amplitude can be adjusted to stay within the stroke capability of the test facility.
- [3] Hardware shall show positive margins for acceptance level Sine vibration using the factors of safety defined in Table 5.7.2-1.
- [4] Acceleration levels applicable to hardware less than 200 kg.

SMBA-250	The instrument-level sine vibration tests shall be conducted to protoflight test levels unless demonstrated on an applicable qualification unit.
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7.4.5 Shock Testing

Shock testing will be performed at the satellite level for each of the shock environments expected during launch/commissioning including payload attach fitting separation shocks, solar array deployment shocks, appendage/mechanism deployment shocks, and instrument-induced shocks (if required).

SMBA-251	The instrument shall be qualified for all externally induced shock sources (i.e., spacecraft separation and any other externally induced source) at the instrument interface. The qualification level is the maximum predicted environment (MPE) at the instrument interface due to each shock source plus 3 dB. Note: The shock levels induced due to spacecraft separation and any other externally induced sources at the instrument interface are specified in the NEON Sounder Spacecraft to SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).
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SMBA-252	A shock susceptibility assessment shall be performed on all instrument components. If the component is not considered shock sensitive and if there is low potential for damage due to the shock environment, then shock testing may be deferred to the satellite-level of assembly.
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The potential for damage for non-shock sensitive items due to shock is quantified based on Figure 7.4.5-1. If the flight shock environment as shown on an SRS plot ($Q=10$) is enveloped by the appropriate curve shown in Figure 7.4.5-1, then the shock environment is considered benign and there is low risk in deferring the shock test to satellite-level. For the case in which the shock levels are above the curve, then instrument component level shock testing should be considered.

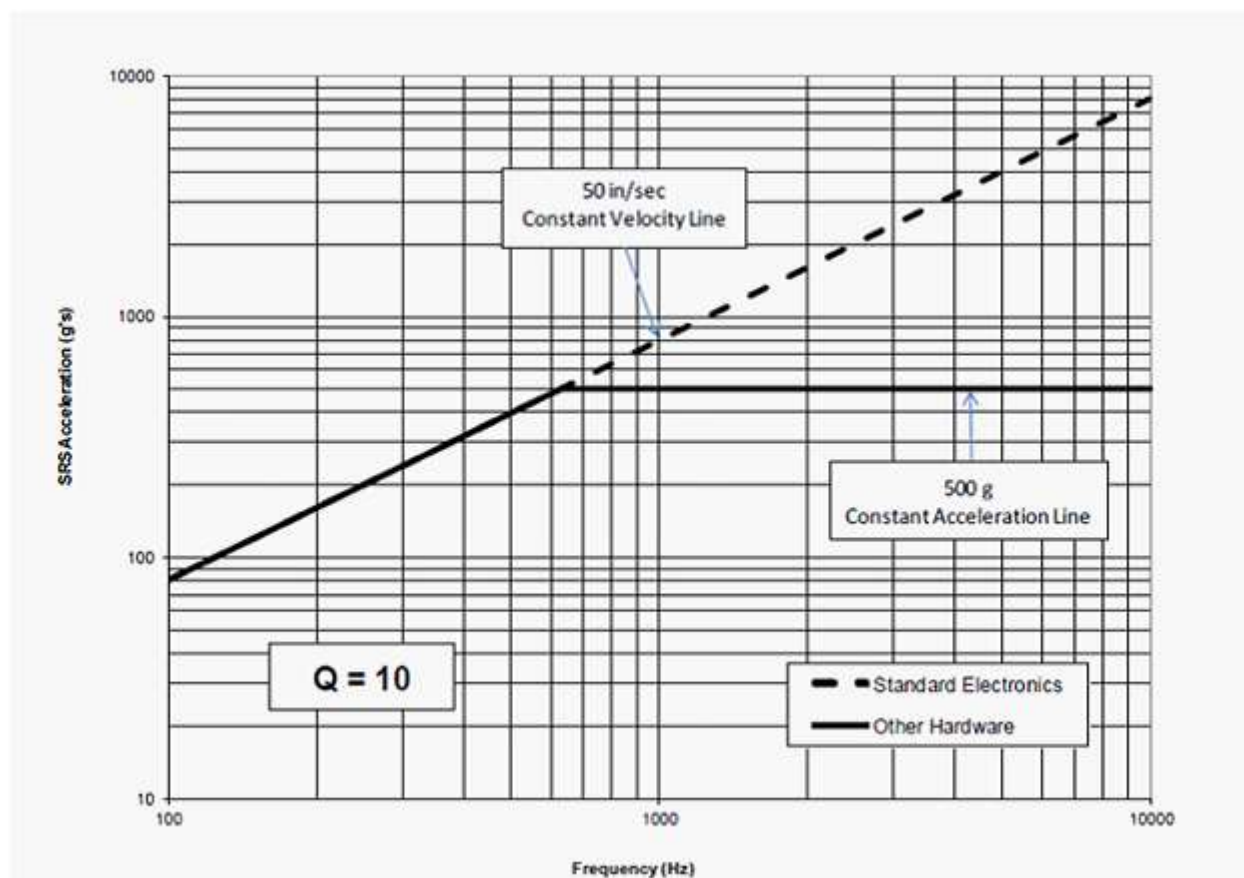


Figure 7.4.5-1. Shock Response Spectrum (SRS) for Assessing Component Test Requirements

SMBA-253 Instrument self-induced shock testing shall be accomplished by two actuations at the instrument level for each self-induced shock source (in order to account for the scatter associated with the actuation of the device) for the first flight unit, and a single actuation on subsequent units.

7.5 Mechanism and Deployment Test Requirements

SMBA-254 Mechanisms shall be life tested to at least two times the number of planned operational cycles plus two times the total number of ground cycles (including assembly, installation; and maintenance, functional, environmental, and run-in cycles) without any failure.

SMBA-255 Mechanisms and deployment and latching devices shall be tested to demonstrate expected functioning both prior to and following exposure to satellite-level environments.

The instrument should be designed to permit full range deployment testing of mechanisms after satellite integration during ground processing at the spacecraft provider facilities.

7.6 Thermal Vacuum Environmental Test Requirements

7.6.1 Thermal Vacuum Cycling

SMBA-256	Instrument thermal vacuum testing shall be performed to verify the thermal design margins to the operational temperature ranges. Note: Thermal design margins may be defined using GSFC-STD-7000 Section 2.6.3 as a guideline.
SMBA-257	All instrument hardware shall be subjected to a minimum equivalent of 8 thermal vacuum temperature cycles before instrument delivery. The instrument should be subjected to ambient air thermal cycles prior to starting instrument-level environmental testing. Following integration of the Government-furnished instruments, the integrated satellite will be subjected to additional thermal vacuum cycles as determined by the satellite integrator.
SMBA-258	To verify survival performance, instrument thermal vacuum testing shall include a minimum of one plateau at each survival temperature extreme for a minimum of 8 hours. Following each survival plateau, the instrument shall be returned to within its operating temperature limits to verify performance.
SMBA-259	Temperature dwell (plateau) periods at the instrument level of assembly shall be greater than or equal to 12 hours at each plateau.
SMBA-260	During the thermal vacuum cycling with exception to the survival plateaus, the instrument shall be operating, and its performance evaluated against specifications during transitions and temperature plateaus.
SMBA-261	All instrument systems that will be ON during launch shall be powered prior to starting chamber pumpdown and their performance verified during transition through the voltage breakdown/electrical discharge region.
SMBA-262	The chamber pressure after the electrical discharge checks are conducted shall be less than 1.33×10^{-3} Pa (1×10^{-5} torr).
SMBA-263	Thermal vacuum tests shall demonstrate satisfactory operation over the range of possible flight voltages.
SMBA-264	Thermal vacuum testing shall demonstrate satisfactory operation of the instrument in all modes including Safe and Survival and include demonstrations of Safe Mode recovery.
SMBA-265	Thermal vacuum testing shall demonstrate satisfactory operation of the instrument survival heaters, both primary and redundant, to extent possible while at survival temperature.

SMBA-266	Hot and cold turn-on shall be demonstrated during thermal vacuum tests.
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SMBA-267	For actively controlled systems such as cold plates, heaters, thermoelectric coolers, loop heat pipes, capillary pumped loops, or other devices with selectable/variable set points, a test temperature margin of 5°C shall be imposed on the respective interface temperature that is under control.
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7.6.2 Thermal Design Margin (Thermal Balance) Testing

SMBA-268	Thermal balance testing shall be performed at worst-case hot operational, worst-case cold operational, cold survival, and nominal operational, using the environmental temperature ranges (FM1 and any flight units with design changes that affect thermal performance). Note: Thermal balance testing may be defined using GSFC-STD-7000 Section 2.6.4 as a guideline.
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SMBA-269	Thermal design margins shall be verified under worst-case hot operational and cold operational, survival, and Safe Mode conditions.
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Stabilization may be considered to have been achieved when all temperature sensors indicate +/- 0.1 °C/hour for at least 4 consecutive hours.

SMBA-270	Acceptance ranges shall be calculated by adding a thermal margin of ± 5 deg C to the maximum expected temperature range per Figure 7.6.2-1. Note: For subsystems and lower-level components, this results in acceptance test levels being ± 10 deg C on top of the analytically determined extreme temperatures predicted from thermal models.
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SMBA-271	Protoflight ranges shall be calculated by adding ± 10 deg C to the maximum expected temperature range per Figure 7.6.2-1. Note: For subsystems and lower-level components, this results in protoflight test levels being ± 15 deg C on top of the analytically determined extreme temperatures predicted from thermal models.
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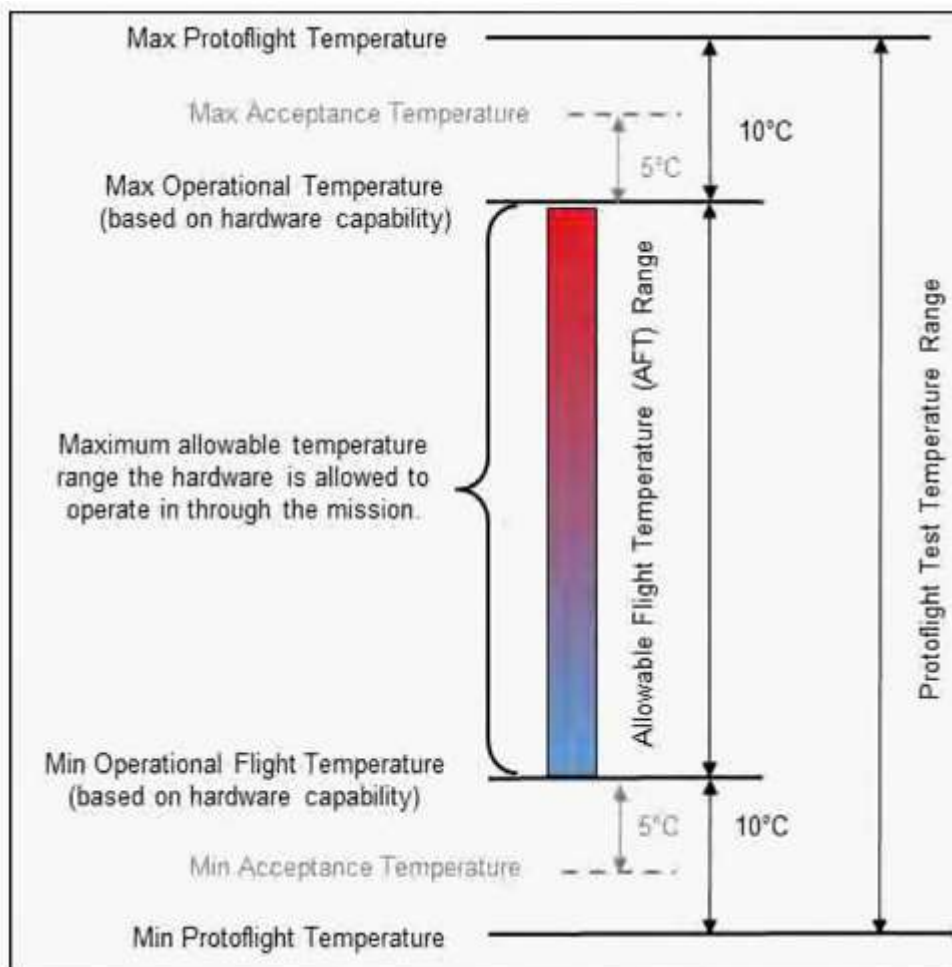


Figure 7.6.2-1. Qualification, Protoflight, and Flight Acceptance Thermal-Vacuum Temperatures

7.6.3 Contamination/Bake-Out Requirements

At the completion of instrument-level thermal vacuum testing, the Instrument should be thermally baked out following normal thermal vacuum multi-cycle functional testing at $< 1\text{E-}5$ torr during the last high temperature soak cycle followed by ambient temperature at test end.

Demonstration that the instrument has outgassed sufficiently should be given by using a minimum of seven Temperature Quartz Crystal Microbalances (either thermal or cryogenic) operating at a collection temperature of $-20\text{ }^{\circ}\text{C}$ to monitor out-gassing rate during bake out. All contamination bakeouts should include a qualitative molecular analysis by scavenger plate or coldfinger assembly.

SMBA-272 During the last hot cycle of the instrument-level thermal vacuum testing, the Instrument shall not outgas at a rate greater than $4.62\text{E-}7$ gram/second (TBR). Outgassing rate should be maintained for at least 4 continuous hours with a resolution of better than 10 minutes.

Multilayer insulation blankets, harnesses, and cables, and GSE electronic boxes/modules to be included in satellite-level thermal vacuum testing, should be baked out separately.

8 REQUIREMENTS VERIFICATION MATRIX

8.1 General Verification Requirements

Each verification will be carried out in accordance with approved procedures. See applicable contract requirements to determine which procedures require Government approval.

In the case of verification by test, the procedure will identify the specific test instrumentation (hardware and/or software) and/or special test equipment used.

8.2 Verification Cross Reference

As summarized in Appendix A - SMBA PRD Verification Cross Reference Matrix (VCRM), each requirement of this specification will be verified by the method (or combination of methods) specified in the CDRL SE-1 System Performance Verification Matrix (NS1-SMBA-CDRL-0001, NEON Series-1 SMBA Instrument CDRL DID). The approved CDRL SE-1 System Performance Verification Matrix will become the basis for the initial Appendix A. Subsequent changes to Appendix A will occur through approved CDRL SE-1 updates as required and the Goddard Space Flight Center (GSFC) Change Control Board (CCB) process.

8.3 Definition of Verification Methods

There are four acceptable requirement verification methods as defined below: Analysis, Inspection, Demonstration, and Test. Verification by Design is not an acceptable verification method. Generally, requirements that contained verification by design should be defined as Inspection of configuration-controlled design documentation, such as drawings or software.

8.3.1 Verification by Analysis

Verification by Analysis is the interpretation, interpolation, or extrapolation of analytical, empirical, or test data to show theoretical compliance with stated requirement. This method also applies to requirements to perform an analysis, or that specify how an analysis is to be performed.

8.3.2 Verification by Inspection

Verification by Inspection is an observation or examination of the item that does not require the item to be powered or operating. This includes:

- Certain requirements verified by simple mechanical measurements that do not require the item to be powered or operated.
- Hardware implementation constraints (such as required or prohibited materials or processes).
- Software implementation constraints (such as coding standards) – verified by examination of source code.
- Requirements for documentation and other requirements that are verified by the examination of documentation. The documentation should be configuration controlled at time of inspection.

8.3.3 Verification by Demonstration

Verification by Demonstration is an exhibition of the operability or supportability of an item under either controlled or intended service-use conditions. These verifications are oriented almost exclusively toward acquisition of qualitative data, not requiring use of special test equipment or processing of data outputs from the hardware.

8.3.4 Verification by Test

Verification by Test is an action that verifies an item's operability, supportability, performance capability, or other specified qualities when subjected to controlled conditions that are real or simulated. These verifications require use of special test equipment and sensors to obtain quantitative data for analysis as well as qualitative data derived from displays and indicators inherent in the item(s) for monitor and control. The Test method includes examination of output data from the unit under test that has been collected and/or processed by special test equipment. This also applies to requirements to perform a test, or that specify how a test is to be performed.

Appendix A Verification Cross Reference Matrix (VCRM)

TBS

Appendix B Acronyms/Abbreviations

Acronym/Abbreviation	Definition
°	Degrees
°C	Degrees Celsius/Centigrade
$\Delta\Phi$	Change in angle
ΔT	Change in time
λ	Wavelength
σ	Sigma
μ	Micro
μF	Microfarad
$\mu g/cm^2$	Micrograms per square centimeter
μm	Micrometer
μs	Microseconds
a	Acceleration
A	Ampere
ADC	Analog-to-Digital Converter
AFT	Allowable Flight Temperature
AR	Attitude Reference
ASD	Acceleration Spectral Density
ASTM	American Society for Testing and Materials
ATBD	Algorithm Theoretical Basis Document
ATMS	Advanced Technology Microwave Sounder
AVMP	Atmospheric Vertical Moisture Profile
AVTP	Atmospheric Vertical Temperature Profile
BIPM	Bureau International des Poids et Mesures
BW	Bandwidth
C	Celsius/Centigrade
C&DH	Command and Data Handling
C&T	Command and Telemetry
CCB	Change Control Board
CCSDS	Consultative Committee for Space Data Systems
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CE	Conducted Emission
CG	Center of Gravity
cm	Centimeter
CM	Common Mode
CMCE	Common Mode Conducted Emissions
CMD	Command
ConOps	Concept of Operations
CPT	Comprehensive Performance Test
CPU	Central Processing Unit

CS	Conducted Susceptibility
CTE	Calibration Test Equipment
d	Distance
DB	Direct Broadcast
dB	Decibel
dBm	Decibels Referenced to 1 milliwatt
dB μ A	Decibels Micro Amp
dB μ V	Decibels Micro Volt
dB μ V/m	Decibels Micro Volt per meter
DC	Direct Current
DID	Data Item Description
DM	Direct Mode
DOC	Department of Commerce
E-Field	Electric Field
ECEF	Earth Centered Earth Fixed
ECI	Earth Centered Inertial
EDR	Environmental Data Record
EEE	Electronic and Electrical Equipment
EEPROM	Electrically Erasable Read Only Memory
ELV	Expendable Launch Vehicle
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
f ₀	Tuned Frequency
FET	Field Effect Transistor
FH	Frequency upper break point
FL	Frequency lower break point
FOV	Field of View
FREQ	Frequency
FS	Factor of Safety
g	gram
g	gravity
G ² /Hz or g ² /Hz	Acceleration spectral density
G	Giga
GCT	NEON Series-1 Ground System Compatibility Test
GEVS	General Environmental Verification Specification
GHz	Gigahertz
GN&C	Guidance, Navigation, and Control
GPS	Global Positioning System
Grms	Root Mean Square Acceleration
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center

Hz	Hertz
I	Moment of Inertia
I&T	Integration and Test
ICD	Interface Control Document
IEST	Institute of Environmental Sciences and Technology
IFOV	Instantaneous Field of View
ISE	Instrument Systems Engineer
ISO	International Standards Organization
ITU	International Telecommunication Union
JPSS	Joint Polar Satellite System
JSC	Johnson Space Flight Center
K	Kelvin
k	Kilo
kg	Kilogram
kHz	Kilohertz
km	Kilometer
krad	Kilorad
L2	L-Band 2 used in navigation communications
lb	Pound
LEO	Low Earth Orbit
LEO&A	Launch, Early Orbit, and Activation
LISN	Line Impedance Stabilization Network
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Test
LTAN	Local Time Ascending Node
LV	Launch Vehicle
m	Meter
M	Mega
Mbps	Mega bits per second
MHz	Megahertz
MICD	Mechanical Interface Control Document
MIL-STD	Military Standard
mm	Millimeter
MPE	Maximum Predicted Environment
msec	millisecond
MUF	Model Uncertainty Factor
MW	LEO Microwave
MWS	Microwave Sounder
N/A	Not Applicable/Available
N-m	Newton-meter
NA	Not Applicable/Available
NASA	National Aeronautics and Space Administration
NEDT	Noise Equivalent Delta Temperature

NEON	Near Earth Orbit Network
NESDIS	National Environmental Satellite, Data, and Information Service (NOAA)
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NS1	NEON Series-1 Project
NSOSA	NOAA Satellite Observation System Architecture Study
NTE	Not to exceed
NWP	Numerical Weather Prediction
OASPL	Overall Sound Pressure Level
Oct	Octave
oz-in	Ounce - inch
Pa	Pascal
PAC	Percent Area Coverage
PDR	Preliminary Design Review
PER	Pre-Environmental Review
PMEF	Primary Mission Essential Functions
PPF	Payload Processing Facility
PROM	Programmable Read-Only Memory
PRD	Performance Requirements Document
Pwr	Power
q	Water content or humidity
Q	Quality
QH	Quasi Horizontal
QV	Quasi Vertical
RAM	Random Access Memory
RDR	Raw Data Record
RE	Radiated Emissions
REC	Receive
RF	Radio Frequency
RFI	Radio Frequency Interference
RMS	Root Mean Square
RS	Radiated Susceptibility
RSS	Root Sum Squared
SC	Spacecraft
S/C	Spacecraft
SDR	Sensor Data Record
SE	Systems Engineering
SEE	Single Event Effect
sec	Second
Si	Silicon
SI	International System of Units (Metric)

SMBA	Sounder for Microwave-Based Applications
SOW	Statement of Work
sqrt	Square root
SPL	Sound Pressure Level
SRF	Spectral Response Function
SRS	Shock Response Spectrum
T	Temperature
T	Torque
T _{avail}	Minimum available torque under worst case conditions at any time of life
TBD	To Be Determined
TBR	To Be Reviewed
TBS	To Be Supplied
TDR	Temperature Data Record
TH	Horizontally polarized brightness temperature
TID	Total Ionizing Dose
TLM	Telemetry
TM	Torque Margin
TMON	Telemetry Monitoring
TNID	Total Non-Ionizing Dose
TOD	Time of Day
TV	Vertically polarized brightness temperature
UTC	Coordinated Universal Time
V	Volt
VCRM	Verification Cross Reference Matrix
VDC	Volts Direct Current
W	Watt
W	Weight
WGS	World Geodetic System